

Cold accelerated sampling with subpolynomial dimension dependence

Abstract

We study sampling from a smooth strongly log-concave probability measure in dimension d , with condition number κ . We construct a cold-start sampler whose number of original-gradient and exact positive-scalar proximal queries is at most $C_\eta \kappa^{2/3} d^\eta$ on every execution, for each fixed $\eta > 0$, at total variation accuracy $1/10$. In the regime $\kappa \leq d^{18}$, the construction combines a conditional Hamiltonian half-turn, a centered randomized quadrature, and a Gaussian-observation posterior step. Centering makes the leading endpoint error cost quadratic in the endpoint perturbation and leaves the averaged transport current unchanged; only a smaller Picard error is divided by the observation variance. A finite-depth noisy-mean construction then replaces every smoothed-score evaluation by original gradients. Standard BPS with windowed thinning covers the remaining condition numbers. A separate padding argument gives independently chosen positive powers of the dimension and inverse accuracy. Only initialization can use an exact proximal query; a supplied near-minimizer makes the algorithm gradient-only for every condition number. The bounds concern exact-real queries, not finite-bit running time.

Preliminary draft. The arguments and exposition remain subject to further scrutiny. No complete independent verification of the proofs is claimed; corrections and suggestions are welcome.

Source and contribution boundary. The analytic noisy-sampling and conditional-mean construction used here is adapted from OpenAI, *Subpolynomial query complexity for well-conditioned log-concave sampling* [14]. Its Theorem 1.1 gives an every-execution $C_\eta d^\eta$ value-and-gradient query bound, for each fixed $\eta > 0$, at total variation accuracy $1/10$ when the minimizer is known and $I \preceq \nabla^2 V \preceq 2I$. The main improvement here is an extension beyond that fixed-conditioning setting: for arbitrary supplied positive curvature bounds with ratio κ , the present sampler has an every-execution $C_\eta \kappa^{2/3} d^\eta$ cap at the same accuracy, counting original gradients and at most one exact positive-scalar proximal query in initialization; with the stated supplied near-center point it uses gradients alone. The Gaussian derivative estimates, stationary noisy-mean representation, baseline finite numerical formulas and recursive evaluation routines, and sampling Gaussian reserve are inherited results. Explicit citations accompany the adapted statements, and [Section J](#) separates them from the additional mean-root, affine-composition, and accelerated outer-sampler arguments.

Contents

1	Introduction	3
1.1	Technical overview	4
2	Preliminaries	5
2.1	Probability metrics and smoothing	5
2.2	A common lifted target	6
2.3	Initialization and energy	7

2.4	Exact-real computation	8
3	The finite conditional Hamiltonian step	8
3.1	Interpolation and Picard iteration	8
3.2	The comparison endpoint	9
4	A residual bound for cold conditional laws	10
4.1	Smoothed derivative estimates	10
4.2	Centered error and path error	12
5	Entropy decay for the finite exact-score chain	12
5.1	A posterior observation with a biased input	13
5.2	The corrected transport current	13
5.3	One loss for the numerical fixed-label step	14
6	Implementation by original gradients	16
6.1	Noisy sampling and noisy means	16
6.2	A finite affine composition	17
6.3	Removing the noise from a small posterior	18
6.4	The phase specialization	19
7	Parameters and the full query bound	20
7.1	Fixed orders and step size	20
7.2	The numerical entropy budget	21
7.3	Checking the original-gradient replacement	22
7.4	Large condition numbers and small dimensions	23
7.5	An arbitrary-accuracy consequence	24
A	Conditional gradient means and noisy sampling	24
A.1	Convexity, moments, and comparisons	25
A.2	Coupling and smoothing conventions	27
B	Gaussian conditional jets and probability transport	28
B.1	Cumulants from the Poincaré inequality	28
B.2	The transport equation	31
C	Adjoint divergence and material derivatives	32
C.1	Material derivatives of the conditional jets	32
C.2	Contractions and the expressions they generate	33
C.3	One final vector dimension factor	37
D	A stationary representation of a noisy mean	39
D.1	Transport Jacobians and the zero endpoint	39
D.2	A kernel for the centered observable	40
D.3	Stationarity and the path integral	41
D.4	Derivatives on a stationary trajectory	43
E	Finite Picard formulas for conditional means	44
E.1	Probability transport by global Picard iteration	45

E.2	The kernel action through harmonic trajectories	46
E.3	The zero endpoint of the kernel integral	47
E.4	A globally Lipschitz approximate centering velocity	49
E.5	The mean formula and the order of parameter choices	50
E.6	Bounds for the finite formulas	52
F	Recursive evaluation using original gradients	54
F.1	Why Gaussian center noise helps	55
F.2	The numerical input and the implementation task	55
F.3	Two routines and two different kinds of depth	56
F.4	Evaluating an expression at a noisy center	57
F.5	Query-preserving shifts and the deterministic cap	58
F.6	Terminal errors and propagation of scales	60
F.7	Stability for arbitrary fixed seeds	61
F.8	Standalone accuracy and the primitive theorem	61
G	The mean-root specialization	63
G.1	The recursive computation	64
G.2	Starting at a mean routine	64
G.3	Reserving a fresh Gaussian	64
H	The translated-posterior inequality	65
I	The momentum-refresh transport estimate	65
I.1	Entropy-sized corrected currents	66
I.2	Common-target compatibility in every dimension	66
I.3	Conditional laws from one endpoint Fisher information	67
I.4	An exact product bridge	68
I.5	Joining the currents	69
I.6	The finite-entropy domain	70
J	Source correspondence and attribution	71

1 Introduction

We consider a probability density $\pi \propto \exp(-V)$ on $\mathbb{R}^d$, where $V \in C^2(\mathbb{R}^d)$ and, for $0 < \alpha \leq \beta$,

$$\alpha I \preceq \nabla^2 V(x) \preceq \beta I, \quad \kappa := \beta/\alpha, \quad x \in \mathbb{R}^d. \quad (1.1)$$

The question is whether accelerated dependence on the condition number can coexist with an arbitrarily small power of the dimension, without a supplied warm start. Gaussian smoothing and high-order Picard approximation address the dimension dependence, but a cold-start entropy proof must also control the numerical transition under its actual nonstationary law.

Our sampler centers the leading endpoint error before applying a Gaussian-observation posterior step. The resulting cold entropy estimate remains valid for a finite Picard approximation.

We count gradient and exact positive-scalar proximal queries in the exact-real model of [Section 2](#).

Theorem 1.1 (Cold sampling). *Assume (1.1), with the positive bounds α, β supplied to the algorithm. For every fixed $\eta > 0$ there is a constant $C_\eta < \infty$ and a randomized procedure, starting at any*

supplied numerical point, whose output law $\hat{\pi}$ satisfies

$$\text{TV}(\hat{\pi}, \pi) \leq 1/10.$$

On every execution, the total number of gradient and exact positive-scalar proximal queries is at most $C_\eta \kappa^{2/3} d^\eta$. Computation between queries is unrestricted and exact-real. At most one exact proximal query is used, in initialization. If a point x_{ref} with $\|\nabla V(x_{\text{ref}})\| \leq \sqrt{\alpha d}$ is supplied, the algorithm uses gradients alone for every κ .

The constant and dimension threshold depend on the fixed exponent η . The theorem fixes the accuracy before choosing the approximation orders. It does not assert a uniform logarithmic dependence on arbitrary inverse accuracy. For large κ , we use standard BPS with windowed thinning [12], extending its event-count estimate to our explicit near-center Gaussian initialization.

Relation to earlier constructions. Smoothed Picard HMC [5] combines Gaussian smoothing, Picard iteration, and higher-order quadrature under only a Hessian bound on the original potential. We use the same distinction between an exact-gradient numerical transition and its original-gradient implementation. We give the Gaussian derivative estimates, stationary mean construction, Picard circuits and recursive noisy sampler with complete proofs, adapting the construction of OpenAI [14]. The mean-root specialization and affine composition then connect these tools to the present cold sampler. The entropy proof uses a corrected optimal-transport current; its dimension-independent momentum-refresh estimate is proved in Section I.

The outer construction developed here is the composition of a centered endpoint formula with the small posterior observation and the corresponding cold entropy estimate. The exact conditional Hamiltonian trajectory in our proof is a comparison process; the executed moderate-regime algorithm has finitely many Picard layers and force evaluations.

A close accelerated predecessor. Chen, Chewi, Rakhlin, and Zhang [6, Definition 3 and Theorem 3.2(ii)] give gradient-only high-accuracy sampling guarantees in Rényi divergence. Their abstract records a two-thirds power of the condition number, a one-third power of dimension, and polylogarithmic inverse-accuracy dependence; the theorem states a high-probability query bound, with its initialization and Rényi-order dependence explicit. The present Theorem 1.1 fixes total variation accuracy and gives an every-execution query cap in the stated gradient–proximal model. Its arbitrary-point initialization uses at most one exact positive-scalar proximal query; with the supplied near-center point specified in the theorem, it uses gradients alone. The accuracy metric, initialization convention, and type of query guarantee differ, so the smaller dimension exponent here is not a uniform domination of the high-accuracy result in [6].

1.1 Technical overview

The first smoothing fixes a target π_a for the whole computation. At a phase entrance, the label $Y = X + hP$ has reference law π_{a+h^2} . Holding Y fixed turns the remaining coordinate into a near-Gaussian conditional law. A half-period Hamiltonian motion in that conditional law moves the physical position while retaining Y as its backward endpoint label; a small momentum Ornstein–Uhlenbeck step then produces the accelerated transport-current decrease. A Gaussian-observation posterior step provides a separate conditional-entropy decrease of order $\Gamma \asymp h/\sqrt{\kappa}$.

The numerical challenge is to retain these two decreases. Let R denote the input conditional entropy. High-order interpolation and an entropy transfer give a residual estimate of order $h^2(R + d^{-K})$, including the derivative of the residual in the incoming conditional variables. Randomizing

only the final quadrature centers its leading error pointwise in those variables. A change-of-variables argument then bounds the entropy increase by $Ch^4(R + d^{-K})$, while the transport current has exactly zero averaged change. Finite Picard iteration leaves a smaller, generally biased endpoint discrepancy with mean square $Ch^8(R + d^{-K})$, plus a separately bounded iteration tail. Only that smaller discrepancy is compared through the observation noise. The two costs relative to the entropy decrease are therefore

$$\frac{h^4}{\Gamma} \asymp h^3 \sqrt{\kappa} \quad \text{and} \quad \frac{h^8}{\Gamma^2} \asymp (h^3 \sqrt{\kappa})^2.$$

Taking h slightly smaller than $\kappa^{-1/6}$ leaves a contraction rate of order $\kappa^{-2/3}$ per phase.

Three laws remain distinct throughout the proof. The exact Hamiltonian flow is used only to analyze the centered endpoint. The finite exact-score phase has its own entropy contraction, proved with the numerical entropy error bound included. Finally, the original-gradient implementation is compared to that finite phase in total variation. Moments used to sum implementation errors are taken only along the entropy-controlled finite exact-score chain.

For bounded condition number and a supplied minimizer, [Theorem 1.1](#) gives the subpolynomial gradient-query upper bound as an immediate special case.

Proof structure. The proof starts from the finite transition in [Algorithm 3.1](#). [Lemma 4.1](#) bounds the interpolation residual and its derivative under a cold conditional law; [Proposition 4.2](#) converts these bounds into a centered entropy cost and a smaller mean-square path error. [Lemma 5.1](#) controls the latter through the posterior observation, and [Proposition 5.4](#) proves entropy decay for the finite exact-score chain. [Propositions 6.2](#) and [6.3](#) replace its score and posterior calls by original gradients. Finally, [Section 7](#) combines these estimates with initialization and the BPS branch to prove the full query cap.

2 Preliminaries

After the change of variables $z = \sqrt{\beta}x$, we may assume

$$\kappa^{-1}I \preceq \nabla^2 V \preceq I. \quad (2.1)$$

A gradient query to $V(z/\sqrt{\beta})$ uses one original gradient query, and its proximal query of width s equals $\sqrt{\beta} \operatorname{prox}_{(s/\beta)V}(z/\sqrt{\beta})$. Thus this normalization preserves the declared oracle model. Below V denotes the normalized potential; the output is returned to the original coordinates by $z \mapsto z/\sqrt{\beta}$.

2.1 Probability metrics and smoothing

For probability measures μ, ν , write $\operatorname{KL}(\mu \parallel \nu) = \int \log(d\mu/d\nu) d\mu$, with value $+\infty$ if μ is not absolutely continuous with respect to ν , and $\operatorname{TV}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|$. The quadratic Wasserstein distance is $W_2^2(\mu, \nu) = \inf_{\lambda \in \mathcal{C}(\mu, \nu)} \int \|x - y\|^2 \lambda(dx, dy)$, where $\mathcal{C}$ is the set of couplings. For a differentiable U , write $D_U(z, x) = U(z) - U(x) - \nabla U(x) \cdot (z - x)$. Vector norms are Euclidean; matrix operator and Hilbert–Schmidt norms carry subscripts op and HS . All optimal maps below are quadratic-cost Brenier maps, not arbitrary transports.

Let $\gamma_d = \mathcal{N}(0, I)$, and define

$$\pi_a = \pi * \mathcal{N}(0, aI), \quad V_a = -\log \pi_a, \quad g_a = \nabla V_a, \quad \Pi_a = \pi_a \otimes \gamma_d. \quad (2.2)$$

We use a for the fixed smoothing variance, reserving η for the arbitrarily small dimension exponent. For $b > 0$ and $y \in \mathbb{R}^d$, the restricted Gaussian distribution is

$$R_{b,y}^V(dx) \propto \exp\left(-V(x) - \frac{\|x - y\|^2}{2b}\right) dx. \quad (2.3)$$

We also write R_b^V for the Markov kernel $y \mapsto R_{b,y}^V$. These densities and smoothed scores are analytic objects; the model does not provide them as oracles.

2.2 A common lifted target

The entropy analysis is stated for a generic base potential, then applied to the fixed-smoothed target. Let U be a base potential with a supplied lower bound $m_U I \preceq \nabla^2 U \preceq I$, where $m_U > 0$. Define $\pi_U \propto e^{-U}$, $\pi_{U,b} = \pi_U * \mathbf{N}(0, bI)$, $U_b = -\log \pi_{U,b}$, and $\Pi_U = \pi_U \otimes \gamma_d$. For conditional width $h > 0$, write

$$q_{h,y}^U(dz) \propto e^{-\|z\|^2/2 - U(y+hz)} dz, \quad m_{\text{obs}} = \frac{m_U}{1 + m_U h^2}, \quad \Gamma_U(h) = h\sqrt{m_{\text{obs}}}. \quad (2.4)$$

The physical entrance state is (X, P) , the label is $Y = X + hP$, and the normalized conditional coordinate is $Z = -P$. The reference law factors exactly as $\pi_{U,h^2}(dy)q_{h,y}^U(dz)$. A conditional evolution keeps Y fixed. If its terminal normalized position is Z' , reconstruction and momentum refresh are

$$X' = Y + hZ', \quad P' = \rho Z' + \sqrt{1 - \rho^2} G, \quad \rho = e^{-\Gamma_U(h)/32}. \quad (2.5)$$

The main construction takes $U = V_a$, so $\pi_U = \pi_a$, $m_U = (\kappa + a)^{-1}$ and $\pi_{U,h^2} = \pi_{a+h^2}$. We abbreviate Π_V to Π and $q_{h,y}^U$ to q_y when the base potential and width have been fixed.

We use the logarithmic Sobolev and transport inequalities for a c -strongly log-concave reference ν :

$$\text{KL}(\mu \| \nu) \leq \frac{I(\mu \| \nu)}{2c}, \quad W_2^2(\mu, \nu) \leq \frac{2}{c} \text{KL}(\mu \| \nu). \quad (2.6)$$

We also write $H(\mu | \nu) = \text{KL}(\mu \| \nu)$. The relative score is $\ell = \nabla \log(d\mu/d\nu)$. Here $I(\mu \| \nu) = \int \|\nabla \log(d\mu/d\nu)\|^2 d\mu$ is relative Fisher information when defined, and is $+\infty$ otherwise. We also use Pinsker's inequality, $\text{TV}^2 \leq \text{KL}/2$, and entropy duality, $\mathbb{E}_\mu f \leq \text{KL}(\mu \| \nu) + \log \mathbb{E}_\nu e^f$. These classical facts are cited in [3, 15]; their applications to nonsmooth densities are by approximation or the variational definition of entropy.

Lemma 2.1 (Smoothing and posterior identities). *Under (2.1), V_a is smooth and*

$$(\kappa + a)^{-1} I \preceq \nabla^2 V_a \preceq I, \quad g_a(y) = \mathbb{E}_{R_{a,y}^V} \nabla V(X). \quad (2.7)$$

Moreover,

$$\|\nabla V(y) - g_a(y)\| \leq a\|g_a(y)\| + \sqrt{ad}. \quad (2.8)$$

For $s > 0$, an exact draw from $R_{s,c}^{V_a}$ is obtained by independently drawing $Z \sim R_{a+s,c}^V$ and $G \sim \gamma_d$, then setting

$$X = \frac{ac + sZ}{a + s} + \sqrt{\frac{as}{a + s}} G. \quad (2.9)$$

In this joint draw, $\text{Law}(Z | X, c) = R_{a,X}^V$. This remains true for an arbitrary law of c and any retained external anchors, provided the displayed draws are fresh conditionally on them. Finally, $Y \sim \pi_a$ followed by $X \sim R_{a,Y}^V$ gives $X \sim \pi$.

Proof. Differentiation of the convolution gives the mean identity and $\nabla^2 V_a = a^{-1}I - a^{-2} \text{Cov}(X | Y = y)$. The posterior curvature is at least $\kappa^{-1} + a^{-1}$, so Poincaré gives the lower bound. Differentiating through the original potential instead gives $\nabla^2 V_a = \mathbb{E}[\nabla^2 V | Y] - \text{Cov}(\nabla V | Y) \preceq I$. The convolution is smooth because V has quadratic growth and the Gaussian factor allows differentiation locally to every order. The posterior mean is $y - ag_a(y)$ and its covariance has trace at most ad . Lipschitzness of ∇V and Jensen's inequality prove (2.8).

Let ϕ_b be the centered Gaussian density of covariance bI . The product identity

$$\phi_a(x - z)\phi_s(c - x) = \phi_{a+s}(c - z)\phi_{as/(a+s)}\left(x - \frac{ac + sz}{a + s}\right) \quad (2.10)$$

shows that the joint density in (2.9), conditional on any c , is proportional to $e^{-V(z)}\phi_a(x - z)\phi_s(c - x)$. Marginalizing z proves the claimed law of X . Conditioning on x cancels the c -dependent factor and proves $\text{Law}(Z | X, c) = R_{a,X}^V$, also after mixing over c and external anchors. The final assertion is the same Gaussian conditioning identity without the second observation. $\square$

2.3 Initialization and energy

The algorithm constructs a near-center point using at most one exact positive-scalar proximal query.

Lemma 2.2 (Cold initialization). *Query $g = \nabla V(y)$. If $g = 0$, put $x_{\text{ref}} = y$. Otherwise set the positive scalar s below and query the displayed proximal point:*

$$s = \kappa^{3/2}\|g\|/\sqrt{d}, \quad x_{\text{ref}} = \text{prox}_{sV}(y).$$

Then $\|\nabla V(x_{\text{ref}})\| \leq \sqrt{d/\kappa}$. For independent $G_0, G_1, G_2 \sim \gamma_d$, set

$$X_* = x_{\text{ref}} + G_0, \quad X_0 = X_* + \sqrt{a}G_1, \quad P_0 = G_2.$$

Then $H_0 := \text{KL}(\text{Law}(X_0, P_0) \| \Pi_a) \leq B_0 := d(1 + \log \kappa)/2$. The unsmoothed Gaussian $\mu_* = \text{Law}(X_*)$ also satisfies

$$D_2(\mu_* \| \pi) := \log \int \left(\frac{d\mu_*}{d\pi} \right)^2 d\pi \leq \Delta_0 := d + \frac{d}{2} \log \kappa. \quad (2.11)$$

The main chain uses X_0 ; the large-condition-number branch uses X_* without the extra smoothing.

Proof. Let x_* be the minimizer. Strong monotonicity gives $\|y - x_*\| \leq \kappa\|g\|$. Proximal optimality and convexity imply $s\|\nabla V(x_{\text{ref}})\| = \|y - x_{\text{ref}}\| \leq \|y - x_*\|$, proving the gradient bound. Strong convexity then gives $V(x_{\text{ref}}) - V(x_*) \leq d/2$. For $X = x_{\text{ref}} + G_0$, smoothness and the upper bound $\int e^{-V} \leq e^{-V(x_*)}(2\pi\kappa)^{d/2}$ yield

$$\text{KL}(\text{Law} X \| \pi) \leq V(x_{\text{ref}}) - V(x_*) + \frac{d}{2} \log \kappa \leq B_0.$$

Convolution is a common Markov kernel, so data processing gives the same bound for X_0 relative to π_a . The independent reference momentum adds no entropy.

For the order-two Rényi bound, let q be the density of $\mathbf{N}(x_{\text{ref}}, I)$ and write $g_0 = \nabla V(x_{\text{ref}})$. The same quadratic upper bound on V and Gaussian completion give

$$\begin{aligned} \int \frac{q(x)^2}{\pi(x)} dx &\leq Z_V(2\pi)^{-d} e^{V(x_{\text{ref}})} \int e^{-\|z\|^2/2 + g_0 \cdot z} dz \\ &\leq \kappa^{d/2} \exp\{V(x_{\text{ref}}) - V(x_*) + \|g_0\|^2/2\}. \end{aligned}$$

The suboptimality and gradient bounds above prove (2.11). $\square$

Lemma 2.3 (Energy from entropy). *If $H = \text{KL}(\mu \parallel \Pi_a) < \infty$ and $W = V_a - \min V_a$, then*

$$\mathbb{E}_\mu \left[W(X) + \frac{\|P\|^2}{2} \right] \leq 2H + 3d \log 2. \quad (2.12)$$

In particular, for $h \leq 1$ and $Y = X + hP$,

$$\mathbb{E}_\mu \|g_a(Y)\|^2 + \mathbb{E}_\mu \|P\|^2 \leq C(H + d).$$

Proof. Convexity around a minimizer and a change of variables give $\mathbb{E}_{\pi_a} e^{tW} \leq (1 - t)^{-d}$ for $0 < t < 1$. The independent Gaussian momentum contributes $(1 - t)^{-d/2}$. Apply entropy duality at $t = 1/2$. A convex 1-smooth function satisfies $\|\nabla V_a\|^2 \leq 2W$ by the descent inequality. Thus $\|g_a(Y)\|^2 \leq 2\|g_a(X)\|^2 + 2h^2\|P\|^2$, giving the last claim. $\square$

2.4 Exact-real computation

A proximal query means $\text{prox}_{sV}(y) = \arg \min_x \{V(x) + \|x - y\|^2/(2s)\}$ with any positive scalar s . The algorithm uses at most one such query, in [Lemma 2.2](#); every subsequent query is an original gradient. With a supplied center satisfying that lemma's gradient bound, no proximal query is used. Random primitives are independent real uniforms, exponentials and Gaussian vectors. Computation between oracle calls is unrestricted and exact-real. No potential value, Hessian, posterior sample or smoothed score is silently supplied. A finite-bit running-time guarantee would require a separate analysis.

3 The finite conditional Hamiltonian step

The physical target stays equal to Π_a throughout the phase chain. At input (x, p) , the fixed label $y = x + hp$ has stationary law π_{a+h^2} , and the conditional coordinate $z = -p$ has density

$$q_y(dz) \propto \exp \left(-\frac{\|z\|^2}{2} - V_a(y + hz) \right) dz. \quad (3.1)$$

The change of variables has unit Jacobian. The Hessian of the negative logarithm of q_y lies between I and $(1 + h^2)I$. Set

$$a = h^2, \quad m_{\text{obs}} = (\kappa + a + h^2)^{-1}, \quad \Gamma = h\sqrt{m_{\text{obs}}}, \quad \rho = e^{-\Gamma/32}. \quad (3.2)$$

The second smoothing in m_{obs} comes from the label distribution. Omitting it changes the current normalization.

3.1 Interpolation and Picard iteration

Fix integers $n \geq 1$, $m \geq 2$, and $J \geq 1$. Divide $[0, \pi]$ into n equal cells. On each cell use the m Chebyshev–Lobatto nodes, obtained by rescaling $(1 - \cos(j\pi/(m-1)))/2$, $0 \leq j \leq m-1$. Let $\mathcal{I}$ be the piecewise degree- $(m-1)$ interpolation operator. Its Lebesgue constant depends only on m . The integrals of its basis polynomials against $\sin(s-t)$ are deterministic scalar weights, and their absolute row sums are bounded by a constant C_m , uniformly in n and s . This follows by integrating the pointwise Lebesgue bound over an interval of length at most π .

For $z_0 = -p$ and a fresh $q \sim \gamma_d$, define simultaneous Picard iterates by

$$z^{[0]}(s) = \cos(s)z_0 + \sin(s)q,$$

$$z^{[j+1]}(s) = \cos(s)z_0 + \sin(s)q - h \int_0^s \sin(s-t) \mathcal{I}[g_a(y + hz^{[j]}(\cdot))](t) dt. \quad (3.3)$$

Only values at the fixed interpolation nodes are needed to construct the next layer. Evaluation at an additional time s uses the same integrated interpolant; it does not solve an extra differential equation. Write $\tilde{z} = z^{[j]}$, and $\tilde{f}_s = g_a(y + h\tilde{z}(s))$.

Draw a clock $U \in [0, \pi]$ with density $\sin(u)/2$, independently of the input and q . For a time-indexed vector f , put

$$\mathcal{Q}(f, U) = \int_0^\pi \sin(s) \mathcal{I}f(s) ds + 2\{f_U - \mathcal{I}f(U)\}. \quad (3.4)$$

The clock can be generated as $U = \arccos(1 - 2B)$ with B uniform on $[0, 1]$. Every coefficient of (3.4) is explicit after this draw.

Algorithm 3.1 (One finite exact-score phase). Given (x, p) , form $y = x + hp$, $z_0 = -p$, and draw independent $q \sim \gamma_d$ and U as above.

1. Compute (3.3), including the partial integral at U , and the endpoint

$$\hat{Z} = -z_0 - h\mathcal{Q}(\tilde{f}, U).$$

2. Draw an independent $G \sim \gamma_d$ and set $O = \hat{Z} + \sqrt{\Gamma}G$.
3. Put $v = \Gamma/(1+\Gamma)$, $s = h^2v$, and $c = y + hO/(1+\Gamma)$. Draw $Z \sim R_{a+s,c}^V$ and $G' \sim \gamma_d$ independently. Set

$$X' = \frac{ac + sZ}{a + s} + \sqrt{\frac{as}{a + s}}G', \quad Z' = \frac{X' - y}{h}.$$

4. Draw an independent $G'' \sim \gamma_d$ and return $(X', \rho Z' + \sqrt{1 - \rho^2}G'')$.

All Gaussian draws named as fresh are independent of all earlier auxiliary randomness.

In Step 3, completing the square shows that Z' is the exact posterior for prior q_y and observation $O = Z' + \sqrt{\Gamma}G$. The representation in original- V coordinates follows from (2.9). The phase above has finitely many exact smoothed-score calls, but it is not yet an original-gradient algorithm. [Section 6](#) implements its whole observation and posterior draw with finite original-gradient procedures.

3.2 The comparison endpoint

For the analysis only, let (z_s, q_s) solve

$$\dot{z}_s = q_s, \quad \dot{q}_s = -z_s - hg_a(y + hz_s), \quad (z_0, q_0) = (z_0, q), \quad 0 \leq s \leq \pi. \quad (3.5)$$

This globally Lipschitz Hamiltonian flow preserves $q_y \otimes \gamma_d$. Put $f_s = g_a(y + hz_s)$ and $r_s = f_s - \mathcal{I}f(s)$. The randomized endpoint evaluated on the exact path is

$$Z_c = -z_0 - h\mathcal{Q}(f, U) = z_\pi + \xi_U, \quad \xi_U = -2h\{r_U - \mathbb{E}_U r_U\}. \quad (3.6)$$

The expectation in (3.6) is conditional on (y, z_0, q) , so $\mathbb{E}_U \xi_U = 0$ pointwise. The algorithm never computes this expectation or the exact path. Its computable endpoint is $\hat{Z} = Z_c + e$, where e is the finite-path error. Separating ξ_U from e is essential: only e will be charged through the small variance Γ .

4 A residual bound for cold conditional laws

The conditional-jet and stationary-derivative inputs used in this section are inherited from OpenAI [14, Lemma 3.1 and Lemma 5.4]. Their specialization and transfer to the conditional laws needed by the outer construction are developed below.

The stationary interpolation estimate must also hold under a cold conditional input. Bochner's identity gives stationary smallness of the residual's derivative, while a dimension-independent Lipschitz bound lets a transport-entropy coupling transfer both estimates to the input law.

4.1 Smoothed derivative estimates

For a tensor, a one-slot split is its operator norm when one tensor slot is treated as the input and all remaining slots as a Euclidean tensor output. The all-split cumulant estimate proved in Lemma B.1 gives for each fixed integer $j \geq 1$,

$$\|D^j g_a\|_{\text{one-slot split}} \leq C_j a^{-(j-1)/2}, \quad \|Dg_a\|_{\text{op}} \leq 1. \quad (4.1)$$

These are bounds on the smoothed potential; no corresponding higher-derivative assumption is made on V . To verify the physical scale directly, let $X \sim R_{a,y}^V$ and consider the joint vector $S = a^{-1/2}(\nabla V(X), X)$. Posterior Poincaré has constant at most a , while the map from X to S is $\sqrt{2/a}$ -Lipschitz. Hence S satisfies the polynomial Poincaré inequality with constant at most two. Its linear functionals have exponential moments in a neighborhood of zero because the posterior has Gaussian tails and ∇V has at most linear growth. Differentiating the Gaussian posterior in its center gives $D_y^j g_a = a^{-j} \text{cum}(\nabla V(X), X, \dots, X)$. Applying the dimension-independent all-split cumulant bound to S yields the factor $a^{-j} a^{(j+1)/2} = a^{-(j-1)/2}$ in (4.1). The sharper first-derivative bound follows from (2.7).

The stationary autonomous-flow estimate is proved in Lemma D.4. Its form used here is as follows. Let e^{-H} on $\mathbb{R}^D$, $D \leq Cd$, have Gaussian tails, bounded Hessian, and positive-order score derivatives bounded in every one-slot split by $C_j R^{-(j-1)}$. Suppose the stationary flow has Lie derivative $\nabla^* A \nabla$, where the adjoint is relative to e^{-H} , A is skew-symmetric, and every derivative $D^j A$ has one-slot split norm at most $C_j \lambda R^{-j}$. If the smooth observable ϕ satisfies $\|D^j \phi\|_F \leq C_j b \sqrt{d} R^{-j}$ for $j \geq 1$, and the fields and derivatives used have polynomial growth, then, for each fixed $k \geq 1$ and $0 < R \leq 1$,

$$\left(\mathbb{E} \left\| \frac{d^k}{ds^k} \phi(Z_s) \right\|^2 \right)^{1/2} \leq C_k b \sqrt{d} (\lambda R^{-2})^k. \quad (4.2)$$

Only this L^2 assertion is used.

For (3.5), take $D = 2d$, $R = 1$, $\lambda = 1$, and $b = h$. Indeed the invariant negative logarithm is

$$H_y(z, q) = \frac{\|z\|^2 + \|q\|^2}{2} + V_a(y + hz).$$

Its Hessian lies between I and $2I$. For $j \geq 2$, its higher score derivatives are bounded by $C_j h^{j+1} a^{-(j-1)/2} \leq C_j h^2$, since $h^2 \leq a$. The observable $\phi(z, q) = g_a(y + hz)$ has one-slot derivatives at most $C_j h$; passing to Frobenius norm costs at most $\sqrt{2d}$ once. The Hamiltonian coefficient A is a constant skew matrix. The possibly large constant $g_a(y)$ is removed by positive-order derivatives. Consequently, uniformly in y ,

$$\|\partial_s^m f_s\|_{L^2(q_y \otimes \gamma_d)} \leq C_m h \sqrt{d}. \quad (4.3)$$

All fields have polynomial growth for each fixed y , so the hypotheses of (4.2) apply.

Lemma 4.1 (Conditional residual). *Fix $K > 0$ and $\omega > 0$. Set $n = \lceil d^\omega \rceil$, and choose a fixed interpolation order m with $\omega m > 2K + 4$, with strict slack to absorb fixed logarithmic factors. For sufficiently large d , every y , every h with $0 < h \leq 1/4$ and $a = h^2$, every input law λ_y , and every deterministic time $s \in [0, \pi]$,*

$$\mathbb{E}_{\lambda_y \otimes \gamma_d} [\|r_s\|^2 + \|D_{(z_0, q)} r_s\|_{\text{HS}}^2] \leq C_m h^2 [\text{KL}(\lambda_y \| q_y) + d^{-K}]. \quad (4.4)$$

The constants and threshold are independent of y, h, κ , and λ_y .

Proof. Step 1: Stationary interpolation error. Taylor's integral remainder, the fixed cell Lebesgue constant, and (4.3) yield $\mathbb{E}_* \|r_s\|^2 \leq C_m d h^2 n^{-2m}$, where $\mathbb{E}_*$ is expectation under $q_y \otimes \gamma_d$.

Step 2: Derivatives in the incoming conditional variables. The Jacobian of the vector field in (3.5) is uniformly bounded. Its first variation is therefore bounded in operator norm on $[0, \pi]$. For a unit incoming direction v , the second variation has Hilbert–Schmidt norm at most $Ch^3/\sqrt{a} \leq Ch^2$: the forcing is $-h^3 D^2 g_a$ with one slot fixed, and (4.1) bounds exactly this split. The variational equation is linear with bounded coefficients. Composing with $g_a(y + h z_s)$ gives

$$\|Df_s\|_{\text{op}} \leq Ch, \quad \|D^2 f_s[v, \cdot]\|_{\text{HS}} \leq Ch^2/\sqrt{a} + Ch^3 \leq Ch.$$

Interpolation is a bounded linear combination on each cell, so the same bounds hold for r_s . Thus $O_1 = \|r_s\|$ and $O_2 = \|Dr_s\|_{\text{HS}}$ are $C_m h$ -Lipschitz in (z_0, q) , with y fixed. Also $\mathbb{E}_* \|D^2 r_s\|_F^2 \leq C_m d h^2$.

More explicitly, write $w = (z_0, q) \in \mathbb{R}^{2d}$. The preceding fixed-direction estimate is

$$\sup_{\|v\|=1} \|D^2 r_s(w)[v, \cdot]\|_{\text{HS}} \leq C_m h.$$

Hence $w \mapsto Dr_s(w)$, valued in the Hilbert space of d -by- $2d$ matrices, is $C_m h$ -Lipschitz, as is its Hilbert–Schmidt norm. The full second-derivative Frobenius norm instead satisfies

$$\|D^2 r_s(w)\|_F^2 = \sum_{j=1}^{2d} \|D^2 r_s(w)[e_j, \cdot]\|_{\text{HS}}^2 \leq 2d C_m^2 h^2.$$

This identifies the single dimension factor used in the next step.

Step 3: Stationary derivative smallness. Write $A = \mathbb{E}_* \|r_s\|^2$, $Y = \mathbb{E}_* \|Dr_s\|_{\text{HS}}^2$, and $Z = \mathbb{E}_* \|D^2 r_s\|_F^2$. For the positive weighted Laplacian $L = -\Delta + \nabla H_y \cdot \nabla$, integration by parts and Bochner's identity give

$$Y = \langle r_s, Lr_s \rangle, \quad \|Lr_s\|_{L^2}^2 \leq Z + 2Y.$$

Hence $Y \leq 2A + \sqrt{AZ}$, and $\mathbb{E}_* F \leq C_m d h^2 n^{-m}$ for $F = O_1^2 + O_2^2$. Gaussian tails and polynomial growth justify these identities by cutoffs.

Step 4: Transfer by transportation. Let $\nu = q_y \otimes \gamma_d$ and $\mu = \lambda_y \otimes \gamma_d$. The vector-valued observable $O = (\|r_s\|, \|Dr_s\|_{\text{HS}})$ is globally $C_m h$ -Lipschitz by Step 2, after enlarging C_m to account for its two coordinates. If $R_y = \text{KL}(\mu \| \nu) < \infty$, the curvature-one transport–entropy inequality gives a coupling (A, B) of μ, ν with $\mathbb{E} \|A - B\|^2 \leq 2R_y$. Minkowski's inequality therefore yields

$$(\mathbb{E}_\mu F)^{1/2} \leq (\mathbb{E}_\nu F)^{1/2} + C_m h \sqrt{2R_y}.$$

Squaring and using Step 3 gives $\mathbb{E}_\mu F \leq C_m h^2 (R_y + d n^{-m})$. The retained choice $\omega m > 2K + 4$ in particular implies $d n^{-m} \leq d^{-K}$ beyond the fixed threshold, including any fixed logarithmic factors. This proves (4.4). When $R_y = \infty$ the assertion is immediate. The independent clock U can also be averaged because the estimate is uniform in deterministic s . $\square$

4.2 Centered error and path error

Let $R_y = \text{KL}(\lambda_y \| q_y)$, and regard the input law of (z_0, q) as $\lambda_y \otimes \gamma_d$. The next statement is the local estimate used by the entropy argument.

Proposition 4.2 (The two endpoint errors). *For $C_m h^2 \leq 1/4$, the centered endpoint and finite endpoint satisfy*

$$\begin{aligned} \text{KL}(\text{Law } Z_c \| q_y) &\leq R_y + S_y, & S_y &\leq C_m h^4 (R_y + d^{-K}), \\ \mathbb{E} \|\hat{Z} - Z_c\|^2 &\leq C_m h^8 (R_y + d^{-K}) + \tau_{J,y}, & \tau_{J,y} &\leq C_J h^4 (C_m h^2)^{2J} E_{0,y}^2. \end{aligned} \quad (4.5)$$

Here $E_{0,y} = \sup_s \|z^{[0]}(s) - z_s\|_{L^2}$. The centered displacement has zero averaged change in every functional affine in the endpoint Z_c with coefficient depending only on y .

Proof. Let $\nu(dz dq) \propto e^{-H_y(z,q)} dz dq$, and let Φ be the exact Hamiltonian endpoint map. It preserves ν , with uniformly bounded derivative and inverse derivative. Put $\mu_* = \Phi_\#(\lambda_y \otimes \gamma_d)$, so $\text{KL}(\mu_* \| \nu) = R_y$, and write the centered perturbation as $\Xi_U = (\xi_U \circ \Phi^{-1}, 0)$. The maps $T_U = \text{Id} + \Xi_U$ are global diffeomorphisms because $\|D\Xi_U\|_{\text{op}} \leq C_m h^2 < 1/2$. Change of variables, followed by convexity of relative entropy, gives

$$\text{KL}(\mathbb{E}_U(T_U)_\# \mu_* \| \nu) - R_y \leq \mathbb{E}_{U,\mu_*} [H_y(w + \Xi_U) - H_y(w) - \log \det(I + D\Xi_U)].$$

Here $\mathbb{E}_U \Xi_U = \mathbb{E}_U D\Xi_U = 0$ pointwise. Taylor's theorem and $|\log \det(I + A) - \text{tr } A| \leq C \|A\|_{\text{HS}}^2$ for $\|A\|_{\text{op}} \leq 1/2$ therefore bound the right side by $C \mathbb{E}(|\Xi_U|^2 + \|D\Xi_U\|_{\text{HS}}^2)$. The bounded derivative of Φ^{-1} and [Lemma 4.1](#) bound this by $C_m h^4 (R_y + d^{-K})$. Discarding momentum proves the entropy bound; pointwise centering also proves the affine-current claim.

For the second line, subtract the exact integral equation from [\(3.3\)](#). Stable interpolation rows and 1-Lipschitzness of g_a give, for the largest nodewise L^2 error E_j ,

$$E_{j+1} \leq C_m h^2 E_j + C_m h \sup_s \|r_s\|_{L^2}.$$

The same bound applies to continuous evaluation, including the independent clock. Summing the geometric series and using [Lemma 4.1](#) yields

$$\sup_s \|\tilde{z}(s) - z_s\|_{L^2} \leq C_m h^2 \sqrt{R_y + d^{-K}} + (C_m h^2)^J E_{0,y}.$$

The coefficient sum of $\mathcal{Q}$ is bounded by C_m . Its force argument changes by $h(\tilde{z} - z)$, and the endpoint multiplies the force by a further h . Squaring proves the second line of [\(4.5\)](#). $\square$

For a physical input law of entropy H relative to Π_a , the free-path equation and Lipschitz stability give $\mathbb{E}_Y E_{0,Y}^2 \leq C h^2 (H + d)$ by [Lemma 2.3](#). In particular we will use the conservative bound $\mathbb{E}_Y E_{0,Y}^2 \leq C(H + d)$. This remainder is not proportional to the conditional entropy $R = \mathbb{E}_Y R_Y$ and cannot be removed when $R = 0$.

5 Entropy decay for the finite exact-score chain

The posterior observation has two roles. It contracts the conditional entropy and controls the change of the transport current. A finite-path bias can increase entropy before this observation, so the relevant nonnegative quantity is the conditional entropy loss augmented by its explicit numerical entropy error bound.

5.1 A posterior observation with a biased input

Lemma 5.1 (Robust Gaussian observation). *Let q be 1-strongly log-concave, let $Z \sim \mu$ with $H_c = \text{KL}(\mu \| q) < \infty$, and let $e = e(Z, A)$ depend on an auxiliary record, with $\mathbb{E}\|e\|^2 < \infty$. For $\Gamma > 0$, draw $G \sim \gamma_d$ independently, set $O = Z + e + \sqrt{\Gamma}G$, and apply the exact reference Bayes decoder for prior q and observation variance Γ . Define*

$$H_O = \text{KL}(\text{Law}(O) \| q * \text{N}(0, \Gamma I)) , \quad B = H_c + \frac{\mathbb{E}\|e\|^2}{2\Gamma} .$$

If H_{out} is the output entropy relative to q , then

$$H_{\text{out}} \leq \frac{H_O}{1 + \Gamma} \leq \frac{B}{1 + \Gamma} . \quad (5.1)$$

There is an analytical coupling of the original and decoded marginals such that

$$\mathbb{E}\|Z_{\text{out}} - Z\|^2 \leq \frac{2\Gamma}{1 + \Gamma}(B - H_O) \leq \frac{2\Gamma}{1 + \Gamma}(B - H_{\text{out}}) . \quad (5.2)$$

Proof. We first prove entropy contraction of the reference decoder. Let $q_t = q * \text{N}(0, tI)$, let P_t be Gaussian convolution of functions with variance t , and let f be an observation density relative to q_Γ . Its decoded density relative to q is $P_\Gamma f$. Put $h_t = P_{\Gamma-t} f$ and $E(t) = \int h_t \log h_t \, dq_t$. The normalization $\int h_t \, dq_t = 1$ is independent of t . Heat differentiation and integration by parts give

$$E'(t) = \frac{1}{2} \int \frac{\|\nabla h_t\|^2}{h_t} \, dq_t \geq \frac{E(t)}{1 + t} ,$$

where logarithmic Sobolev applies because q_t is $(1 + t)^{-1}$ -strongly log-concave. Thus $E(t)/(1 + t)$ is nondecreasing, and $E(0) \leq E(\Gamma)/(1 + \Gamma)$.

Give the reference pair (Z, A) law $q(dz)\mu(dA \mid Z = z)$ and its observation the unshifted conditional law $\text{N}(Z, \Gamma I)$. The full actual/reference joint entropy is exactly B . Marginalization followed by decoder contraction proves (5.1). The same chain rule, then discarding A conditionally on O , gives

$$\mathbb{E}_O \text{KL}(\text{Law}(Z \mid O) \| q(\cdot \mid O)) \leq B - H_O .$$

Every reference posterior has curvature at least $1 + \Gamma^{-1}$. Its conditional T_2 inequality therefore gives the first bound in (5.2); the second follows from $H_O \geq H_{\text{out}}$. This is a comparison of marginal laws and does not change or reveal the executed transition's auxiliary randomness.

The heat calculation is first made for bounded densities bounded away from zero, using cutoffs and interior times. Normalized truncation, heat continuity and lower semicontinuity of entropy extend it to finite-entropy observation laws. No input Fisher-information assumption is needed. $\square$

5.2 The corrected transport current

Use the generic base potential U , width $h \leq 1/4$ and rate $\Gamma = \Gamma_U(h)$ from Section 2.2. For a phase law μ of finite entropy, write

$$H = \text{KL}(\mu \| \Pi_U) , \quad H_Y = \text{KL}(\text{Law}(X + hP) \| \pi_{U, h^2}) , \quad R = H - H_Y = \mathbb{E}_Y \text{KL}(\text{Law}(-P \mid Y) \| q_{h, Y}^U) . \quad (5.3)$$

Let T be the quadratic-cost Brenier map from the actual label law to π_{U, h^2} . Set $\zeta(Y) = Y - T(Y)$, $\Psi = U_{h^2}$, and

$$J(\mu) = \sqrt{m_{\text{obs}}} \{ \mathbb{E}[P \cdot \zeta(Y)] - h \mathbb{E}[\Psi(Y) - \Psi(T(Y))] \} . \quad (5.4)$$

The transport is an analytical object and is never computed by the algorithm.

Lemma 5.2 (Current comparison and momentum-refresh dissipation). *For every finite-entropy phase law, $|J| \leq 2H$. At the exit of a fixed-label block, use the backward current*

$$J^- = \sqrt{m_{\text{obs}}} \{ \mathbb{E}[Z \cdot \zeta(Y)] + h \mathbb{E}[\Psi(Y) - \Psi(T(Y))] \}.$$

Let μ' be the phase law after the momentum-refresh step $P' = \rho Z + \sqrt{1 - \rho^2}G$ at fixed $X = Y + hZ$, and let H', R' be its entropy and conditional entropy defined by (5.3). Let H_{before} be the physical entropy immediately before that OU update. Then

$$J(\mu') - J^- \leq -\frac{\Gamma}{4}H' + 81\Gamma R' + 2066D, \quad D = H_{\text{before}} - H' \geq 0. \quad (5.5)$$

Proof. The proof for this generic base potential is given in Section I, specifically (I.4) and (I.12). There, (X, P) denotes an arbitrary pre-refresh physical law and $P' = \rho P + \sqrt{1 - \rho^2}G$. Apply that proof with the pre-refresh momentum $P = Z$ and position $X = Y + hZ$. Its labels $X - hP$ and $X + hP'$ are exactly the old fixed label Y and the new forward label. Thus the signs in both current definitions agree, with no additional momentum reflection. The finite-entropy extension is part of that proof. $\square$

For the remainder of this section take $U = V_a$. Then $\Pi_U = \Pi_a$, $q_{h,Y}^U$ is (3.1), and $\Gamma_U(h)$ is precisely (3.2).

Lemma 5.3 (Conditional Hamiltonian current). *For the exact conditional flow (3.5), the entrance and exit currents satisfy*

$$J_{\text{HMC}}^- - J_{\text{in}} \leq 12\Gamma h^2 R. \quad (5.6)$$

Proof. Variation of constants for $z'' + z = -hg_a(Y + hz)$ gives $z_\pi + z_0 = -h \int_0^\pi \sin s g_a(Y + hz_s) ds$. Consequently, with the label transport fixed,

$$J_{\text{HMC}}^- - J_{\text{in}} = -\Gamma \int_0^\pi \sin s \mathbb{E}[g_a(Y + hz_s) \cdot \zeta(Y) - \{\Psi(Y) - \Psi(T(Y))\}] ds.$$

Reference preservation of the extended Hamiltonian flow and data processing bound its conditional position entropy at every time by the entrance conditional entropy R . Lemma H.1 therefore bounds the expectation in brackets below by $-6h^2 R$ (discard its nonnegative Bregman term). Since $\int_0^\pi \sin s ds = 2$, the claim follows. All integrals are finite by finite-entropy second moments, the Lipschitz force and flow, and label T_2 ; no density approximation or differentiation of the input law is required. $\square$

5.3 One loss for the numerical fixed-label step

Let $H_k, H_{Y,k}, R_k, J_k$ refer to the input of phase k . Average the two endpoint bounds over Y and set

$$A_k = S_k + I_k, \quad I_k = \frac{\mathbb{E}\|\hat{Z} - Z_c\|^2}{\Gamma}.$$

Denote the post-observation conditional entropy by $R_{\text{out},k}$, and define the augmented loss

$$L_k = R_k + A_k - R_{\text{out},k}. \quad (5.7)$$

Lemma 5.1 and Proposition 4.2 give $R_{\text{out},k} \leq (R_k + S_k + I_k/2)/(1 + \Gamma) \leq (R_k + A_k)/(1 + \Gamma)$. Since $0 < \Gamma \leq 1$, this implies $L_k \geq 0$ and $\Gamma R_k \leq 2L_k$. If D_{k+1} is the momentum OU refresh entropy loss, the exact chronology is

$$H_{k+1} = H_k + A_k - L_k - D_{k+1}. \quad (5.8)$$

The fixed-label step can increase entropy; no nonnegative numerical entropy drop is assumed. Finite entrance entropy and the finite entropy error bounds S_k, I_k imply finite $R_{\text{out},k}$ by [Lemma 5.1](#). The unchanged label entropy $H_{Y,k}$ then makes the actual reconstructed post-observation phase law finite-entropy. Thus [Lemma 5.2](#) applies to that law, without an exact-HMC input, initial Fisher information, or a stationary density ceiling.

Proposition 5.4 (Finite-score entropy contraction). *Put $\varpi = 2^{-16}$, $A_F = 1 - 83\varpi$, and $B_F = 1 + 2\varpi$. Suppose at each phase*

$$A_k \leq \delta_0 \Gamma R_k + \beta_0 \Gamma H_k + b, \quad \delta_0 \leq 1/16, \quad \beta_0 \leq \varpi/32, \quad b \geq 0. \quad (5.9)$$

Then the chain in [Algorithm 3.1](#) satisfies

$$H_N \leq \frac{B_F}{A_F} \left(1 + \frac{\varpi \Gamma}{16 B_F} \right)^{-N} H_0 + \frac{16 B_F b}{A_F \varpi \Gamma}. \quad (5.10)$$

In particular $\sup_k H_k \leq C(H_0 + b/\Gamma)$ for an absolute constant.

Proof. Centering preserves the backward current. The observation coupling has mean-square displacement at most $2\Gamma L_k$; label T_2 gives $m_{\text{obs}} \mathbb{E}|\zeta|^2 \leq 2H_{Y,k}$. Together with the exact Hamiltonian current bound, Young's inequality yields

$$\varpi \Delta J_{\text{fixed}} \leq 12\varpi \Gamma h^2 R_k + L_k/32 + 32\varpi^2 \Gamma H_{Y,k}.$$

Set $F_k = H_k + \varpi J_k - 81\varpi \Gamma R_k$. Current comparison gives

$$A_F H_k \leq F_k \leq B_F H_k. \quad (5.11)$$

Combining the current bounds with $H_{k+1} = H_k + A_k - L_k - D_{k+1}$ cancels the new conditional entropy:

$$F_{k+1} - F_k \leq A_k - \frac{31}{32} L_k - (1 - 2066\varpi) D_{k+1} + \varpi \Gamma (81 + 12h^2) R_k + 32\varpi^2 \Gamma H_{Y,k} - \frac{\varpi \Gamma}{4} H_{k+1}.$$

Use $\Gamma R_k \leq 2L_k$, $81 + 12h^2 \leq 82$, and $H_{Y,k} \leq H_k \leq H_{k+1} + L_k + D_{k+1}$. The inequalities

$$\frac{31}{32} - 164\varpi - 32\varpi^2 \geq \frac{15}{16}, \quad 1 - 2066\varpi - 32\varpi^2 \geq \frac{15}{16}, \quad \frac{1}{4} - 32\varpi \geq \frac{1}{8}$$

therefore imply

$$F_{k+1} - F_k \leq A_k - \frac{15}{16} (L_k + D_{k+1}) - \frac{\varpi \Gamma}{8} H_{k+1}. \quad (5.12)$$

Assumption (5.9) gives $A_k \leq L_k/8 + \beta_0 \Gamma (H_{k+1} + L_k + D_{k+1}) + b$. Since $\beta_0 \leq \varpi/32$ and $\Gamma \leq 1$, absorb the two loss terms to obtain

$$F_{k+1} \leq F_k - \frac{\varpi \Gamma}{16} H_{k+1} + b \leq F_k - \frac{\varpi \Gamma}{16 B_F} F_{k+1} + b.$$

Iterate this scalar recursion and use $A_F H_k \leq F_k \leq B_F H_k$. This proves the stated entropy bound. $\square$

We next compare this finite exact-score chain to its original-gradient implementation in total variation.

6 Implementation by original gradients

The finite phase can be written as a triangular scalar-affine computation in conditional gradient means. Replacing each mean independently at its exact center would not implement that computation: the centers are themselves outputs of earlier approximate evaluations. We therefore supply each center with Gaussian noise and absorb that noise into the random seeds of the next mean evaluation. An independent final Gaussian converts a Wasserstein comparison of the whole output into total variation.

Throughout this section, fix $e \in (0, 1)$, an integer $K_M \geq 2$, and $t > 0$ with $40(K_M + 2)t < 1/4$, and put $\delta = d^{-e}$. Take d above the fixed threshold in [Theorem A.1](#), in particular large enough that $\delta \leq 1/2$. We write $B_d = C(1 + \log d)^C$, allowing the fixed constants to depend on these orders but not on the input center or oracle answers.

6.1 Noisy sampling and noisy means

For $F \in C^2(\mathbb{R}^d)$ with $\text{Lip}(\nabla F) \leq \delta$, define

$$\nu_{x,r}(\mathrm{d}z) \propto e^{-\|z\|^2/2 - F(x+rz)} \mathrm{d}z, \quad m(x) = \mathbb{E}_{\nu_{x,r}} \nabla F(x + rZ), \quad D_F(x) = \sqrt{d} + \|\nabla F(x)\|, \quad (6.1)$$

where $0 < r \leq 1$. In the technical appendices the same mean is denoted $u_{x,r}$ to retain its scale argument. No minimizer is supplied. When invoking the technical construction we set its order $K_0 = K_M$. The noisy near-quadratic sampling theorem, [Theorem A.1](#), returns an approximation to $Z + G$, $Z \sim \nu_{x,r}$, with pointwise total variation error at most $B_d \delta^{K_M} D_F(x)$ and every-execution query cap $B_d \delta^{-t}$. It retains a final independent Gaussian reserve of covariance $I/4$, with a pre-reserve W_2 bound of the same order against $\nu_{x,r} * \mathcal{N}(0, 3I/4)$.

The next version starts the recursion at its mean routine. Its proof is in [Section G](#); the change of root is separate from the sampling statement in [Theorem A.1](#).

Construction provenance. This is the mean-root specialization of OpenAI’s conditional-mean construction [14, Sections 6–7]; the additional mean-root statement is proved in [Section G](#).

Proposition 6.1 (A standalone noisy mean). *Fix $C_0, q_0 \geq 0$ and $\sigma > 0$. Suppose*

$$\max\{\delta r^2, \delta r/\sigma\} \leq C_0(1 + \log d)^{q_0} \delta. \quad (6.2)$$

There is a finite gradient procedure $M_\sigma(x; G)$ with deterministic seed slots and query cap $B_d \delta^{-t}$ such that

$$W_2(\text{Law}(M_\sigma(x)), \mathcal{N}(m(x), \sigma^2 I)) \leq B_d \sigma \delta^{K_M} D_F(x). \quad (6.3)$$

Its fixed-seed Lipschitz constant in x is $L_M \leq B_d \delta$. It has a deterministic query-preserving shift vector v_σ , $\|v_\sigma\| \leq D_s := \delta^{-t}$, satisfying

$$M_\sigma(x + \Delta; z + v_\sigma \otimes \Delta/r) = M_\sigma(x; z). \quad (6.4)$$

It can reserve a final independent Gaussian of covariance $\sigma^2 I/4$, with pre-reserve W_2 error at most $B_d \sigma \delta^{K_M} D_F(x)$ against $\mathcal{N}(m(x), 3\sigma^2 I/4)$. Its total variation error is therefore at most $2B_d \delta^{K_M} D_F(x)$. The constants in B_d and the dimension threshold may also depend on C_0, q_0 .

6.2 A finite affine composition

Condition on an external anchor record, which may include the initial state and the clock U . Suppose the exact centers form a triangular computation

$$x_i = b_i + \sum_{j < i} a_{ij} m(x_j), \quad \sum_j |a_{ij}| \leq A.$$

All coefficients are deterministic given the anchors. Let the dependency depth be at most L , a fixed positive integer, and maximum number of direct mean dependencies (fan-in) at most M , including the final expression. We seek only the declared output

$$b + \sum_i C_i m(x_i) + \mathbf{N}(0, \Sigma), \quad \Sigma \succ 0. \quad (6.5)$$

Internal query records are not included in this law. Put $R_\Sigma = \sum_i \|\Sigma^{-1/2} C_i\|_{\text{op}}$, using a deterministic upper bound when necessary.

Construction provenance. Gaussian center-noise absorption and recursive evaluation use OpenAI [14, Lemma 7.1 and Section 7]. The outer affine composition and its joint covariance accounting are the specialization stated here.

Proposition 6.2 (Affine implementation). *For fixed $C_0, q_0 \geq 0$, assume that all mean calls use the common scale r and satisfy (6.2) with these same constants, including the output call with $\sigma_{\text{out}} = 1/(4R_\Sigma)$ when $R_\Sigma > 0$. Then (6.5) has a finite original-gradient implementation with conditional total variation error at most*

$$B_d \delta^{K_M} \max_i D_F(x_i) \quad (6.6)$$

and every-execution query cap $B_d(1 + M)^{L+1} \delta^{-t}$. The Gaussian comparison law retains the full covariance Σ in (6.5), including its cross-covariances. No equality between Σ and the covariance of the approximate output is asserted. The fixed constants in B_d may also depend on C_0, q_0 and the depth L .

Proof. If $R_\Sigma = 0$, draw the displayed Gaussian directly. Otherwise put $s_c = r/(2D_s)$ and supply each internal center with target noise $\mathbf{N}(0, s_c^2 I)$. Expand every repeated mean occurrence as a separate copy of its entire recursive center computation. For a scalar row with coefficient sum $A_E \leq A$, request each mean with noise $\sigma_E = s_c/(2A)$, ignoring term-free rows. For its child shift vector v , use fresh seeds transformed by

$$P_v = \left(I - \frac{vv^\top}{4D_s^2} \right)^{1/2}.$$

Add residual Gaussian variance $s_c^2 - \sum_j a_j^2 \sigma_E^2 \geq 3s_c^2/4$ to the row. Term-free rows return the direct term plus $s_c G$.

At a center with ideal supplied value $x + s_c Z$, identity (6.4) gives

$$M_\sigma(x + s_c Z; P_v G) = M_\sigma \left(x; P_v G - \frac{v \otimes Z}{2D_s} \right) \stackrel{\text{law}}{=} M_\sigma(x; G).$$

The last equality uses independence of Z and G , and covariance $P_v P_v^\top + vv^\top/(4D_s^2) = I$. It is an averaged identity, not one conditional on the auxiliary Z .

Couple an actual supplied center to the ideal noisy center first. Fixed-seed stability costs $L_M e_i$ in W_2 , where e_i is its center error. Then use seed absorption, followed by (6.3). Independent tree copies admit product couplings. Put $P_d = C_0(1 + \log d)^{q_0}$. If all centers are term-free, then $e_i = 0$ and we set $\chi = 0$. Otherwise the internal noise $\sigma_E = r/(4AD_s)$ and its root condition give $4AD_s \leq P_d$. Consequently $\chi = L_M A \leq B_d P_d \delta^{1+t}/4$ is bounded beyond a fixed threshold. Put $G_L(\chi) = \sum_{j=0}^{L-1} \chi^j$. The row errors satisfy

$$e_i \leq L_M \sum_j |a_{ij}| e_j + B_d \delta^{K_M} D_* \sum_j |a_{ij}| \sigma_E \leq \chi \max_{j < i} e_j + \frac{1}{2} B_d s_c \delta^{K_M} D_* ,$$

where $D_* = \max_i D_F(x_i)$ over exact centers. Term-free supplied centers have zero error. Induction on the fixed dependency depth gives $e_i \leq \frac{1}{2} B_d s_c \delta^{K_M} D_* G_L(\chi)$. Because L is fixed and χ is bounded, this finite geometric polynomial is bounded. No independent contraction or stability assumption is required.

For the final output, request each mean with $\sigma_{\text{out}} = 1/(4R_\Sigma)$ using independent recursive center computations and transformed seed blocks. Put $S = \sum_i \sigma_{\text{out}}^2 C_i C_i^\top \preceq \Sigma/16$. Add independent noise of covariance $3\Sigma/4 - S$, then an independent final reserve of covariance $\Sigma/4$. Before that reserve the whitened W_2 error is bounded by

$$\begin{aligned} \sum_i \|\Sigma^{-1/2} C_i\|_{\text{op}} \{L_M e_i + B_d \sigma_{\text{out}} \delta^{K_M} D_*\} \\ \leq B_d \delta^{K_M} D_* \left\{ \frac{1}{4} + \frac{1}{2} R_\Sigma L_M s_c G_L(\chi) \right\} . \end{aligned}$$

The output root condition gives $4rR_\Sigma \leq P_d$. Since $L_M \leq B_d \delta$ and $s_c = r/(2D_s)$, it follows that $R_\Sigma L_M s_c \leq B'_d \delta^{1+t}$, which is bounded beyond a fixed dimension threshold. Together with the fixed-depth estimate this absorbs the bracket into B_d . The ideal pre-reserve law is Gaussian of covariance $3\Sigma/4$. Convolution by the independent reserve bounds total variation by a universal constant times the whitened W_2 error: couple the means and use the Gaussian shift bound, then average. This proves (6.6). A depth- L , fan-in- M expanded tree has at most $C(1+M)^{L+1}$ terms. Every term has a deterministic query cap, so the claimed total cap holds even on erroneous intermediate histories. $\square$

6.3 Removing the noise from a small posterior

The primitive in Theorem A.1 returns a posterior plus independent Gaussian noise. The following finite descent removes that noise at arbitrary supplied centers.

Construction provenance. The following routine adapts the conditional-descent construction and ideal-prefix replacement argument of OpenAI [14, Section 8.2]. The arbitrary-center bounds and the terminal Gaussian rule are analyzed here; related posterior descent appears in [5, Algorithm 3.3 and Section 6.3].

Proposition 6.3 (Small-width posterior sampling). *Let $0 < w \leq \delta$, and let V satisfy (2.1). For every fixed $A_0 > 0$ and input-center distribution with*

$$\mathbb{E}[\sqrt{d} + \sqrt{\delta} \|\nabla V(Y)\|] \leq C d^{M_0} , \tag{6.7}$$

there is an original-gradient procedure approximating $R_{w,Y}^V$ with averaged total variation error at most d^{-A_0} and cap $B_d \delta^{-t}$ times a fixed logarithmic factor, provided $eK_M > A_0 + M_0 + 2$. The orders and threshold depend on the fixed parameters, not on w, Y , or κ .

Proof. Set $F(z) = V(\sqrt{\delta}z)$, $x = Y/\sqrt{\delta}$, and $r = \sqrt{w/\delta}$. Then $\text{Lip}(\nabla F) \leq \delta$ and $D_F(x) = \sqrt{d} + \sqrt{\delta}\|\nabla V(Y)\|$. Let $(x_0, r_0) = (x, r)$. For $j = 0, \dots, L_0 - 1$, apply the noisy sampling routine of [Theorem A.1](#) to (x_j, r_j) , obtaining Y_j , and set

$$x_{j+1} = x_j + r_j Y_j / 2, \quad r_{j+1} = r_j / \sqrt{2}.$$

At the terminal center, draw $Z_{L_0} \sim \gamma_d$ independently. Return the physical variable

$$\sqrt{\delta}(x_{L_0} + r_{L_0} Z_{L_0}).$$

The exact conditioning identity is $Z = Y_j/2 + Z_{j+1}/\sqrt{2}$, with $Z_{j+1} \mid Y_j \sim \nu_{x_j + r_j Y_j/2, r_j/\sqrt{2}}$. Thus the reconstruction is exact if all noisy observations and the terminal conditional sample are exact.

We use a Gaussian comparison valid at every center and scale:

$$\text{TV}(\gamma_d, \nu_{x,r}) \leq C\{r\|\nabla F(x)\| + \delta r^2 \sqrt{d}\}, \quad (\mathbb{E}_{\nu_{x,r}} \|Z\|^2)^{1/2} \leq C\{\sqrt{d} + r\|\nabla F(x)\|\}. \quad (6.8)$$

Indeed the target curvature is at least $1/2$, and the relative score under $G \sim \gamma_d$ is $r\nabla F(x + rG)$, with L^2 norm at most $r\|\nabla F(x)\| + \delta r^2 \sqrt{d}$. Logarithmic Sobolev followed by Pinsker gives the first bound. Logarithmic Sobolev followed by T_2 bounds $W_2(\gamma_d, \nu_{x,r})$ by the same score norm; a coupling and Minkowski give the second bound.

Condition on a deterministic initial center x . Realize the ideal descent using one latent $W = x + rZ$, $Z \sim \nu_{x,r}$, and fresh independent Gaussian observations

$$Y_j = (W - x_j)/r_j + N_j, \quad x_{j+1} = (x_j + W + r_j N_j)/2.$$

Gaussian conditioning gives the required posterior at each level. Expanding the linear recursion yields, marginally for every j ,

$$x_j \stackrel{\text{law}}{=} x + (1 - 2^{-j})rZ + r\sqrt{2^{-j} - 4^{-j}} G_j, \quad G_j \sim \gamma_d, \quad G_j \perp Z.$$

The G_j need not be independent across levels. Using the second bound in [\(6.8\)](#), uniformly in j ,

$$(\mathbb{E}\|\nabla F(x_j)\|^2)^{1/2} \leq \|\nabla F(x)\| + C\delta r(\sqrt{d} + r\|\nabla F(x)\|) \leq CD_F(x).$$

The first bound in [\(6.8\)](#) controls the terminal Gaussian approximation. Averaging it with the preceding ideal-center bound requires no terminal optimization.

A forward replacement of noisy observations, using a common exact prefix each time, then gives the pointwise total error

$$CL_0 B_d \delta^{K_M} D_F(x) + Cr2^{-L_0/2} D_F(x) + C\delta r^2 2^{-L_0} \sqrt{d}. \quad (6.9)$$

For example, take $L_0 = \lceil 2(A_0 + \max\{M_0, 1/2\} + 3) \log_2 d \rceil$. Average [\(6.9\)](#) using [\(6.7\)](#). The specified accuracy order proves the assertion beyond a fixed dimension threshold. The deterministic work is $L_0 B_d \delta^{-t}$. No moment bound on an implemented erroneous prefix is used. $\square$

6.4 The phase specialization

Use $F(z) = V(\sqrt{\delta}z)$ and $r = \sqrt{a/\delta}$. Then

$$\sqrt{\delta} g_a(X) = m(X/\sqrt{\delta}).$$

The observation in physical coordinates is $O_{\text{phys}} = Y + h\hat{Z} + h\sqrt{\Gamma}G$. The internal physical force-row sum and final physical force-row sum are at most $C_m h^2$, while scaled-center rows have sum $A \leq C_m h^2/\delta$. The output mean coefficients have sum at most $C_m h^2/\sqrt{\delta}$, with covariance $\Sigma = h^2 \Gamma I$. Hence

$$R_\Sigma \leq \frac{C_m h}{\sqrt{\delta} \sqrt{\Gamma}}, \quad \delta r R_\Sigma \leq \frac{C_m h^2}{\sqrt{\Gamma}}, \quad \delta r / \sigma_E \leq C_m h^2 \delta^{-t}, \quad L_M A \leq B_d C_m h^2. \quad (6.10)$$

Conditionally on the anchors, the ideal physical observation has covariance $\Sigma = h^2 \Gamma I$. [Proposition 6.2](#) compares against this whole observation law, with $\Sigma/4$ reserved as independent final noise. Internal center noises are not retained as outputs. Applying the common Bayes decoder and momentum refresh therefore bounds the total variation error of the full phase.

There are $O_m(n)$ direct children in each interpolated row. Counting the J simultaneous layers, evaluation at U , the final force layer, and output dependencies conservatively gives the cap

$$Q_{\text{phase}} \leq B_d (1+n)^{J+3} \delta^{-t}. \quad (6.11)$$

The posterior decoder adds, rather than multiplies, its own cap from [Proposition 6.3](#) because $a + s \leq 2a \leq \delta$.

To average (6.6), we need exact-center moments. Bounded absolute row sums and $C_m h^2 \leq 1/2$ imply, simultaneously over the finite exact-score centers,

$$\max_i \|X_i - Y\| \leq C_m \{h(\|P\| + \|q\|) + h^2 \|g_a(Y)\|\}.$$

This follows by subtracting Y in (3.3), using Lipschitzness of g_a , and absorbing the resulting $C_m h^2$ multiple of the maximum. There is no node-count factor. [Lemma 2.3](#) and (2.8) therefore give

$$\mathbb{E} \max_i \{\sqrt{d} + \sqrt{\delta} \|\nabla V(X_i)\|\} \leq C_m \sqrt{d + \delta(H + d)}. \quad (6.12)$$

The same estimate, with the additional fresh observation Gaussian, controls the decoder center.

7 Parameters and the full query bound

We prove one inverse-dimension accuracy bound and derive both stated accuracy guarantees from it. The approximation orders, horizons, and implementation cutoffs are fixed before any oracle answer; only the initialization proximal width uses the first queried gradient.

Proposition 7.1 (Inverse-dimension accuracy). *Under the hypotheses and oracle model of [Theorem 1.1](#), for every fixed $\eta > 0$ there is an algorithm whose output law and query count satisfy*

$$\text{TV}(\hat{\pi}, \pi) \leq \frac{1}{10d}, \quad Q \leq C_\eta \kappa^{2/3} d^{\eta/4} \quad (7.1)$$

with the query bound holding on every execution. It uses at most one exact positive-scalar proximal query, and no such query when a center satisfying [Lemma 2.2](#) is supplied.

7.1 Fixed orders and step size

Fix $\eta > 0$ and, initially, sufficiently large d with $1 \leq \kappa \leq d^{18}$. Set

$$\theta = \frac{\min\{\eta, 1\}}{20}, \quad e = \theta/2, \quad \gamma = (2\kappa)^{-1/2}, \quad \delta = d^{-e}, \quad J = \lceil 4/\theta \rceil + 2. \quad (7.2)$$

Choose $\omega = \theta/[4(J+3)]$, $K = 12$, $n = \lceil d^\omega \rceil$, and a fixed interpolation order m with $\omega m > 32$. Then choose an integer $K_M \geq \lceil 20/e \rceil$ and

$$0 < t < \min\{1/4, 1/[320(K_M + 2)]\}.$$

The recursive procedure's internal orders are then selected according to [Theorem A.1](#). Finally take a sufficiently small fixed constant $c > 0$, depending only on all previously chosen orders, and put

$$h = c\gamma^{1/3}d^{-\theta}, \quad a = h^2, \quad \Gamma = h(\kappa + a + h^2)^{-1/2}. \quad (7.3)$$

Thus $\Gamma \asymp \gamma h$, with absolute comparison constants. Decrease c to meet every dimension-independent condition such as $h \leq 1/4$ and $C_m h^2 \leq 1/4$. Increase the fixed dimension threshold to meet the recursive primitive's conditions and to ensure $2a \leq \delta$. These operations change no exponent or approximation order.

7.2 The numerical entropy budget

The averaged endpoint estimates give

$$A_k \leq C_m \left(h^4 + \frac{h^8}{\Gamma} \right) (R_k + d^{-K}) + \frac{C_J h^4 (C_m h^2)^{2J}}{\Gamma} (H_k + d). \quad (7.4)$$

The conservative free-path envelope at the end of [Section 4](#) suffices here. Dividing the coefficients of R_k by the conditional-entropy gap Γ gives

$$C_m \left(\frac{h^4}{\Gamma} + \frac{h^8}{\Gamma^2} \right) \leq C'_m (c^3 d^{-3\theta} + c^6 d^{-6\theta}).$$

This is at most $1/16$ by the fixed smallness choice. The associated stationary entropy floor, after division by Γ , is at most $C_\eta d^{-12}$.

For the finite Picard tail, define

$$\beta_J = \frac{C_J h^{4J+4}}{\Gamma^2} \leq C'_J \frac{h^{4J+2}}{\gamma^2} = C'_J c^{4J+2} \gamma^{(4J-4)/3} d^{-\theta(4J+2)}.$$

The conditioning exponent is nonnegative. Also $\theta(4J+2) > 16$, so $\beta_J d \leq C_\eta d^{-15}$ after enlarging a fixed constant. Consequently (7.4) meets (5.9) with $\beta_0 \leq \varpi/32$ and b/Γ arbitrarily small beyond a fixed dimension threshold. This retains both the H_k and the stationary d part of the Picard tail.

Algorithm 7.2 (The complete moderate-regime sampler). Choose (7.2)–(7.3) and all recursive routine orders. Use [Lemma 2.2](#) and set

$$N = \lceil C\Gamma^{-1} \log\{1000d^2(B_0 + 1)\} \rceil, \quad B_0 = d(1 + \log \kappa)/2, \quad (7.5)$$

with the absolute constant prescribed by [Proposition 5.4](#).

1. Execute $N - 1$ full phases, replacing the observation computation of [Algorithm 3.1](#) by [Proposition 6.2](#), and its posterior draw by [Proposition 6.3](#).
2. In the last phase implement O_{phys} , set $s = h^2\Gamma/(1 + \Gamma)$ and $c = (\Gamma Y + O_{\text{phys}})/(1 + \Gamma)$, and return the posterior sample Z produced at width $a + s$ and center c .

The final Gaussian reconstruction and momentum refresh are used only in the comparison process.

By [Proposition 5.4](#), the virtual complete finite exact-score chain at time N is within $1/(100d)$ TV of Π_a : the entropy floor $O_\eta(d^{-12} + d^{-15})$ is smaller than a sufficiently small multiple of d^{-2} beyond the fixed dimension threshold. Throughout this chain, $H_k \leq C(B_0 + 1)$. Furthermore,

$$N \leq B_d \kappa^{2/3} d^\theta \leq B_d d^{12+\theta}. \quad (7.6)$$

This is a deterministic horizon, not a stopping time inferred from an approximate law.

7.3 Checking the original-gradient replacement

By (6.10), the output and internal root ratios are at most $C'_m c^{3/2} d^{-3\theta/2}$ and $C'_m d^{-2\theta+et}$, respectively. Both are bounded by $B_d \delta$, since $e = \theta/2$ and $e(1+t) < 2\theta$. The remaining root-size condition follows from $a \leq \delta$. Thus every hypothesis of [Proposition 6.2](#) holds, with its positive independent Gaussian reserve and output stability budget.

Since $\kappa \leq d^{18}$, (6.12) and the ideal entropy bound give $\mathbb{E}D_* \leq B_d \sqrt{d}$. The total observation implementation error over all ideal prefixes is therefore at most

$$NB_d \delta^{K_M} \sqrt{d} \leq B_d d^{25/2+\theta-eK_M} \leq B_d d^{25/2+\theta-20}.$$

For each posterior decoder, use [Proposition 6.3](#) with accuracy d^{-16} , increasing its fixed logarithmic cutoffs if needed. Its ideal observation center has the same envelope $B_d \sqrt{d} \leq Cd$ beyond a fixed threshold. Thus we may take $M_0 = 1$ and $A_0 = 16$, with the strict inequality $eK_M \geq 20 > 16 + 1 + 2 = 19$ required by [Proposition 6.3](#). The sum of posterior errors is at most $B_d d^{-4+\theta}$. Both sums are below their assigned $1/(100d)$ budgets beyond a fixed threshold.

For completeness, the forward kernel-TV replacement used here can be proved by writing

$$\mu \hat{K}_1 \cdots \hat{K}_N - \mu K_1 \cdots K_N = \sum_{j=1}^N \mu K_1 \cdots K_{j-1} (\hat{K}_j - K_j) \hat{K}_{j+1} \cdots \hat{K}_N.$$

Taking TV and using data processing bounds each term by the error averaged at an exact prefix. In particular, no entropy, warmth, or unbounded moment is transferred through TV proximity to an implemented law.

For the output, retain the final exact posterior latent Z_N and reconstruct (X_N, P_N) only virtually. By (2.10), its law is $\mu_N R_a^V$, where μ_N is the virtual physical-position law; this identity holds for the nonstationary observation-center law. Since $\pi_a R_a^V = \pi$, the ideal output is within $1/(100d)$ of π .

The posterior routine controls its posterior sample Z , so appending the common virtual Gaussian reconstruction preserves its TV bound. In the last phase, replace the observation using the common approximate posterior as downstream kernel, then replace that posterior under the ideal observation-center law. Thus the same forward telescope compares the terminal latent laws, with exactly N observation and N posterior errors. Adding these two budgets to the mixing error gives TV less than $1/(10d)$, without a separate recovery call.

The posterior costs are additive. By (6.11) and (7.6),

$$Q \leq B_d \kappa^{2/3} d^{\theta+\omega(J+3)+et}. \quad (7.7)$$

Here $\omega(J+3) = \theta/4$ and $et < \theta/8$, so the displayed exponent is below $11\theta/8 \leq 11\eta/160$. All remaining fixed logarithmic factors fit below the exponent $\eta/4$. The complete tree shapes, horizons, and cutoffs are fixed before their queries, so the resulting cap $C_\eta \kappa^{2/3} d^{\eta/4}$ holds on every execution.

7.4 Large condition numbers and small dimensions

For large condition numbers, standard BPS with windowed thinning is sufficient. We give the short extension of the cold-start estimate to the common initializer, which need not be centered at the minimizer.

Proposition 7.3 (BPS from a cold Gaussian). *Under (2.1), initialize the windowed-thinning BPS algorithm of [12, Algorithm 1] from $\mu_* \otimes \gamma_d$, with μ_* and Δ_0 as in Lemma 2.2. For $0 < \varepsilon \leq 1/2$, use refreshment rate $\gamma_B = \sqrt{d/\kappa}$, window length $\tau = d^{-1/2}$, and horizon*

$$T = C\sqrt{\kappa d}[\Delta_0 + \log(1/\varepsilon) + 1]. \quad (7.8)$$

For a sufficiently large universal C , the position is within ε TV of π , and the expected number of original-gradient queries satisfies

$$\mathbb{E}Q_{\text{BPS}} \leq 5\sqrt{d}T \leq C\sqrt{\kappa}d[\Delta_0 + \log(1/\varepsilon) + 1]. \quad (7.9)$$

No minimizer, potential value or further proximal query is supplied.

Proof. The initial joint χ^2 divergence is at most $e^{\Delta_0} - 1$ by (2.11). The general-initial-law contraction in [12, Theorem 1] gives (7.8). Only the event-count estimate needs an extension to our near-center initialization.

Write $x_* = \arg \min V$, $M = \mathbb{E}\|X_0 - x_*\|^2 \leq 2\kappa d$, $\psi(x, p) = (x - x_*) \cdot p$, and $\lambda(x, p) = (p \cdot \nabla V(x))_+$. Convexity and unit smoothness give $r(x) = (x - x_*) \cdot \nabla V(x) / \|\nabla V(x)\|^2 \geq 1$ away from x_* ; set $r(x_*) = 1$. Between events $\dot{X} = P$; bounces reflect P across the gradient hyperplane, and refreshes draw from γ_d . The generator therefore satisfies

$$\mathcal{L}\psi = \|p\|^2 - 2\lambda^2 r - \gamma_B \psi.$$

Here $\mathbb{E}\psi(X_0, P_0) = 0$ and $\mathbb{E}\|P_t\|^2 = d$. Integrate this identity, using $\int_0^T \mathbb{E}\psi \, dt = (\mathbb{E}\|X_T - x_*\|^2 - M)/2$ and $\|X_T - x_*\|_{L^2} \leq \sqrt{M} + \sqrt{dT}$, to obtain

$$\int_0^T \mathbb{E}\lambda_t^2 \, dt \leq dT + \gamma_B M/4 + \sqrt{dM}/2 \leq 4dT.$$

The final inequality uses $T \geq \sqrt{\kappa d}$. Finite-horizon localization is justified by the Gaussian initial position and refresh velocities: reflections preserve speed and $\|X_t\| \leq \|X_0\| + T \sup_{s \leq T} \|P_s\|$, giving all needed moments. Cauchy–Schwarz bounds the expected bounce count by $2\sqrt{dT}$. The general windowed-thinning query estimate [12, Proposition 9, equation (26)] now gives

$$\mathbb{E}Q_{\text{BPS}} \leq T/\tau + 1 + 2\sqrt{d}T + \tau Td \leq 5\sqrt{d}T.$$

The minimizer occurs only in this proof, not in the gradient-only simulation. $\square$

Apply this procedure with $\varepsilon = 1/(40d)$ and cap the entire execution at forty times d times its proved expectation bound, returning a fixed point if that cap is reached. Markov’s inequality adds at most $1/(40d)$ TV, so the output error is at most $1/(20d)$. Its query cap is

$$C\sqrt{\kappa}d^3[1 + \log(d\kappa)].$$

For $\kappa \geq d^{18}$, write $\kappa = d^{18}u$, $u \geq 1$. Then

$$\frac{\sqrt{\kappa}d^3}{\kappa^{2/3}} = u^{-1/6}.$$

This factor absorbs $\log u$, while $d^{\eta/4}$ absorbs $1 + \log d$. For the finitely many dimensions below the moderate-branch threshold, use this same capped BPS procedure for every κ ; the dimension factors are fixed and $\sqrt{\kappa} \text{polylog } \kappa \leq C\kappa^{2/3}$. This proves [Proposition 7.1](#), hence [Theorem 1.1](#).

Both branches use only original gradients after the common center is constructed. Consequently a supplied x_{ref} with $\|\nabla V(x_{\text{ref}})\| \leq \sqrt{d/\kappa}$ gives a gradient-only algorithm for all condition numbers. Otherwise the center requires at most one exact positive-scalar proximal query.

7.5 An arbitrary-accuracy consequence

To obtain arbitrary accuracy, add independent Gaussian coordinates until the inverse-dimension error is below the requested tolerance. No approximation order needs to depend on accuracy.

Corollary 7.4 (Independent dimension and accuracy powers). *Assume (1.1) with supplied positive curvature bounds. For every fixed $\eta_1, \eta_2 > 0$ and every $0 < \varepsilon \leq 1/2$, there is a sampler in the same exact-real gradient-proximal model satisfying*

$$\text{TV}(\hat{\pi}, \pi) \leq \varepsilon, \quad Q \leq C_{\eta_1, \eta_2} \kappa^{2/3} (d^{\eta_1} + \varepsilon^{-\eta_2})$$

on every execution. The constant is uniform in d, κ, ε for fixed η_1, η_2 .

Proof. Normalize as in (2.1), put $\eta = \min\{\eta_1, \eta_2, 1\}$, and set

$$D = \max\{d, \lceil \varepsilon^{-1} \rceil\}, \quad \bar{V}(x, z) = V(x) + \|z\|^2/2, \quad z \in \mathbb{R}^{D-d}.$$

The augmented target is $\pi \otimes \gamma_{D-d}$ with the same curvature bounds. One augmented gradient costs one original gradient; one augmented scalar proximal query is the original proximal point together with $z/(1+s)$. Apply [Proposition 7.1](#) in dimension D and project onto the original coordinates. Projection cannot increase TV, so the error is at most $1/(10D) \leq \varepsilon$. On every execution its query count is at most

$$C_{\eta} \kappa^{2/3} D^{\eta/4} \leq C_{\eta_1, \eta_2} \kappa^{2/3} (d^{\eta_1} + \varepsilon^{-\eta_2}).$$

The last inequality uses $D \leq d + 2\varepsilon^{-1}$ and $\eta/4 \leq \min\{\eta_1, \eta_2\}$. No quantitative growth bound on the constant, or polylogarithmic inverse-accuracy complexity, is asserted. $\square$

A Conditional gradient means and noisy sampling

This and the following five appendices prove the noisy sampling construction and its analytic ingredients, adapting OpenAI [14]. [Theorem A.1](#) states the sampling guarantee; its proof concludes in [Section F](#). The mean-root specialization used in the cold sampler is proved in [Section G](#).

We retain the law $\nu_{x,r}$ and envelope $D_F(x)$ from (6.1), and write $u_{x,r} = m(x)$ to display the mean's scale. The Gaussian interpolation variance is $\nu_{\rho} = 1 - \rho^2$; the sampling noise standard deviation is τ . For tensor-valued random variables, $\|\cdot\|_2$ is the L^2 norm of the displayed pointwise norm, and $\|\cdot\|_{\text{HS},2}$ uses the Hilbert-Schmidt norm. All kernels and oracle algorithms below are measurable in the exact-real model of [Section 2](#).

Fix $0 < e < 1$, and put $\delta = d^{-e}$. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}$ be C^2 , with gradient $g = \nabla F$ of Lipschitz constant at most the declared bound $\lambda = \delta$. Values of F need not be available. For $x \in \mathbb{R}^d$ and $0 < r \leq 1$, define

$$\nu_{x,r}(dz) = \frac{\exp(-|z|^2/2 - F(x + rz)) dz}{\int_{\mathbb{R}^d} \exp(-|v|^2/2 - F(x + rv)) dv}, \quad f(z) = g(x + rz), \quad u = \mathbb{E}_{\nu_{x,r}} f. \quad (\text{A.1})$$

We often write $\nu = \nu_{x,r}$, and use

$$L = \lambda r, \quad l = \lambda r^2 = rL, \quad D_F(x) = \sqrt{d} + |g(x)|. \quad (\text{A.2})$$

Here x is the actual input center; it need not be a minimizer of F . The positive number λ is a declared upper bound, so the notation also covers affine or constant F . The full potential in (A.1) has Hessian between $(1-l)I$ and $(1+l)I$. Since $l \leq \delta < 1$ for $d \geq 2$, the normalizing integral is finite, even when F is not convex. For the quantitative construction we take d large enough that $l \leq 1/2$.

Throughout the construction, $\text{polylog}(d)$ denotes a factor bounded by $C(1+\log d)^C$. Its constants may depend on the fixed parameters e , accuracy orders, and other fixed orders introduced below, but never on F, x, r , or the oracle responses. Different occurrences may have different constants. Every sufficiently-large-dimension threshold has the same uniformity.

Source attribution. Adapted from OpenAI [14, Theorem 2.1].

Theorem A.1 (Noisy near-quadratic sampling). *Fix $0 < e < 1$, an integer $K_0 \geq 2$, and $t > 0$ satisfying*

$$40(K_0 + 2)t < \frac{1}{4}. \quad (\text{A.3})$$

There are constants and a dimension threshold depending only on e, K_0, t with the following property. For every integer d above that threshold, put $\delta = d^{-e}$. One measurable randomized oracle procedure works for every $F \in C^2(\mathbb{R}^d)$ with $\text{Lip}(\nabla F) \leq \delta$, every $x \in \mathbb{R}^d$, and every $0 < r \leq 1$. Given evaluation access to $g = \nabla F$, it uses at most $\text{polylog}(d)\delta^{-t}$ evaluations of g on every execution and returns

$$X = X_{\text{pre}} + \frac{1}{2}G,$$

where $G \sim \mathcal{N}(0, I_d)$ is independent of X_{pre} and of all queries, and

$$W_2(\text{Law}(X_{\text{pre}}), \nu_{x,r} * \mathcal{N}(0, \frac{3}{4}I_d)) \leq \text{polylog}(d)\delta^{K_0}D_F(x). \quad (\text{A.4})$$

Consequently both

$$W_2(\text{Law}(X), \nu_{x,r} * \mathcal{N}(0, I_d)) \quad \text{and} \quad \|\text{Law}(X) - \nu_{x,r} * \mathcal{N}(0, I_d)\|_{\text{TV}}$$

are at most $\text{polylog}(d)\delta^{K_0}D_F(x)$. All constants in these bounds are uniform in F, x, r . The procedure's scalar parameters, shapes of its computations, and query cap are independent of x and of the primitive. It uses no evaluations of F itself.

Theorem A.1 is proved at the end of Section F. Its final Gaussian is generated independently after the query computation. A common copy of that Gaussian preserves the coupling error, and Lemma A.3 converts the pre-noise W_2 bound to TV. The comparison law is $\nu_{x,r} * \mathcal{N}(0, I_d)$. The posterior application in Proposition 6.3 removes this added noise by conditioning.

The factor $D_F(x)$ is part of the guarantee. Uniformity of the constants in x does not make the error uniformly small over centers. At every random-center application, we will either retain this factor or establish a moment bound for it.

A.1 Convexity, moments, and comparisons

We will use the following consequences of uniform convexity. We include the C^2 approximation argument to record the regularity needed later; see [3] for the curvature method.

Source attribution. Adapted from OpenAI [14, Lemma 2.2].

Lemma A.2 (Uniformly convex laws). *Let $H \in C^2(\mathbb{R}^n)$ satisfy $mI \preceq \nabla^2 H \preceq MI$, with $0 < m \leq M < \infty$. Write $\mu(dx) \propto e^{-H(x)}dx$, and let z_0 be its unique mode. Then*

$$\mathbb{E}_\mu |X - z_0|^2 \leq n/m, \quad \text{Var}_\mu h \leq m^{-1} \mathbb{E}_\mu |\nabla h|^2.$$

The latter applies, in particular, to C^1 functions whose value and gradient have polynomial growth. The diffusion

$$dX_t = -\nabla H(X_t) dt + \sqrt{2} dB_t$$

has invariant law μ , and synchronous solutions contract in distance by e^{-mt} .

If $\bar{H}$ is another strongly convex C^2 potential with bounded Hessian and law χ , then

$$W_2(\mu, \chi) \leq \frac{1}{m} (\mathbb{E}_\chi |\nabla H - \nabla \bar{H}|^2)^{1/2}. \quad (\text{A.5})$$

Proof. We first prove the moment bound and contraction. Strong convexity gives Gaussian tails and a unique minimizer. Integration by parts yields $\mathbb{E}_\mu (X - z_0) \cdot \nabla H(X) = n$; strong monotonicity of ∇H gives the moment bound. The globally Lipschitz drift gives a nonexplosive diffusion. For synchronous solutions, differentiating their squared distance and using strong monotonicity gives the contraction.

To establish invariance and Poincaré, first assume that H is smooth and all derivatives of order at least two are bounded. For a smooth compactly supported h , let $P_t h(x) = \mathbb{E}[h(X_t) \mid X_0 = x]$. Differentiating the synchronous flow in the initial state gives bounded first and second derivatives on bounded time intervals. Itô's formula and the semigroup identity yield

$$\partial_t P_t h = P_t \mathcal{A} h = \mathcal{A} P_t h, \quad \mathcal{A} = \Delta - \nabla H \cdot \nabla.$$

The equality with $\mathcal{A} P_t h$ also follows by differentiating $P_{t+s} h = P_s(P_t h)$ at $s = 0$. Its at-most-linear growth is integrable against μ . Integration by parts gives $\int \mathcal{A} P_t h d\mu = 0$, hence invariance. Coupling to an invariant starting point and using contraction shows $P_t h \rightarrow \mu h$ in $L^2(\mu)$. Therefore

$$\text{Var}_\mu h = 2 \int_0^\infty \int |\nabla P_t h|^2 d\mu dt.$$

The flow derivative has operator norm at most e^{-mt} . Jensen's inequality and invariance thus bound the inner integral by $e^{-2mt} \int |\nabla h|^2 d\mu$, proving the Poincaré inequality.

For C^2 potentials, convolve with smooth compactly supported mollifiers of radii tending to zero. Both Hessian bounds are preserved; all higher derivatives are bounded for each mollification. The normalized laws converge in total variation by the uniform quadratic tail bounds (subtract the potential's value at zero). The gradients converge uniformly since the original gradient is globally Lipschitz. Synchronous solutions of the mollified and original diffusions consequently converge uniformly in their common starting point on each bounded time interval, by Gronwall's inequality. Testing invariance against bounded Lipschitz functions passes it to the limit. Poincaré first passes on smooth compact tests, then by cutoff and approximation to the stated class.

For (A.5), start the two diffusions with their respective invariant marginals, with any initial coupling, and use common Brownian motion. Writing $a(t) = (\mathbb{E}|X_t - Y_t|^2)^{1/2}$ and $\beta = (\mathbb{E}_\chi |\nabla H - \nabla \bar{H}|^2)^{1/2}$, strong monotonicity and Cauchy-Schwarz give, in the usual upper-derivative sense,

$$a'(t) \leq -ma(t) + \beta.$$

The marginals remain fixed, so $W_2(\mu, \chi) \leq a(t)$. Letting t tend to infinity proves the claim. $\square$

For the auxiliary law, the quadratic term supplies strong convexity even when F is nonconvex. Its mode solves $z_0 = -rg(x + rz_0)$. In particular, for $l \leq 1/2$,

$$|z_0| \leq \frac{r|g(x)|}{1-l}, \quad (\mathbb{E}_\nu |Z|^2)^{1/2} \leq C(\sqrt{d} + r|g(x)|), \quad |u| \leq CD_F(x). \quad (\text{A.6})$$

Indeed, $|g(x + rz_0) - g(x)| \leq L|z_0|$ proves the first bound, and the lemma gives $\|Z - z_0\|_2 \leq \sqrt{d/(1-l)}$. The triangle inequality and $|u| \leq |g(x)| + L\|Z\|_2$ give the remaining bounds. Thus $D_F(x)$ controls the required moments uniformly over all centers, including those with very large gradients.

A.2 Coupling and smoothing conventions

We measure numerical error by the L^2 norm under the stated input law. We compare randomized output laws in W_2 . The following rules pass from conditional couplings to unconditional laws and from a pre-noise coupling to total variation.

Source attribution. Adapted from OpenAI [14, Lemma 2.3].

Lemma A.3 (Conditional comparisons and smoothing). *Let (μ_z) and (χ_z) be probability kernels on finite-dimensional Euclidean spaces, and let Z have law ζ . Whenever the relevant second moments are finite,*

$$W_2^2\left(\int \mu_z d\zeta(z), \int \chi_z d\zeta(z)\right) \leq \int W_2^2(\mu_z, \chi_z) d\zeta(z).$$

For finite families, independent sums satisfy

$$W_2\left(\text{Law}\left(\sum_i a_i X_i\right), \text{Law}\left(\sum_i a_i Y_i\right)\right) \leq \sum_i |a_i| W_2(\text{Law}(X_i), \text{Law}(Y_i)),$$

where independence is required within each of the two families, not between them. For $s > 0$,

$$\|\mu * \mathbf{N}(0, s^2 I) - \chi * \mathbf{N}(0, s^2 I)\|_{\text{TV}} \leq s^{-1} W_2(\mu, \chi). \quad (\text{A.7})$$

Moreover, applying a common probability kernel cannot increase total variation, and

$$\left\| \int \mu_z d\zeta(z) - \int \chi_z d\zeta(z) \right\|_{\text{TV}} \leq \int \|\mu_z - \chi_z\|_{\text{TV}} d\zeta(z).$$

Proof. For the first inequality, condition on Z and use near-optimal couplings. For independent sums, take their product and apply the L^2 triangle inequality. Measurability of such conditional comparisons can also be seen without an exact optimal-kernel selection: quantize each output law on a finite grid inside a box, sending the complement to zero. A minimizing plan on the finite sets can be chosen measurably by a fixed ordering of the finitely many basic feasible solutions. Lift it using the conditional measures in the cells, and refine the grids and boxes. Finite unconditional second moments and the triangle inequality give the asserted limit.

For deterministic centers x, y , the relative entropy of $\mathbf{N}(x, s^2 I)$ from $\mathbf{N}(y, s^2 I)$ is $|x - y|^2/(2s^2)$. Pinsker's inequality gives TV at most $|x - y|/(2s)$. Integrate this estimate over any coupling and apply Cauchy–Schwarz to obtain (A.7), with room in its constant. The total-variation rules follow directly by testing measurable functions taking values in $[0, 1]$. $\square$

We will also repeatedly combine deterministic root-mean-square errors with absolute coefficient sums. This uses the ordinary L^2 triangle inequality and does not require independence.

B Gaussian conditional jets and probability transport

The conditional-gradient mean is smooth after Gaussian observation even when F is only C^2 . We establish all-split derivative bounds, then identify its probability flow. These estimates supply both the finite noisy-mean computation and the derivative bounds used in the cold residual analysis. The argument in this section is adapted from [14].

Retain the primitive notation from Section A, and assume $l \leq 1/2$, as holds for all sufficiently large d . For independent $Z \sim \nu$ and $N \sim \mathcal{N}(0, I_d)$, put

$$v_\rho = 1 - \rho^2, \quad Y = \rho Z + \sqrt{v_\rho} N, \quad \xi_\rho = \text{Law}(Y), \quad 0 \leq \rho < 1.$$

Write $\xi_\rho(dy) = e^{-H_\rho(y)} dy$, incorporating the normalizing constant in H_ρ , and define

$$M_\rho(y) = \mathbb{E}[f(Z) \mid Y = y].$$

We use the version defined at every y by the conditional density. Completing its quadratic gives

$$\text{Law}(Z \mid Y = y)(dz) \propto \exp\left(-\frac{|z - \rho y|^2}{2v_\rho} - F(x + rz)\right) dz. \quad (\text{B.1})$$

Under the change of variables $z = \rho y + \sqrt{v_\rho} w$, $M_\rho(y)$ is exactly the primitive mean with center $x + r\rho y$ and scale $r\sqrt{v_\rho}$. In particular, $M_0 = u$. Thus conditioning preserves the auxiliary form and specifies its new center and scale. Equation (B.1) identifies the mean exactly. Its finite-query implementation is deferred to Section F.

B.1 Cumulants from the Poincaré inequality

We use Poincaré to control cumulants across every split of their coordinate slots. This strengthens the type of derivative estimate needed here beyond bounds tested only on products of directions. Related Gaussian-smoothing estimates appear in [5, Appendix C.2].

A tensor $T \in (\mathbb{R}^D)^{\otimes n}$ can be viewed as a matrix by partitioning its n slots into two nonempty sets A and B . Contraction on the A slots then defines a linear map $(\mathbb{R}^D)^{\otimes |A|} \rightarrow (\mathbb{R}^D)^{\otimes |B|}$, where both tensor spaces carry their Euclidean, or Hilbert–Schmidt, norm. We call the operator norm of this matrix a *split norm*. Equivalently, it is the supremum of $|T : (V \otimes W)|$ over coefficient tensors V, W of Hilbert–Schmidt norm at most one, with slots in A, B , respectively. The colon denotes contraction over all paired slots. A bound on $|T|_{\text{split}}$ means that bound for every such partition. Permuting slots simply permutes these norms. The Hilbert–Schmidt norm of a tensor itself is denoted by $|T|_{\text{HS}}$.

We use the multivariate Appell polynomials described in [2, Section 2.3]. Their vanishing expected lower derivatives connect Poincaré inequalities to tensor covariance bounds; see [1, Proposition 3.2 and Theorem 3.3] and [10, Section 4.2]. We give the direct L^2 argument, with its dependence on the polynomial degree, and then use a logarithm with two groups of variables to bound every split of the cumulant tensor.

We use the following consequence of Poincaré, including for random vectors whose laws need not have densities in their ambient spaces.

Source attribution. Adapted from OpenAI [14, Lemma 3.1].

Lemma B.1 (All-split cumulant bound). *Let S be a random vector in $\mathbb{R}^D$ whose moment-generating function is finite in a neighborhood of zero. Suppose that, for some $\beta \geq 0$,*

$$\text{Var}(P(S)) \leq \beta \mathbb{E}|\nabla P(S)|^2$$

for every polynomial P on $\mathbb{R}^D$. For each integer $n \geq 2$, define its cumulant tensor by $\kappa_n(S) = \nabla_\theta^n \log \mathbb{E} e^{\theta \cdot S} \big|_{\theta=0}$. Then, for an absolute constant C ,

$$|\kappa_n(S)|_{\text{split}} \leq C^n n^{Cn} \beta^{n/2}$$

for every nontrivial split. The constant is independent of D and of the mean of S .

Proof. We reduce the cumulant estimate to a covariance estimate for Appell polynomials. If $\beta = 0$, the coordinate variances vanish and S is almost surely constant. All cumulants in the statement are then zero. Assume henceforth that $\beta > 0$.

Define the symmetric Appell tensor polynomials by

$$P_j(s) = \nabla_\theta^j \frac{\exp(\theta \cdot s)}{\mathbb{E} \exp(\theta \cdot S)} \bigg|_{\theta=0}.$$

For a coefficient tensor T of order j , let $Q_T(s) = T : P_j(s)$, where the colon denotes contraction on all slots. Differentiating the generating function in s multiplies it by copies of θ . Consequently

$$\mathbb{E} \nabla^h Q_T(S) = 0 \quad (0 \leq h < j), \quad \nabla^j Q_T = j! \text{Sym}(T).$$

Here $\text{Sym}(T)$ is the average over all permutations of the slots; hence $|\text{Sym}(T)|_{\text{HS}} \leq |T|_{\text{HS}}$. Apply Poincaré to each component of $\nabla^h Q_T$ and sum. Iterating from $h = 0$ to $h = j - 1$ yields

$$\mathbb{E}|T : P_j(S)|^2 \leq \beta^j (j!)^2 |T|_{\text{HS}}^2. \tag{B.2}$$

Only first derivatives in the displayed Poincaré inequality are used at each application.

In particular, if T and V have orders $j, k \geq 1$, then

$$|\mathbb{E}[(T : P_j(S))(V : P_k(S))]| \leq \beta^{(j+k)/2} j! k! |T|_{\text{HS}} |V|_{\text{HS}}. \tag{B.3}$$

The coefficient tensors in (B.3) are arbitrary. Consequently the estimate controls the operator norm of the indicated matrix unfolding.

Fix a split of the slots. Assign one group of variables to each side and define

$$\mathcal{G}(\theta, \omega) = \frac{\mathbb{E} \exp((\theta + \omega) \cdot S)}{\mathbb{E} \exp(\theta \cdot S) \mathbb{E} \exp(\omega \cdot S)}.$$

Its derivative of orders $j \geq 1$ in θ and $k \geq 1$ in ω at $(0, 0)$ is the cross-moment tensor $\mathbb{E}[P_j(S) \otimes P_k(S)]$. Across the split between the θ and ω slots its operator norm is at most $\beta^{(j+k)/2} j! k!$ by (B.3).

Both $\mathcal{G}(\theta, 0)$ and $\mathcal{G}(0, \omega)$ are identically one. Thus every nonzero term of $\mathcal{G} - 1$ contains variables from both groups. In the formal identity

$$\log \mathcal{G} = \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} (\mathcal{G} - 1)^m,$$

only $m \leq n$ can contribute at total order n . Every differentiated product is a tensor product of the preceding cross-moment operators, up to permutations within the two sides. Its norm is therefore at most the product of their norms. To bound the total coefficient, distribute the n labeled derivative slots among at most n ordered nonempty factors. There are at most n^{n+1} such assignments; the product of the factorials within any assignment is at most $n!$. Enlarging an absolute constant bounds their total by $C^n n^{C^n}$.

The mixed derivatives of $\log \mathcal{G}$ are precisely the corresponding derivatives of $\log \mathbb{E} \exp((\theta + \omega) \cdot S)$, since the other two logarithms have only one variable group. They are the cumulants, and the two groups can represent any prescribed nontrivial split of the n slots. This proves the claim. $\square$

The dependence on y in (B.1) is a linear tilt with coefficient ρ/v_ρ . We normalize the spatial derivatives by ρ^q . The cumulant formula below defines their extension at zero.

Source attribution. Adapted from OpenAI [14, Proposition 3.2].

Proposition B.2 (Gaussian jet bounds). *Under the assumptions above, in particular $l \leq 1/2$, define for each integer $q \geq 1$*

$$U_q(\rho, y) = \rho^{-q} \nabla^q M_\rho(y) \quad (\rho > 0).$$

These tensors extend smoothly to $\rho = 0$. There is an absolute constant C such that, for every $0 \leq \rho < 1$, $y \in \mathbb{R}^d$ and every split,

$$|U_q(\rho, y)|_{\text{split}} \leq C^q q^{C^q} L(\sqrt{v_\rho})^{-(q-1)}. \quad (\text{B.4})$$

Here and below the vector-output slot is included among the tensor slots. The constant is uniform in the dimension, the primitive, its center and scale, and the derivative order q . Moreover, DM_ρ and U_1 are symmetric, and

$$\nabla H_\rho(y) = y + r\rho M_\rho(y), \quad \|\nabla^2 H_\rho\|_{\text{op}} \leq C. \quad (\text{B.5})$$

For $j \geq 1$, every split of $\nabla^j \nabla^2 H_\rho$ is bounded by

$$C^{j+1} (j+1)^{C(j+1)} (\sqrt{v_\rho})^{-j}. \quad (\text{B.6})$$

Proof. We apply the cumulant bound to a joint vector containing the state and the gradient. The conditional potential has Hessian between $(v_\rho^{-1} - l)I$ and $(v_\rho^{-1} + l)I$, so Lemma A.2 gives Poincaré constant at most Cv_ρ . Under this conditional law the map

$$z \mapsto S(z) = (z, f(z)/L)$$

has derivative norm at most $\sqrt{2}$, since f is L -Lipschitz. Pulling back an ambient test function shows that the possibly singular law of S satisfies Poincaré with constant Cv_ρ . It has exponential moments of every linear functional: f has at most linear growth and the conditional law has Gaussian tails. Lemma B.1 therefore applies. This use of repeated Poincaré does not differentiate f repeatedly. Each application pulls back a new ambient polynomial and uses only Df .

In the conditional density, the linear tilt parameter for z is $\rho y/v_\rho$. Repeated differentiation of an expectation with respect to a linear tilt gives joint cumulants. In coordinates this says

$$(U_q)_{i,k_1,\dots,k_q} = v_\rho^{-q} \text{cum}(f_i(Z), Z_{k_1}, \dots, Z_{k_q} \mid Y = y). \quad (\text{B.7})$$

It can also be checked directly by differentiating the logarithm of the conditional moment-generating function with one auxiliary parameter coupled to f_i . Restrict the cumulant of S to one f/L

coordinate and q ordinary coordinates. Lemma B.1 gives a factor $Lv_\rho^{(q+1)/2}$, and the prefactor v_ρ^{-q} gives (B.4).

All the conditional integrals used here are smooth in (ρ, y) for $\rho < 1$. This follows by differentiating their densities: on compact parameter sets, every derivative inserts a polynomial with an integrable Gaussian majorant. Formula (B.7) gives the smooth extension at zero without division by ρ . For later integrations by parts we record a slightly stronger regularity fact. On each compact interval in $\rho < 1$, conditional moments of any fixed order grow at most polynomially in y . Indeed, the conditional mode satisfies

$$z_0 = \rho y - v_\rho r g(x + rz_0),$$

so $|z_0| \leq C(|y| + r|g(x)|)$. The upper and lower conditional Hessian bounds compare the density and its normalizing integral around this mode to Gaussian integrals. This proves the claimed moment growth. The same argument applies to time derivatives of the conditional integrals. Dimension-dependent constants are harmless for this regularity justification; the quantitative estimates above remain dimension-uniform.

Conditional integration by parts gives $\mathbb{E}[Z \mid Y = y] = \rho y - v_\rho r M_\rho(y)$. Differentiating the marginal density then yields the first identity in (B.5). For $\rho > 0$ it implies symmetry of DM_ρ , hence of U_1 ; continuity handles zero. Its derivative, together with (B.4), gives the bounded Hessian and (B.6). In particular, all these conclusions use only the original C^2 regularity of F . $\square$

B.2 The transport equation

Source attribution. Adapted from OpenAI [14, Proposition 3.3].

Proposition B.3 (Probability transport). *The ordinary differential equation*

$$\frac{dY_\rho}{d\rho} = -rM_\rho(Y_\rho) \tag{B.8}$$

transports $\xi_0 = \mathcal{N}(0, I_d)$ to ξ_ρ for every $\rho < 1$. Its velocity is globally Cl -Lipschitz in space. On every compact probability-time interval it has a smooth flow in both directions.

Proof. We identify the conditional velocity of the interpolation and then use uniqueness of the induced transport. Differentiation gives

$$\frac{d}{d\rho}(\rho Z + \sqrt{v_\rho} N) = \frac{Z - \rho Y}{v_\rho}.$$

The conditional mean of this derivative is $-rM_\rho(Y)$ by the conditional integration-by-parts identity above. Thus, for compactly supported smooth ϕ ,

$$\frac{d}{d\rho} \int \phi d\xi_\rho = -r \int M_\rho \cdot \nabla \phi d\xi_\rho.$$

On the other hand, (B.4) with $q = 1$ gives $r\|DM_\rho\|_{\text{op}} \leq Cl$. The velocity has at most linear growth, so its smooth ordinary differential equation has a global-in-space flow and inverse on every compact interval in $\rho < 1$. To identify its pushforward, fix a smooth compactly supported terminal test and transport it backward along this flow. The transported test has zero material derivative. Its supports remain in a bounded set on the interval, by the linear-growth bound. The last display then shows its expectation under ξ_ρ is constant, which proves the pushforward assertion. $\square$

The flow is now defined and its spatial derivatives are controlled. For high-order quadrature we next need repeated derivatives along its trajectories. We keep the dimension dependence visible in that step.

C Adjoint divergence and material derivatives

We now differentiate the conditional mean along its probability flow. The tensor calculation retains matrix split norms until the last vector-valued divergence; only that last step incurs a square-root dimension factor. We give the complete argument of [14], including its high-moment trace estimate, since ordinary directional derivative bounds would not suffice for the numerical formulas.

Retain the primitive parameters $L = \lambda r$ and $l = rL \leq 1/2$, and the interpolation $\xi_\rho = \text{Law}(\rho Z + \sqrt{v_\rho} N)$, where $v_\rho = 1 - \rho^2$ and $0 \leq \rho < 1$. Expectations and L^p norms in this section use ξ_ρ at the displayed probability time. Along a solution of $Y'_\rho = -rM_\rho(Y_\rho)$, the chain rule gives

$$\frac{d}{d\rho} Q(\rho, Y_\rho) = (\mathcal{D}Q)(\rho, Y_\rho), \quad \mathcal{D} = \partial_\rho - rM_\rho \cdot \nabla.$$

Thus the required path derivatives are iterates of $\mathcal{D}$.

We express the material identities using the spatial adjoint under ξ_ρ . For a tensor field V with distinguished slot k , define its adjoint divergence on that slot by

$$(\nabla^* V)_I = \sum_{k=1}^d \{-\partial_k V_{I,k} + (\partial_k H_\rho) V_{I,k}\}.$$

Here I lists the slots that remain; the distinguished slot need not have been last before a permutation. For a compactly supported smooth test function ϕ , the defining integration-by-parts relation is

$$\int (\nabla^* V)_I \phi \, d\xi_\rho = \int \sum_{k=1}^d V_{I,k} \partial_k \phi \, d\xi_\rho.$$

The next identities express material derivatives using this operation and contractions of the normalized spatial jets $U_q = \rho^{-q} \nabla^q M_\rho$ from Proposition B.2.

C.1 Material derivatives of the conditional jets

Source attribution. Adapted from OpenAI [14, Lemma 4.1].

Lemma C.1. *The mean and its jets satisfy*

$$\mathcal{D}M = \nabla^* U_1, \tag{C.1}$$

$$(\mathcal{D}U_q)_{i,I} = (\nabla^* U_{q+1})_{i,I} + 2r\rho \sum_{\emptyset \neq S \subseteq I} (U_{|S|})_{k,S} (U_{q-|S|+1})_{i,k,I \setminus S}, \quad q \geq 1. \tag{C.2}$$

In the second identity I is a labeled list of q derivative slots; subsets refer to positions in that list and the index k is summed. For any smooth tensor field V ,

$$\mathcal{D}(\nabla^* V) = \nabla^*(\mathcal{D}V) + r\nabla^*((DM_\rho)V), \tag{C.3}$$

where the matrix acts on the slot removed by the divergence. These identities hold also at $\rho = 0$ by continuous extension.

Proof. Differentiate $\mathbb{E}[f_i(Z)\phi(Y)] = \int (M_\rho)_i \phi \, d\xi_\rho$. The right side differentiates using the transporting velocity from Proposition B.3. On the left the random velocity is $(Z - \rho Y)/v_\rho$. Subtract its conditional mean from the flux multiplying $\nabla \phi$. The remaining coefficient is $\text{Cov}(f_i(Z), Z_k \mid Y)/v_\rho = (U_1)_{i,k}$ by (B.7). Integration by parts proves (C.1).

For the higher identity, coordinate differentiation gives

$$\mathcal{D}\partial_I M_i = \partial_I \mathcal{D}M_i + r \sum_{\emptyset \neq S \subseteq I} (\partial_S M_k) \partial_k \partial_{I \setminus S} M_i.$$

Insert (C.1) and multiply by ρ^{-q} . When all derivatives pass inside the adjoint divergence, the result is $\nabla^* U_{q+1}$. In the other terms they hit the score $y + r\rho M_\rho$. A derivative hitting its linear part contributes qU_q/ρ , canceled by differentiating the prefactor ρ^{-q} . The nonlinear score terms and the material/spatial commutator terms are identical after scaling: each is

$$r\rho \sum_{\emptyset \neq S \subseteq I} (U_{|S|})_{k,S} (U_{q-|S|+1})_{i,k,I \setminus S}.$$

This proves (C.2). Each contraction retains at least one free slot in each tensor.

Finally differentiate the adjoint identity against a transported scalar test, componentwise in the other slots. The terms involving $\mathcal{D}\phi$ cancel. The gradient commutator is

$$\mathcal{D}\partial_k \phi - \partial_k \mathcal{D}\phi = r(\partial_k M_j) \partial_j \phi.$$

Moving it to the other side by the adjoint relation gives (C.3). All coefficients in the final formulas are nonsingular at zero, so the extensions follow from Proposition B.2. $\square$

C.2 Contractions and the expressions they generate

The product terms in the material identities contract one slot from each of two tensors. We need this operation to preserve split-norm bounds. The next lemma gives the matrix-composition interpretation; it requires both factors to retain a free slot.

Source attribution. Adapted from OpenAI [14, Lemma 4.2].

Lemma C.2 (A single contraction). *Let A and B be tensors, and contract one slot of A with one slot of B , leaving at least one free slot in each. For any nontrivial split of the resulting free slots there are nontrivial splits of A and B such that*

$$|A * B|_{\text{split}} \leq |A|_{\text{split}} |B|_{\text{split}}.$$

Proof. Fix the desired split of the resulting tensor. We orient the contracted index so that each factor acts between nonempty input and output groups. If all free slots of one factor lie on the same side, its contracted slot must be placed on the opposite side of that factor's unfolding. These requirements are compatible: incompatibility would put every free slot of the result on one side, contrary to the hypothesis. If a factor already has free slots on both sides, it imposes no requirement on the orientation. With the resulting orientation, the contraction is composition of the two matrix actions, with identities on the unused input and output coordinates and with coordinate permutations. Identities and these permutations have operator norm one. Submultiplicativity proves the assertion. $\square$

We now define the class of expressions generated by the material identities. Its contraction rule does not permit a trace within a single factor.

Source attribution. Adapted from OpenAI [14, Definition 4.3].

Definition C.3 (Tensor expressions and their weights). A tensor expression is a finite formula built by the following rules.

1. For each integer $q \geq 1$, U_q/L is an expression with $q + 1$ free slots and weight $q - 1$.
2. A permutation of the free slots preserves the weight.
3. Two expressions may be contracted on one chosen slot each, provided each factor retains at least one free slot. The resulting weight is the sum of their weights.
4. An adjoint divergence on a chosen free slot is allowed when at least two free slots remain. It increases the weight by one.

The formula specifies its leaves, their orders, and the slots used at each operation. This finite syntactic data is independent of d, ρ, x, r and of the potential. Repeated occurrences are treated as separate factors, even when their values agree.

For example, $\nabla^*(U_2/L)$ has two free slots and weight two, whereas $\nabla^*(U_1/L)$ is not an expression under these rules: it has vector output. The latter is precisely the operation we will reserve for the final $\sqrt{d}$ estimate. If E is an expression, $\nabla^j E$ has j additional free derivative slots, and a split may place any of these slots on either side. The next theorem bounds all such splits simultaneously. Its dependence on j and p is explicit because its proof must later take a moment order growing with $\log d$.

For a related Gaussian result, [4, Theorem 1.1 and Corollary 1.4] prove sharp divergence inequalities with Hilbert–Schmidt derivative norms. Here we require estimates under the generally non-Gaussian law ξ_ρ that retain every matrix split through the contractions and adjoint divergences in Definition C.3.

Source attribution. Adapted from OpenAI [14, Theorem 4.4].

Theorem C.4 (Tensor divergence bound). *Let E be a tensor expression of weight k in Definition C.3. There is a constant $C_E \geq 1$, depending only on that finite formula, such that for every integer $j \geq 0$, every real $2 \leq p < \infty$, every $0 \leq \rho < 1$, and every nontrivial split,*

$$\left\| |\nabla^j E|_{\text{split}} \right\|_{L^p(\xi_\rho)} \leq C_E^{j+1} (p + j + \log(d + 1))^{C_E(j+1)} (\sqrt{v_\rho})^{-(k+j)}. \quad (\text{C.4})$$

The same C_E works for all j, p, d, ρ, x, r . In particular it is not restricted to fixed derivative or moment orders.

Proof. We induct on the construction of E , simultaneously in the derivative and moment orders. The only new estimate is the adjoint-divergence step. For that step, we expand a high trace moment and remove the adjoints by integration by parts. We will arrange each resulting contraction as a matrix composition followed by one trace.

For a normalized jet, $\nabla^j U_q = \rho^j U_{q+j}$, so Proposition B.2 proves the assertion after enlarging a constant depending on q . Permutations cause no change. For a contraction, distribute the derivatives by Leibniz, use Lemma C.2 on each summand, and use the inductive estimates at exponent $2p$ on both factors. The number of Leibniz summands is at most 2^j . Their weights and derivative orders add, giving (C.4) with an enlarged fixed constant.

It remains to consider $E = \nabla^* V$, where V has weight k_V and E has at least two slots. Commute the j derivatives past the divergence. Apart from

$$T = \nabla^*(\nabla^j V),$$

the terms are single contractions of $\nabla^{j-h}V$ with an order- h derivative of the score, $1 \leq h \leq j$. Both factors retain slots. The score derivative is bounded pointwise by a constant of the form $C^h h^{Ch} (\sqrt{v_\rho})^{-(h-1)}$, by (B.5)–(B.6). Lemma C.2, the inductive hypothesis, and Leibniz multiplicities therefore give the required bound for all these terms, even with exponent $k_V + j - 1$ in place of the larger allowed exponent $k_V + j + 1$. It remains to bound T .

Step 1: integration by parts in a trace moment. Fix an unfolding of T , with both sides nonempty, and an integer $s \geq 1$. Pointwise,

$$\|T\|_{\text{op}}^{2s} \leq \text{tr}((TT^T)^s).$$

Expand the trace by coordinate indices. It contains $2s$ labeled primary factors, each an adjoint divergence of $\nabla^j V$. Their free slots are joined in the alternating trace cycle; the two bundles incident to each primary factor are nonempty.

Remove adjoint divergences one at a time by integration by parts. For a chosen primary, its adjoint differentiates the product of the other factors, never its own inside field. If a derivative hits a remaining adjoint, use

$$\partial_i(\nabla^* W) = \nabla^*(\partial_i W) + (\partial_i \partial_k H_\rho) W_k.$$

The first term leaves that adjoint to be processed later. The second consumes it and creates a Hessian factor joining the two divergence indices. A later derivative may also hit a previously created Hessian factor. After all adjoints have been removed, a typical term has primary factors $\nabla^{j+e_i} V$, $1 \leq i \leq 2s$, and n_H factors $\nabla^{h_v} \nabla^2 H_\rho$. The bookkeeping identity is

$$\sum_{i=1}^{2s} e_i + \sum_{v=1}^{n_H} h_v + 2n_H = 2s. \quad (\text{C.5})$$

Indeed, a removed adjoint either adds one derivative or pairs with another adjoint to create a Hessian. Each step has at most Cs possible target factors and at most two commutator terms, so there are at most $(Cs)^{Cs}$ resulting terms.

We use a graph only to describe these explicit coordinate sums. Its primary vertices are the $2s$ labeled tensor factors; a Hessian vertex represents one factor $\nabla^{h_v} \nabla^2 H_\rho$. An edge represents a pair of tensor slots summed over a common coordinate, and a bundle represents several such edges. Three properties will control its contraction. All original trace bundles remain. Every new direct edge connects distinct primary factors, because an adjoint never differentiates its own inside field. Finally, each Hessian factor has at least two distinct primary neighbors, the pair that created it, and there are no Hessian-to-Hessian edges. Later derivatives only add further primary-to-Hessian edges. These statements remain true if some derivatives had already been commuted inside an unprocessed adjoint.

Step 2: one closing trace. Choose one bundle in the original trace cycle and cut it, leaving its $b \geq 1$ indices unsummed at the two ends. All remaining original bundles form a chain through the $2s$ primary factors. Order these factors along that chain. For each Hessian factor, its two distinct creating primaries have different positions, so we can place it strictly between its earliest and latest primary neighbors. No edge joins two Hessians, so these placements impose no competing condition. Order Hessians occupying the same interval arbitrarily, and orient every uncut edge from its earlier factor to its later factor. The b cut indices enter the first primary and leave the last one.

Every tensor has both an input and an output in this ordering. An interior primary receives one original chain bundle and sends the next; at the two endpoints the cut bundle supplies the

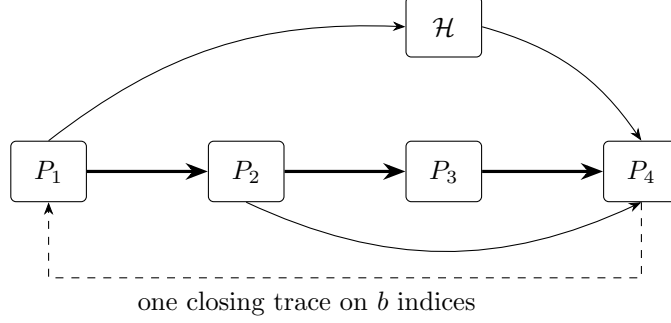


Figure 1: A schematic term with four primary tensors $P_i = \nabla^{j+e_i} V$ after removing the adjoint divergences. Thick arrows are the surviving trace-chain bundles; thin arrows are new contracted indices. Here $\mathcal{H} = \nabla^2 H_\rho$ is the Hessian factor, inserted between its first and last primary neighbors. The arrows specify matrix inputs and outputs, so every factor acts across a nontrivial split. The dashed closure is the one trace in (C.6); its cost d^b becomes bounded after taking a sufficiently high moment root.

missing side. A Hessian receives an edge from an earlier neighbor and sends an edge to a later one. Thus all the matrix unfoldings used below are nontrivial splits, exactly as required by the induction hypothesis.

Here is an explicit matrix-composition interpretation. Immediately before processing a factor, retain one copy of $\mathbb{R}^d$ for each index whose earlier endpoint has been processed but whose later endpoint has not. The initial space is $(\mathbb{R}^d)^{\otimes b}$, belonging to the cut inputs. Permute these tensor coordinates so that the incoming indices of the next factor are adjacent. Apply that factor's matrix unfolding to them and the identity to all other retained coordinates; then permute the outputs into the new retained list. Matrix multiplication performs precisely the sums over the incoming indices. Induction over the factors therefore accounts for every uncut index once and leaves only the b cut outputs at the end. If Q denotes the resulting map from $(\mathbb{R}^d)^{\otimes b}$ to itself, then

$$\|Q\|_{\text{op}} \leq \prod_{i=1}^{2s} |\nabla^{j+e_i} V|_{\text{split}} \prod_{v=1}^{n_H} |\nabla^{h_v} \nabla^2 H_\rho|_{\text{split}}.$$

Indeed coordinate permutations and the intervening identity maps have operator norm one. Closing the original cut is now exactly a single matrix trace, so the absolute value of this completely contracted term is at most

$$|\text{tr } Q| \leq d^b \|Q\|_{\text{op}}. \quad (\text{C.6})$$

This argument explains why other undirected cycles introduce no further dimension factor: their coordinate sums have already been performed by the matrix compositions. Figure 1 shows a possible ordering.

Step 3: choice of the moment order. The coordinate expansion and its contraction estimate are now complete. It remains to verify that their constants allow s to grow with d and j . Apply the induction hypothesis to every primary factor with moment exponent $2s$, and use the pointwise Hessian estimates on all other factors. There are exactly $2s$ random primary factors, so Hölder's inequality uses no larger exponent. The power of $1/\sqrt{v_\rho}$ before taking the $2s$ -th root is at most

$$2s(k_V + j) + \sum_i e_i + \sum_v h_v \leq 2s(k_V + j + 1).$$

Since $e_i, h_v \leq 2s$, all polynomial bases in the inductive estimates are at most $C(s + j + \log(d + 1))$. The total exponent of those bases is at most

$$C \left(2s(j + 1) + \sum_i e_i + \sum_v (h_v + 1) \right) \leq C's(j + 1),$$

using (C.5). The term count $(Cs)^{Cs}$ is absorbed in the same bound. The cut bundle uses only the original free slots of $T = \nabla^*(\nabla^j V)$, not the new indices introduced by integration by parts. Consequently $b \leq j + C_V$, where C_V depends only on the fixed number of slots of V , and not on s . We have proved

$$\mathbb{E}\|T\|_{\text{op}}^{2s} \leq d^{j+C_V} [C(s + j + \log(d + 1))]^{Cs(j+1)} (\sqrt{v_\rho})^{-2s(k_V+j+1)}.$$

Choose s to be the smallest positive integer satisfying $2s \geq p$ and $s \geq (j + C_V) \log(d + 1)$, increasing C_V if necessary. Then $d^{(j+C_V)/(2s)} \leq e^{1/2}$ and

$$s \leq C_V(p + (j + 1) \log(d + 1)).$$

The expression on the right is bounded by a fixed multiple of $(p + j + \log(d + 1))^2$. Taking the $2s$ -th root, using monotonicity of probability-space L^p norms, and increasing C_E now gives (C.4) with $k = k_V + 1$.

All integration-by-parts identities above are legitimate for each finite j, s, d and each $\rho < 1$. Indeed, the conditional-integral regularity in Proposition B.2, together with the score formula, gives polynomial growth for every field and derivative appearing in the finite expansion. The laws ξ_ρ have Gaussian tails. Cutoffs therefore justify the identities with vanishing boundary terms. Uniformity in growing s is supplied by the explicit estimates after the identities, not by an unproved uniform regularity assertion. This completes the induction. $\square$

C.3 One final vector dimension factor

We now pass from split norms to the vector output. All preceding operations retained at least two slots. The last divergence is estimated in Hilbert–Schmidt norm, using the following L^2 identity.

Source attribution. Adapted from OpenAI [14, Lemma 4.5].

Lemma C.5 (Hilbert–Schmidt divergence estimate). *Let $\mu(dy) \propto e^{-H(y)} dy$, where $H \in C^2(\mathbb{R}^d)$ has bounded Hessian and μ has Gaussian tails. Let V be a smooth tensor field with a distinguished divergence slot and with $V, \nabla V \in L^2(\mu)$ in Hilbert–Schmidt norm. Then*

$$\|\nabla^* V\|_{\text{HS},2} \leq C(\|V\|_{\text{HS},2} + \|\nabla V\|_{\text{HS},2}), \tag{C.7}$$

where C depends only on $\|\nabla^2 H\|_{\text{op},\infty}$, and is independent of d and the number of remaining tensor slots. For $H = H_\rho$ it is an absolute constant.

Proof. First take a compactly supported smooth vector field, so that the output is scalar. Integrating by parts twice gives

$$\int (\nabla^* V)^2 d\mu = \int \sum_{i,j} [(\partial_i V_j)(\partial_j V_i) + V_i(\partial_{ij} H)V_j] d\mu.$$

The first term is bounded in absolute value by $|\nabla V|_{\text{HS}}^2$, and the second by $\|\nabla^2 H\|_{\text{op}}|V|^2$. For tensor output, apply the scalar identity to each fixed collection of remaining indices and sum. Taking a square root proves (C.7) for compactly supported fields. For general V , multiply by smooth cutoffs χ_R equal to one on the ball of radius R , with $|\nabla \chi_R| \leq C/R$. Then $\chi_R V \rightarrow V$ and $\nabla(\chi_R V) \rightarrow \nabla V$ in $L^2(\mu)$, while their adjoint divergences converge pointwise. Fatou's lemma proves the stated estimate. All fields to which we apply it have polynomially growing derivatives and hence satisfy the required integrability. $\square$

Source attribution. Adapted from OpenAI [14, Corollary 4.6].

Corollary C.6 (Derivatives along transported paths). *For each fixed integer $m \geq 1$,*

$$\|\mathcal{D}^m M_\rho\|_{L^2(\xi_\rho)} \leq \text{polylog}(d) L \sqrt{d} v_\rho^{-m}. \quad (\text{C.8})$$

For fixed integers $q \geq 1$, $m \geq 0$, and fixed real $2 \leq p < \infty$, every nontrivial split satisfies

$$\| |\mathcal{D}^m U_q|_{\text{split}} \|_{L^p(\xi_\rho)} \leq \text{polylog}(d) L (\sqrt{v_\rho})^{-(q-1+2m)}. \quad (\text{C.9})$$

The constants hidden in $\text{polylog}(d)$ may depend on m, q, p when these parameters occur, but are uniform in ρ , the primitive center, scale, and oracle.

Proof. We first prove (C.9), and then isolate the final divergence needed for (C.8). Iterate Lemma C.1 and use the product rule on each contraction. Replacing U_q by $\nabla^* U_{q+1}$ raises weight by two. Replacing it by one of the contracted products in (C.2) does not raise weight: the two jet weights sum to $q - 1$. The commutator (C.3) inserts $DM_\rho = \rho U_1$ of weight zero on the divergence slot. Every additional jet factor L is accompanied by a factor r , so, after retaining one overall L , the extra factors are powers of $l = rL \leq 1$. Scalar coefficients are fixed polynomials in ρ and remain bounded under the fixed number of differentiations. Starting from a jet U_q , all divergences retain at least two free slots, and all contractions retain a slot of each factor. The resulting terms are therefore expressions of Theorem C.4, of weight at most $q - 1 + 2m$. Its case $j = 0$ proves (C.9).

Starting instead from M , the first identity is $DM = \nabla^* U_1$. Its outermost divergence remains a divergence to a vector under further material differentiations. Inside it, the same construction gives finitely many normalized expressions of weight at most $2(m - 1)$, each with one overall factor L . Apply Lemma C.5 at this outermost operation. For a tensor with at least two slots, its Hilbert–Schmidt norm is at most $\sqrt{d}$ times the operator norm of an unfolding that puts one coordinate slot on one side: that matrix has at most d singular values. Thus Theorem C.4, for $j = 0, 1$ and $p = 2$, bounds the right side of (C.7) by

$$\text{polylog}(d) L \sqrt{d} (\sqrt{v_\rho})^{-(2m-1)} \leq \text{polylog}(d) L \sqrt{d} v_\rho^{-m}.$$

This is (C.8). $\square$

To see the quantitative connection with quadrature, start the transport at its correct marginal and put $Q_\rho = M_\rho(Y_\rho)$. Proposition B.3 ensures that $Y_\rho \sim \xi_\rho$, so (C.8) bounds $\|Q_\rho^{(m)}\|_2$ for every probability time. On a cell $[\rho_0, \rho_1] \subset [0, 1)$ of length h_c , Taylor's integral remainder and the L^2 triangle inequality give, for every $\rho \in [\rho_0, \rho_1]$,

$$\left\| Q_\rho - \sum_{h=0}^{m-1} \frac{(\rho - \rho_0)^h}{h!} Q_{\rho_0}^{(h)} \right\|_2 \leq \frac{h_c^m}{m!} \sup_{t \in [\rho_0, \rho_1]} \|Q_t^{(m)}\|_2 \leq C_m \text{polylog}(d) L \sqrt{d} h_c^m (1 - \rho_1^2)^{-m}.$$

In particular, for a mesh ratio $\theta > 0$, the condition $h_c \leq \theta(1 - \rho_1)$ makes this remainder at most $C_m \text{polylog}(d)L\sqrt{d}\theta^m$. This is a supremum of deterministic RMS bounds; it does not require an expected supremum over a random trajectory. The proof of Proposition E.1 uses this cancellation between cell length and endpoint singularity in polynomial quadrature. Before constructing those finite formulas, we use the material identity itself to produce a stationary flow whose path integral represents a noisy mean.

D A stationary representation of a noisy mean

The mean formula comes from an invariant flow whose integrated observable has exactly the desired Gaussian law. We construct its centering kernel, prove global existence and invariance, and derive the stationary time-derivative estimates used by both Picard quadrature and the cold residual argument. This section develops the construction of [14] with the conditional notation fixed above.

Retain the primitive and interpolation notation of Sections A–C. Thus $L = \lambda r$, $l = \lambda r^2 \leq 1/2$, $M_0 = u$, and the symmetric first jet satisfies $DM_\rho = \rho U_1$. Fix

$$0 < R < 1/2, \quad b_* = (1 - R^2)^{1/2},$$

and use a star for evaluation at $\rho = b_*$. The positive smoothing radius R keeps all coefficients smooth, although F itself is only C^2 . In this section ρ is the Gaussian-interpolation parameter; the new stationary flow will use a separate time variable t .

D.1 Transport Jacobians and the zero endpoint

Let $T_\rho = T_{\rho \rightarrow b_*}$ denote the probability transport (B.8) from time ρ to time b_* . Given a terminal position y , define

$$Y_\rho(y) = T_\rho^{-1}(y), \quad B_\rho(y) = DT_\rho(Y_\rho(y)) = (DY_\rho(y))^{-1}.$$

In particular, $Y_{b_*}(y) = y$. The backward path satisfies $Y'_\rho = -rM_\rho(Y_\rho)$ with this terminal condition. Its spatial Jacobian $J_\rho = DY_\rho$ solves

$$J'_\rho = -rDM_\rho(Y_\rho)J_\rho, \quad J_{b_*} = I_d.$$

Differentiating its inverse gives the order of multiplication in

$$B'_\rho = rB_\rho DM_\rho(Y_\rho) = r\rho B_\rho U_1(\rho, Y_\rho), \quad B'_0 = 0. \quad (\text{D.1})$$

The equality $B'_0 = 0$ will be used in the quadrature at zero. Gronwall also gives $\|B_\rho\|_{\text{op}} \leq C$, since $r\|DM_\rho\|_{\text{op}} \leq Cl$ on an interval of length at most one. We record the higher spatial and time derivative bounds together.

Source attribution. Adapted from OpenAI [14, Lemma 5.1].

Lemma D.1 (Transport derivatives). *For each fixed $k \geq 1$, the order- k spatial derivatives of any transport between two times in $[0, b_*]$, after subtracting the identity map, have all nontrivial split norms at most*

$$C_k l R^{-(k-1)}.$$

For each fixed $m \geq 1$ and fixed finite $p \geq 1$, uniformly in $0 \leq \rho \leq b_$,*

$$\|B_\rho^{(m)}(y)\|_{\text{op}}\|_{L^p(y \sim \xi_*)} \leq \text{polylog}(d) l (1 - \rho^2)^{-m}.$$

The constants and the polylogarithmic factor may depend on the fixed orders and on p , but not on the primitive center or scale.

Proof. For the spatial estimate, differentiate the transport in its initial position. Gronwall bounds its first derivative and gives an $O(l)$ bound for the difference from I_d . For order $k > 1$, the chain rule has terms consisting of an order- j derivative of the velocity and derivatives of the state of orders $i_1, \dots, i_j \geq 1$, where $i_1 + \dots + i_j = k$. By (B.4), the outer tensor has every split norm at most $C_j l R^{-(j-1)}$. Attach the inner tensors one at a time. Each attachment is a single contraction retaining the outer output and the inner derivative slots, so Lemma C.2 applies without a dimension factor. Bound a first state derivative by a constant and use the induction hypothesis for all higher state derivatives. Since $l \leq 1$, the total power of R^{-1} is at most

$$(j-1) + \sum_{\nu=1}^j (i_\nu - 1) = k - 1,$$

and each forcing term retains the outer factor l . The term involving the unknown order- k state derivative is linear, with coefficient bounded by Cl , and this derivative starts at zero. Gronwall proves the assertion. The same argument with absolute interval lengths applies to transport in the reverse direction.

For the second assertion, write $C_\rho = rDM_\rho(Y_\rho) = r\rho U_1(\rho, Y_\rho)$, so that $B'_\rho = B_\rho C_\rho$. The product rule expresses $B_\rho^{(m)}$ as a finite sum of ordered products of B_ρ and $k \geq 1$ factors $C_\rho^{(j_1)}, \dots, C_\rho^{(j_k)}$, with $j_1 + \dots + j_k = m - k$. The zero-order jet bound and (C.9) imply, for every fixed moment order,

$$\| \| C_\rho^{(j)} \|_{\text{op}} \|_{L^p} \leq \text{polylog}(d) l (1 - \rho^2)^{-j}.$$

Here the path estimates apply because inverse transport sends $y \sim \xi_*$ to $Y_\rho(y) \sim \xi_\rho$. Apply Hölder with sufficiently high fixed moments to each product. Using $l^k \leq l$ and $(1 - \rho^2)^{-(m-k)} \leq (1 - \rho^2)^{-m}$ gives the claimed conservative exponent. Moments below one, if desired, follow from the L^1 bound. $\square$

D.2 A kernel for the centered observable

For a matrix field K , define its row-wise adjoint divergence by

$$(\nabla_{\xi_*}^* K)_i = \sum_j \{ -\partial_j K_{ij} + (\partial_j H_*) K_{ij} \}.$$

Thus $\nabla_{\xi_*}^* K = h$ means $\mathbb{E}_{\xi_*}[h_i \phi] = \mathbb{E}_{\xi_*} \sum_j K_{ij} \partial_j \phi$ for smooth compactly supported scalar tests ϕ . We seek this identity with $h = M_* - u$.

This is a Stein-type integration-by-parts identity. A classical coordinate Stein kernel for a law ξ represents the observable $y - \mathbb{E}_\xi y$ instead. Courtade, Fathi and Pananjady construct such kernels from spectral-gap assumptions [8, arXiv version 2, Definition 2.1 and Theorem 2.4]. Our observable is $M_* - u$; its mean is zero by conditional expectation. The transport formula below supplies its particular adjoint-divergence representation together with pointwise derivative bounds. Those bounds are needed for quadrature and do not follow from the cited existence statement. We do not require K to be symmetric or positive.

Define

$$K(y)^T = \int_0^{b_*} B_\rho(y) U_1(\rho, Y_\rho(y)) d\rho = \frac{1}{r} \int_0^{b_*} B'_\rho(y) \frac{d\rho}{\rho}. \quad (\text{D.2})$$

The first integral defines the integrand at zero in the second formula: $B'_\rho/(r\rho) = B_\rho U_1(\rho, Y_\rho)$ extends smoothly there. The transpose is essential: the adjoint divergence will act on the rows of K , while the position velocity will use K^T .

Source attribution. Adapted from OpenAI [14, Lemma 5.2].

Lemma D.2 (The centering kernel). *The field K is smooth and satisfies*

$$\nabla_{\xi_*}^* K = M_* - u, \quad \|K(y)\|_{\text{op}} \leq CL.$$

For each fixed $j \geq 1$, every nontrivial split norm of $\nabla^j K$ is bounded pointwise by $C_j LR^{-j}$.

Proof. We prove the adjoint-divergence identity against a scalar test. Fix an output coordinate i and a smooth compactly supported ϕ . The function $\phi \circ T_\rho$ is constant along the probability flow, and therefore has zero material derivative. Differentiate its product with $M_{\rho,i}$ under the transported laws. Lemma C.1 gives $\mathcal{D}M_\rho = \nabla_{\xi_\rho}^* U_1$, so integration by parts and the chain rule yield

$$\frac{d}{d\rho} \mathbb{E}_{\xi_\rho} [M_{\rho,i} \phi \circ T_\rho] = \mathbb{E}_{\xi_\rho} \sum_{j,k} (U_1)_{ik} (DT_\rho)_{jk} (\partial_j \phi) \circ T_\rho.$$

Smooth transport on the compact time interval and Gaussian tails justify differentiation; alternatively the test can first be cut off and the resulting identities passed to the limit. Push the expectation forward to $y \sim \xi_*$. Since U_1 is symmetric, its coefficient of $\partial_j \phi(y)$ is

$$\sum_k (U_1)_{ik}(\rho, Y_\rho(y)) (B_\rho(y))_{jk} = (B_\rho(y) U_1(\rho, Y_\rho(y)))_{ji}.$$

The terminal expectation is $\mathbb{E}_{\xi_*} [M_{*,i} \phi]$. At zero, $M_0 = u$ and $(T_0)_\# \xi_0 = \xi_*$, so the initial expectation is $u_i \mathbb{E}_{\xi_*} \phi$. Integrating the displayed derivative in ρ proves

$$\mathbb{E}_{\xi_*} [(M_{*,i} - u_i) \phi] = \mathbb{E}_{\xi_*} \sum_j K_{ij} \partial_j \phi.$$

This proves the adjoint-divergence identity weakly, and smoothness makes it a pointwise identity.

For the bounds, $\|B_\rho\|_{\text{op}} \leq C$ and $\|U_1\|_{\text{op}} \leq CL$ immediately give $\|K\|_{\text{op}} \leq CL$. For a positive-order spatial derivative, use Lemma D.1 on both DT_ρ and Y_ρ . An order- j derivative of DT_ρ has all split norms at most $C_j l R^{-j}$, and an order- j derivative of U_1 has all split norms at most $C_j LR^{-j}$ by (B.4). When these fields are composed with Y_ρ , attach each chain-rule tensor by a single edge as in the preceding lemma. First derivatives of Y_ρ are bounded, and a derivative of order $i > 1$ costs $C_i l R^{-(i-1)}$. Every order- j term in the product has total cost at most $C_j LR^{-j}$, since $l \leq 1$. Each factor retains free slots, so the single-contraction bound preserves every nontrivial split. Integrating over an interval of length at most one proves the assertion. All orders are smooth by Gaussian conditional integration and the smooth transport on $\rho \leq b_* < 1$; this uses no higher derivatives of the original unsmoothed gradient. $\square$

D.3 Stationarity and the path integral

The kernel K defines a skew-matrix flow of the form studied by [13, Section 2.1, equation (3), Theorem 1]. We verify completeness before using the stationary-density identity. This ensures that invariance holds for the full integration interval.

Source attribution. Adapted from OpenAI [14, Proposition 5.3].

Proposition D.3 (A noisy mean from a path integral). *For $s > 0$, let (Y_0, G_0) have law $\xi_* \otimes \mathbf{N}(0, I_d)$ and solve*

$$\dot{Y}_t = -K(Y_t)^T G_t/s, \quad \dot{G}_t = (M_*(Y_t) - u)/s. \quad (\text{D.3})$$

The solution exists for every finite time, and its law remains $\xi_ \otimes \mathbf{N}(0, I_d)$. In particular,*

$$sG_0 + \int_0^1 M_*(Y_t) dt \stackrel{\text{law}}{=} u + sG, \quad G \sim \mathbf{N}(0, I_d). \quad (\text{D.4})$$

Proof. We first establish a complete flow, then invariance, and finally the output law. The vector field v in (D.3) is smooth, so it is locally Lipschitz. The bound on K and the global Lipschitz bound on M_* imply $|v(y, g)| \leq C_{x,r,s}(1 + |y| + |g|)$. Gronwall's inequality therefore prevents a solution from escaping to infinity in finite positive or negative time. Local uniqueness gives a complete smooth flow Φ_t , with inverse Φ_{-t} . We only need local Lipschitz continuity here; the factor G does not in general give a globally Lipschitz joint vector field.

Write the product density as

$$p(y, g) = (2\pi)^{-d/2} \exp(-H_*(y) - |g|^2/2).$$

Differentiating in the position and momentum variables separately, with the row convention of the kernel lemma, gives

$$\text{div}_y(pv_Y) = \frac{p}{s} \sum_i g_i (\nabla_{\xi_*}^* K)_i = \frac{p}{s} g \cdot (M_* - u), \quad \text{div}_g(pv_G) = -\frac{p}{s} g \cdot (M_* - u).$$

Thus $\text{div}(pv) = 0$. The Jacobian formula for the complete flow now yields, pointwise,

$$\frac{d}{dt} \{p(\Phi_t(z)) \det D\Phi_t(z)\} = \text{div}(pv)(\Phi_t(z)) \det D\Phi_t(z) = 0.$$

The determinant starts at one and stays positive. Change of variables therefore proves that Φ_t preserves the probability density p .

For later differentiation it is useful to record the same cancellation as an operator identity. Relative to the product law, the Lie derivative is

$$\mathcal{L} = \nabla^* A \nabla, \quad A = \begin{pmatrix} 0 & K^T/s \\ -K/s & 0 \end{pmatrix}. \quad (\text{D.5})$$

Indeed the second-order part vanishes by skew symmetry. The first-order coefficients are $-K^T g/s$ in position and $\nabla_{\xi_*}^* K/s = (M_* - u)/s$ in momentum, exactly as in (D.3).

Finally, integrating the momentum equation over $[0, 1]$ gives the pathwise identity

$$sG_0 + \int_0^1 M_*(Y_t) dt = u + sG_1.$$

Stationarity makes G_1 a standard Gaussian, proving (D.4). $\square$

Stationarity also makes G_1 independent of Y_1 . It does not assert that G_1 is independent of the initial anchors (Y_0, G_0) . The exact mean $u = M_0$ remains in the momentum velocity, and the proposition assumes an ideal sample from ξ_* . Thus the proposition is a comparison construction, not yet an oracle routine. Section E will express its approximation by finitely many exact mean evaluations; Section F will replace every such evaluation, including M_0 . The displayed output itself needs no separate addition of u .

D.4 Derivatives on a stationary trajectory

For quadrature, it suffices to bound each time derivative in L^2 at a fixed time. Stationarity identifies these norms with norms under the invariant product measure. The following lemma gives the estimate in a form that also records the required coefficient assumptions.

Source attribution. Adapted from OpenAI [14, Lemma 5.4].

Lemma D.4 (Stationary time derivatives). *Fix $0 < R \leq 1$. Let a smooth probability density on $\mathbb{R}^n$, with $n \leq Cd$, have Gaussian tails and negative logarithm H . Assume its Hessian is bounded and that every positive-order score derivative $\nabla^j \nabla H$ has every one-slot split norm at most $C_j R^{-(j-1)}$, for $j \geq 1$. Here a one-slot split isolates one tensor slot as the matrix input and groups all remaining slots as its output. Suppose a stationary autonomous flow has Lie derivative $\mathcal{L} = \nabla^* A \nabla$, where the adjoint is for this density, A is skew symmetric, and*

$$|\nabla^j A|_{\text{one-slot split}} \leq C_j \alpha R^{-j} \quad (j \geq 0)$$

for every one-slot split. Let ϕ_0 be smooth and scalar- or vector-valued, with

$$|\nabla^j \phi_0|_{\text{HS}} \leq C_j a_0 \sqrt{d} R^{-j} \quad (j \geq 1).$$

Assume the fields and all derivatives used have polynomial growth. Then, for each fixed $k \geq 1$, the order- k time derivative of ϕ_0 on the stationary flow has RMS norm at most

$$C_k a_0 \sqrt{d} (\alpha R^{-2})^k. \quad (\text{D.6})$$

Proof. Define $\phi_k = \mathcal{L}^k \phi_0$. We induct on k , proving simultaneously for $j \geq 0$ the stronger estimate

$$\|\nabla^j \phi_k\|_{\text{HS},2} \leq C_{j,k} a_0 \sqrt{d} \alpha^k R^{-(j+2k)}. \quad (\text{D.7})$$

For $k = 0$ the same bound is assumed only for $j \geq 1$, which suffices: the iteration differentiates ϕ_{k-1} before multiplying by A . All L^2 norms below are for the invariant density.

Set $V = A \nabla \phi_{k-1}$. Leibniz's rule and the induction hypothesis give, for each $h \geq 0$,

$$\|\nabla^h V\|_{\text{HS},2} \leq C_{h,k} a_0 \sqrt{d} \alpha^k R^{-(h+1+2(k-1))}.$$

To see why no dimension factor appears, in each coefficient tensor $\nabla^v A$ isolate the slot contracted against the gradient of ϕ_{k-1} . Its one-slot operator norm multiplies the Hilbert–Schmidt norm of the other factor. Summing the finitely many Leibniz terms preserves the displayed estimate.

Now commute j derivatives past the adjoint divergence in $\phi_k = \nabla^* V$. The main term is $\nabla^*(\nabla^j V)$. The Hilbert–Schmidt estimate (C.7) bounds its L^2 norm by a constant times $\|\nabla^j V\|_{\text{HS},2} + \|\nabla^{j+1} V\|_{\text{HS},2}$. These two terms have powers $R^{-(j+2k-1)}$ and $R^{-(j+2k)}$, respectively. Each remaining commutator contracts an order- v derivative of the score, $1 \leq v \leq j$, with $\nabla^{j-v} V$. The coefficient's one-slot norm again suffices; the total exponent of R^{-1} is

$$(v-1) + (j-v+1+2(k-1)) = j+2k-2.$$

All these terms are bounded by (D.7) because $0 < R \leq 1$. This closes the induction, including $j = 0$ for $k \geq 1$. Gaussian tails and polynomial growth justify the integrations by parts through cutoffs.

Along an autonomous trajectory, the k th time derivative of ϕ_0 is ϕ_k evaluated at the current state. Stationarity identifies its RMS norm at every time with the fixed-measure norm in (D.7) for $j = 0$. $\square$

For (D.3), the ambient dimension is $2d$, and (D.5) satisfies the lemma's coefficient bounds with $\alpha = L/s$. The product potential is $H_*(y) + |g|^2/2$; its score-derivative bounds follow from (B.4) and the Gaussian factor. For the joint state take $a_0 = 1$: its first derivative has Hilbert–Schmidt norm $\sqrt{2d}$ and all higher derivatives vanish. For M_* take $a_0 = L$. Indeed, isolating any derivative slot in (B.4) gives a matrix with input dimension d , so passage from its operator norm to its Hilbert–Schmidt norm costs at most $\sqrt{d}$. Consequently, at every stationary time,

$$\left\| \frac{d^k}{dt^k}(Y_t, G_t) \right\|_2 \leq C_k \sqrt{d} ((L/s)R^{-2})^k, \quad \left\| \frac{d^k}{dt^k} M_*(Y_t) \right\|_2 \leq C_k L \sqrt{d} ((L/s)R^{-2})^k.$$

The strict radius choice $0 < R < 1/2$ made for this construction satisfies the abstract lemma's $0 < R \leq 1$ hypothesis.

We also need an initial displacement estimate for the later iteration. Conditional expectation and the primitive Poincaré inequality give

$$\mathbb{E}_{\xi_*} |M_* - u|^2 \leq \mathbb{E}_\nu |f - u|^2 \leq CL^2 d.$$

At every stationary time, G_t is independent of Y_t , hence

$$\mathbb{E} |K(Y_t)^T G_t|^2 = \mathbb{E} \|K(Y_t)\|_{\text{HS}}^2 \leq CL^2 d.$$

Integrate the velocity and apply Minkowski's inequality to obtain

$$\|(Y_t, G_t) - (Y_0, G_0)\|_2 \leq Ct(L/s)\sqrt{d} \leq C(L/s)\sqrt{d} \quad (0 \leq t \leq 1). \quad (\text{D.8})$$

These are fixed-time RMS estimates for the exact stationary flow. They give the derivative and starting-error bounds used in the next section to approximate its path integral by finite formulas.

E Finite Picard formulas for conditional means

Global Picard iteration and high-order quadrature also build on the numerical frameworks of [5, Appendix E, Algorithm E.1 and Theorem E.2] and [11, Algorithm 2 and Theorem 2.3]. The finite conditional-mean circuits and dependency-depth bounds reproduced below are adapted from OpenAI [14, Section 6].

We turn the probability and stationary flows into deterministic finite formulas. The central requirement is a mean-dependency depth fixed before the mesh and interpolation orders. The resulting coefficient, error, and depth bounds are proved explicitly below, following [14]; they are used directly by the recursive implementation and the mean-root specialization.

Fix an integer $K_0 \geq 2$ and $t > 0$ such that

$$40(K_0 + 2)t < \frac{1}{4}, \quad R = \delta^t, \quad \psi = \delta^{4t}, \quad h_0 = \delta^w, \quad 0 < w < t. \quad (\text{E.1})$$

The iteration count will first fix a depth bound. We then choose w and finally the interpolation orders; increasing those orders will not change the depth bound. For the centering flow set $s = \sigma/2$, where $\sigma > 0$ is the parent mean-noise parameter. Thus the centering formula itself will approximate noise of standard deviation $\sigma/2$. Throughout its numerical analysis assume

$$\max\{l, L/\sigma\} \leq \text{polylog}(d)\delta^b, \quad b \geq \frac{1}{2}. \quad (\text{E.2})$$

For probability transport alone only the corresponding bound on l is required. All probability-time intervals lie in $[0, (1 - cR^2)^{1/2}]$, for a fixed $c > 0$. Constants may depend on fixed interpolation orders and fixed parameters, but not on the center, dimension, primitive scale r , noise parameter, or oracle. A symbol B_d below denotes a bound $C(1 + \log d)^C$ with precisely this uniformity; the fixed exponent and constant in a size hypothesis are part of the fixed parameters. All assertions are for sufficiently large d .

E.1 Probability transport by global Picard iteration

Partition a probability-time interval using equal steps of size at most h_0 in $-\log(1 - \rho)$. For a nonzero interval take the ceiling of its transformed length divided by h_0 as the number of cells. Cell lengths ℓ_c then obey $\ell_c \leq Ch_0(1 - \rho)$ on that cell, and the number of cells is

$$O(1 + h_0^{-1}(1 + |\log R|)).$$

A zero-length interval requires no operations. On each cell interpolate at $m \geq 2$ distinct equally spaced nodes, including both ends, and integrate the resulting degree- $(m - 1)$ polynomial. Include partial integrals to every state node. By scaling the fixed Lagrange basis on $[0, 1]$, its absolute value sum is at most C_m , and its order- k derivative absolute sum is at most $C_m \ell_c^{-k}$. Consequently every partial-integral row has absolute weight sum at most C_m , uniformly in the number of cells.

Let w_{ij} be these partial-integral weights from the starting time ρ_{in} to node ρ_i , with signed weights for backward integration. Duplicate shared endpoints as labeled nodes if convenient. Given a starting position y_{in} , use simultaneous updates

$$y_i^{[0]} = y_{\text{in}}, \quad y_i^{[k+1]} = y_{\text{in}} - r \sum_j w_{ij} M_{\rho_j}(y_j^{[k]}). \quad (\text{E.3})$$

All node values at layer $k + 1$ use layer k across the full interval. Thus the dependency depth is the number of Picard layers, independent of the number of cells.

Source attribution. Adapted from OpenAI [14, Proposition 6.1].

Proposition E.1 (Probability-flow circuit). *If the input position has marginal law $\xi_{\rho_{\text{in}}}$, then after N_p iterations the maximum, over all state nodes, of the RMS error relative to the exact transport is at most*

$$C_m \text{polylog}(d) l \sqrt{d} h_0^m + (C_m l)^{N_p} C r D_F(x). \quad (\text{E.4})$$

Every computed state is globally Lipschitz in its initial position with constant C_m . After subtracting the initial position, its Lipschitz constant is at most $C_m l$.

Proof. We estimate the quadrature residual on the exact trajectory and then propagate it through Picard iteration. Write Y_ρ for that trajectory and $Q_\rho = M_\rho(Y_\rho)$. The marginal identity in Proposition B.3 and (C.8) give

$$\|Q_\rho^{(m)}\|_2 \leq B_d L \sqrt{d} (1 - \rho^2)^{-m}.$$

On a cell $I = [a_c, b_c]$ of length ℓ_c , let P be the Taylor polynomial of Q of degree $m - 1$ at a_c . Minkowski's inequality applied to the integral remainder gives, at each $v \in I$,

$$\|Q_v - P(v)\|_2 \leq \frac{1}{(m - 1)!} \int_{a_c}^v (v - z)^{m-1} \|Q_z^{(m)}\|_2 dz \leq C_m B_d L \sqrt{d} h_0^m. \quad (\text{E.5})$$

The last step uses $\ell_c \leq Ch_0(1-z)$ on that cell and $1-z^2 \geq 1-z$. The same estimate holds at the interpolation nodes. Because interpolation reproduces P and its basis has bounded absolute sum, $\|Q_v - \mathcal{I}_c Q(v)\|_2$ obeys the same bound. Integrate this pointwise estimate over the cells, whose total length is at most one. This proves the required quadrature error, for partial integrals and for either orientation. No supremum of the random path has been taken.

Let $e_k = \max_i \|y_i^{[k]} - Y_{\rho_i}\|_2$. The global bound $\text{Lip}(M_\rho) \leq CL$, together with the absolute row sums, gives

$$e_{k+1} \leq C_m l e_k + C_m B_d l \sqrt{d} h_0^m.$$

On each exact transported marginal, conditional expectation and the primitive moment bound give $\|M_\rho\|_2 \leq CD_F(x)$. Integrating the velocity shows $e_0 \leq CrD_F(x)$. Since $C_m l < 1/2$ for large d , summing the geometric recurrence proves (E.4). The norm here is the maximum of the nodewise RMS errors; it is not the RMS of their maximum.

For deterministic inputs the same recurrence, now in maximum node norm, gives a bounded Lipschitz constant for every iterate. In the correction $y_i^{[k+1]} - y_{\text{in}}$, the factor r and the CL -Lipschitz mean give the stronger constant $C_m l$. $\square$

E.2 The kernel action through harmonic trajectories

We need a globally Lipschitz approximation of the vector $K(y)^T G$. An approximation of K alone would leave its error multiplied by an unbounded momentum. We therefore approximate the action directly through harmonic motion and a forward transport correction.

For the kernel integral use the mesh on $[0, b_*]$ above. Its cell lengths also satisfy $\ell_c \geq c'h_0(1-\rho)$ for all sufficiently large d ; this follows from the ceiling construction because the transformed interval length tends to infinity. Set $C_\rho(y) = B_\rho(y) - I$. We first describe a circuit for $C_\rho(y)G$ at one probability node ρ .

Backward position. Apply Proposition E.1 backward from input position y at time b_* to obtain an approximation $\hat{Y}_\rho$ to $Y_\rho(y)$.

Short harmonic trajectories. At fixed probability time ρ , the Hamiltonian equations with potential H_ρ have position integral equation

$$W_z = \cos(z)Y_\rho + \sin(z)G - r\rho \int_0^z \sin(z-v)M_\rho(W_v)dv. \quad (\text{E.6})$$

For every requested angle z , interpolate on the single cell $[0, z]$. Let z_i be its nodes and $\ell_j(v)$ its Lagrange basis, and set

$$w_{ij}^{\sin} = \int_0^{z_i} \sin(z_i - v)\ell_j(v)dv.$$

Use the simultaneous Picard updates

$$\begin{aligned} W_i^{[0]} &= \cos(z_i)\hat{Y}_\rho + \sin(z_i)G, \\ W_i^{[k+1]} &= \cos(z_i)\hat{Y}_\rho + \sin(z_i)G - r\rho \sum_j w_{ij}^{\sin} M_\rho(W_j^{[k]}). \end{aligned} \quad (\text{E.7})$$

At angle zero there is no integral. The free sine and cosine terms are retained exactly.

Differentiate the transport correction. Let d_j be the weights differentiating at zero the polynomial interpolant on angle nodes $0, \psi, \dots, (m-1)\psi$. Their absolute sum is $C_m\psi^{-1}$. At every such angle node, compute $T_\rho(W_z) - W_z$ by the forward probability-flow circuit, keeping only its integral correction at the output. For exact paths,

$$\left. \frac{d}{dz}(T_\rho - \text{id})(W_z) \right|_{z=0} = (B_\rho - I)G = C_\rho G, \quad (\text{E.8})$$

since $W_0 = Y_\rho$ and $W'_0 = G$. The weighted sum with coefficients d_j therefore approximates $C_\rho G$ using exact mean nodes only.

Here and below the interpolation orders denoted by m may be increased independently. Using a common sufficiently large order merely simplifies the error displays. For the next estimates use the same number $N_p \geq 1$ of Picard layers in the backward, harmonic, and forward computations.

Source attribution. Adapted from OpenAI [14, Lemma 6.2].

Lemma E.2 (Errors in the angle computation). *Let $y \sim \xi_*$ and $G \sim \mathbf{N}(0, I_d)$ independently. The exact-input angle differentiation in (E.8) has RMS truncation error at most*

$$C_m l \sqrt{d} \psi^{m-1} R^{-2m} = C_m l \sqrt{d} \delta^{2tm-4t}. \quad (\text{E.9})$$

Before applying the angle differentiation weights, the RMS errors of the computed forward correction values are at most

$$C_m \text{polylog}(d) l \sqrt{d} (h_0^m + \delta^{2tm}) + (C_m l)^{N_p} Cr D_F(x). \quad (\text{E.10})$$

Every such approximate correction is globally Lipschitz in (y, G) with constant at most $C_m l$.

Proof. For exact inputs $(Y_\rho(y), G)$, harmonic motion $\dot{W} = P, \dot{P} = -\nabla H_\rho(W)$ preserves $\xi_\rho \otimes \mathbf{N}(0, I_d)$. The vector field is globally Lipschitz, and its weighted divergence vanishes. Its Lie derivative is $\nabla^* A \nabla$ with constant skew coefficient whose (W, P) block is $-I$. Lemma D.4 therefore applies with $\alpha = 1$, with $a_0 = l$ for $T_\rho - \text{id}$, and with $a_0 = L$ for M_ρ . Indeed Lemma D.1 and (B.4) give the required spatial bounds; changing a one-slot split bound to a Hilbert–Schmidt bound costs only $\sqrt{d}$. To apply the angle stencil, subtract the Taylor polynomial of $z \mapsto (T_\rho - \text{id})(W_z)$ of degree $m-1$ at zero. Its derivative at zero is exact, and its remainder at each stencil node has RMS norm at most $C_m l \sqrt{d} \psi^m R^{-2m}$. Multiplication by the absolute stencil sum C_m/ψ proves (E.9). This uses pointwise RMS remainders on the stationary harmonic path.

For the harmonic state equation, interpolate $M_\rho(W_v)$ in (E.6). Since $z \leq (m-1)\psi$, its Taylor remainder, multiplied by the prefactor r , is bounded by $C_m l \sqrt{d} \psi^m R^{-2m} = C_m l \sqrt{d} \delta^{2tm}$; the additional integration factors are bounded and can be discarded. The initial free-path error is at most $Cr D_F(x)$, and Picard contracts by $C_m l$. The computed harmonic state is globally Lipschitz with bounded constant in its two inputs, by the same row-sum argument as before.

Next compare the forward probability correction started at the exact harmonic state. That state has marginal law ξ_ρ , so Proposition E.1 applies. Errors in its starting state then propagate through the approximate correction with factor $C_m l$, not a factor of order one. This also controls the backward-position error transmitted through the harmonic computation. Combining these estimates proves (E.10) and the asserted global Lipschitz bound. $\square$

E.3 The zero endpoint of the kernel integral

It remains to integrate the derivative of $C_\rho G$ against $1/\rho$. The exact identity $B'_0 = 0$ removes the apparent singularity, and we impose that identity on the interpolant as well.

On all but the first probability cell use ordinary degree- $(m - 1)$ interpolation. On the first cell use $m - 1$ equally spaced value nodes including both endpoints and the additional Hermite condition that the derivative at zero is zero, for a total of m conditions, with $m \geq 3$. This uniquely determines a polynomial of degree at most $m - 1$: a polynomial in the null space would have a double zero at zero and the remaining $m - 2$ distinct zeros. Let q_i be the resulting weights on values when the derivative of this piecewise interpolant is integrated against $1/\rho$.

Source attribution. Adapted from OpenAI [14, Lemma 6.3].

Lemma E.3 (Derivative quadrature at zero). *Let $y \sim \xi_*$ and $G \sim \mathbf{N}(0, I_d)$ independently. For exact values of $C_\rho(y)G$, the RMS error of this quadrature for $\int_0^{b_*} B'_\rho(y)G d\rho/\rho$ is at most*

$$C_m \text{polylog}(d) l \sqrt{d} h_0^{m-2}.$$

The value weights satisfy

$$\sum_i |q_i| \leq C_m h_0^{-1} (1 + |\log R| + |\log h_0|). \quad (\text{E.11})$$

These weights are well defined even if the supplied values are approximate: the derivative condition at zero remains exact.

Proof. By Lemma D.1 and independence of G , the RMS norm of the order- m derivative of $C_\rho(y)G$ is at most $\text{polylog}(d) l \sqrt{d} (1 - \rho^2)^{-m}$. Here multiplication by the independent Gaussian converts the matrix norm to a Hilbert–Schmidt norm and costs at most $\sqrt{d}$.

On an ordinary cell, Taylor expansion and scaling the interpolation basis bound the derivative error by this derivative bound times $C_m \ell_c^{m-1}$. It is therefore at most $C_m \text{polylog}(d) l \sqrt{d} h_0^{m-1} / (1 - \rho)$. After integration against $1/\rho$, all ordinary cells contribute at most this prefactor times $C(1 + |\log R| + |\log h_0|)$, which fits the claimed more conservative h_0^{m-2} bound.

For the first cell, subtract the Taylor polynomial at zero of degree $m - 1$. Its derivative at zero agrees with the exact zero derivative. The second derivative of the remainder, and that of the Hermite interpolant of the remainder, are bounded in RMS by $C_m \text{polylog}(d) l \sqrt{d} \ell_c^{m-2}$. Both first derivatives vanish at zero. If E is the difference between the exact function and its Hermite interpolant on this cell, then

$$\|E'(\rho)\|_2 \leq \int_0^\rho \|E''(v)\|_2 dv \leq C_m B_d l \sqrt{d} \rho \ell_c^{m-2}.$$

Consequently the integral of $E'(\rho)/\rho$ over the first cell has RMS norm at most $C_m B_d l \sqrt{d} \ell_c^{m-1}$. This is within the stated, more conservative h_0^{m-2} bound.

On an ordinary cell the absolute coefficient sum of the interpolant derivative is at most C_m/ℓ_c . Integrating and using $\ell_c \geq c' h_0 (1 - \rho)$ gives a total bounded by

$$C_m h_0^{-1} \int_{\rho_{\text{first}}}^{b_*} \frac{d\rho}{\rho(1 - \rho)}.$$

This gives the logarithms in (E.11). On the first cell, scale to the unit interval. The polynomial derivative vanishes at zero and is bounded by distance to zero times the second derivative, so the integral of its absolute value divided by that distance is bounded. Scaling back gives a weight sum $C_m/\ell_c \leq C_m h_0^{-1}$. This proves the claim. $\square$

E.4 A globally Lipschitz approximate centering velocity

Let $\text{corr}_{ij}(y, G)$ denote the forward correction computed at probability node i and angle node j by the three operations above. The approximate position velocity is the explicit expression

$$\hat{v}_Y(y, G) = -\frac{1}{rs} \sum_i q_i \sum_j d_j \text{corr}_{ij}(y, G). \quad (\text{E.12})$$

Each correction retains only the integral part of the forward Picard formula. Writing its weights as w_{ijk} and its mean values as M_{ijk} gives

$$\text{corr}_{ij} = -r \sum_k w_{ijk} M_{ijk},$$

and hence

$$\hat{v}_Y = \frac{1}{s} \sum_{i,j,k} q_i d_j w_{ijk} M_{ijk}. \quad (\text{E.13})$$

The apparent factor $1/r$ has canceled algebraically. There is no additive forward state left in this formula, and all coefficients multiplying mean values are deterministic scalars. Use the exact mean expression $\hat{v}_G(y, G) = (M_*(y) - M_0)/s$ for the momentum velocity. In particular, the momentum occurs inside angle states; it is never multiplied by a separately approximated matrix at the output. *Source attribution.* Adapted from OpenAI [14, Proposition 6.4].

Proposition E.4 (Kernel-action circuit). *On the stationary product input law, the RMS error of $\hat{v} = (\hat{v}_Y, \hat{v}_G)$ relative to (D.3) is bounded by $\text{polylog}(d)$ times*

$$\begin{aligned} & \frac{L}{\sigma} \sqrt{d} [h_0^{m-2} + h_0^{-1} \delta^{-4t} (h_0^m + \delta^{2tm})] \\ & + \delta^{-4t-w} (C_m l)^{N_p-1} r \frac{L}{\sigma} D_F(x). \end{aligned} \quad (\text{E.14})$$

It is globally Lipschitz, with constant at most

$$C_m \text{polylog}(d) (L/\sigma) \psi^{-1} h_0^{-1}. \quad (\text{E.15})$$

For every fixed $J \geq 1$, its RMS error can be made at most $\text{polylog}(d) \delta^J D_F(x)$ with $N_p = O(J+1)$, independently of w and the interpolation orders.

Proof. Combine Lemmas E.2 and E.3, and divide by rs . The value errors are amplified only by $C_m \psi^{-1}$ and the weight sum (E.11). The exact stencil truncation has factor $\psi^{m-1} R^{-2m} = \delta^{2tm-4t}$ and the same probability-quadrature weight sum. Crucially,

$$\frac{l}{r\sigma} = \frac{L}{\sigma}, \quad \frac{l^{N_p} r D_F(x)}{r\sigma} = l^{N_p-1} r \frac{L}{\sigma} D_F(x).$$

Thus neither division introduces a negative power of the potentially small scale r . The resulting estimate is (E.14).

Each forward correction has Lipschitz constant $C_m l$ in its inputs, and the backward and harmonic maps have bounded constants. Applying the same absolute weight sums and using $l/(r\sigma) = L/\sigma$ gives (E.15); the momentum component has the smaller Lipschitz bound CL/σ .

For the last assertion, increase interpolation orders until the first line of (E.14) is of the desired order. Under (E.2), sufficiently many $O(J+1)$ Picard layers make the second line of that order as well. The explicit common choice given below verifies independence from interpolation orders and from w . $\square$

E.5 The mean formula and the order of parameter choices

We can now discretize the centering flow itself. Interpolate on the single time cell $[0, 1]$, using ordinary degree- $(m - 1)$ interpolation and partial-integral rows w_{ij} . With anchor $Z_0 = (Y_0, G_0)$, set

$$Z_i^{[0]} = Z_0, \quad Z_i^{[k+1]} = Z_0 + \sum_j w_{ij} \widehat{v}(Z_j^{[k]}).$$

At the end keep the position components y_j and output

$$sG_0 + \sum_j \omega_j M_*(y_j), \quad \sum_j |\omega_j| \leq C_m, \quad (\text{E.16})$$

where ω_j are the full-interval integration weights.

Source attribution. Adapted from OpenAI [14, Proposition 6.5].

Proposition E.5 (Mean-flow circuit). *For ideal independent anchors $Y_0 \sim \xi_*$ and $G_0 \sim \mathcal{N}(0, I_d)$, and every fixed $J \geq 1$, the circuit (E.16) can be chosen to differ in RMS by at most $\text{polylog}(d)\sigma\delta^J D_F(x)$ from the exact path integral (D.4). The approximate joint states have RMS errors at most $\text{polylog}(d)\delta^J D_F(x)$. Every integration uses $O(J + 1)$ Picard layers, with a bound independent of w and all interpolation orders.*

Proof. First compare the exact flow to polynomial integration of its exact velocity. Lemma D.4, applied to the state with one additional derivative, bounds this quadrature error by

$$C_m \sqrt{d} ((L/s)R^{-2})^{m+1}.$$

The approximation error of $\widehat{v}$ at every exact trajectory node is bounded by Proposition E.4, by stationarity. Previous state errors propagate with its global Lipschitz constant and the absolute row sums. The true centering velocity need not be globally Lipschitz in the joint variables; it is the approximate velocity's bound that is used in this recurrence.

By (E.1), $b \geq 1/2$ and $w < t$ give $b - 4t - w > 1/3$. For sufficiently large d , fixed constants and polylogarithms in (E.15) can therefore be absorbed so that the recurrence contracts by at most $\delta^{1/3}$. Its initial error is at most $C(L/s)\sqrt{d}$ by (D.8). Choose the quadrature and velocity error orders sufficiently high, and then $O(J + 1)$ Picard layers yield the state error claimed in the statement.

For the output, apply Lemma D.4 to M_* with $a_0 = L$. Its path quadrature error is at most $C_m L \sqrt{d} ((L/s)R^{-2})^m$, which can be made at most $\text{polylog}(d)\sigma\delta^J D_F(x)$. The additional error from evaluating on the approximate positions is bounded by their RMS error times $C_m L$, using the CL -Lipschitz bound for M_* . Since L/σ is small, this is within the same budget. $\square$

We collect the numerical accuracy and dependency-depth estimates before expanding the affine center formulas. This fixes the Picard depth without yet choosing the mesh exponent.

Source attribution. Adapted from OpenAI [14, Corollary 6.6].

Corollary E.6 (Accuracy without increasing dependency depth). *Fix $J \geq 1$. In all the probability, harmonic, and centering integrations above, one may take at most*

$$N = \lceil 10(J + 2) \rceil$$

Picard layers. After any fixed $w \in (0, t)$ is chosen, a common fixed interpolation order satisfying

$$w(m - 3) > J + 10, \quad t(m - 3) > J + 10, \quad m > 4(J + 10) \quad (\text{E.17})$$

suffices for the following conclusions, for all sufficiently large d . Probability transport from its exact starting marginal has endpoint RMS error at most $\text{polylog}(d)\delta^J D_F(x)$; the same holds after division by an endpoint parameter bounded away from zero. The mean circuit with ideal anchors has RMS error at most $\text{polylog}(d)\sigma\delta^J D_F(x)$ relative to (D.4).

The number of layers of exact mean dependencies within either circuit is $O((N + 1)^2)$, independently of m , w , and the number of mesh cells. All mesh nodes, interpolation weights, scalar parameters, and iteration counts are deterministic functions of the prescribed parameters and dimension.

Proof. For large d , probability and harmonic Picard contractions $C_m l \leq \text{polylog}(d)\delta^b$ can be bounded by $\delta^{1/3}$. The only additional inverse powers in kernel evaluation are δ^{-4t-w} , besides the divisions already canceled in Proposition E.4. The centering iteration also contracts by $\delta^{1/3}$. The probability iteration remainder is at most $C\delta^{N/3}D_F(x)$; after the kernel weights the iteration remainder is at most $C\delta^{N/3-4t-w}D_F(x)$. Both exponents exceed J with a fixed positive margin. For the truncation terms it suffices to check the exponents

$$wm, \quad w(m - 2), \quad w(m - 1) - 4t, \quad 2tm - 4t - w, \quad (m + 1)/3.$$

The first four come respectively from probability quadrature, the zero-endpoint rule, its amplified probability-value error, and the harmonic and angle errors. For the last one use $(L/s)R^{-2} \leq \delta^{1/3}$ for large d in both outer quadratures, dividing the mean-output error by σ and using $L/\sigma \leq \delta^{1/3}$. Every displayed exponent exceeds J by (E.17). The outer Picard remainder contracts by the same $\delta^{1/3}$ factor. Thus fixed constants and polylogarithms, including those depending on m , fit within the strict exponent margins as $d \rightarrow \infty$.

Within a probability or harmonic Picard layer, all nodes depend on only the preceding layer, regardless of the number of nodes. A path through one kernel evaluation passes through a backward probability computation, a harmonic computation, and a forward probability computation, each having at most N layers. Such a path traverses at most N layers of the outer centering iteration. This gives depth $O((N + 1)^2)$; final quadratures add a bounded number of layers. All weights integrate known polynomials, or polynomials against sine or $1/\rho$ with the stated zero-endpoint condition. They require no oracle information and are allowable real constants in the model. $\square$

We choose parameters in the order

$$\text{accuracy order} \longrightarrow \text{Picard depth} \longrightarrow w \longrightarrow m.$$

The interpolation order affects the number of mean evaluations, while the depth bound remains fixed. The recursive implementation will use this separation when it chooses w .

E.6 Bounds for the finite formulas

To display a primitive's parameters, for $x' \in \mathbb{R}^d$ and $0 < r' \leq 1$ write

$$u_{x',r'} := \int_{\mathbb{R}^d} g(x' + r'z) \nu_{x',r'}(dz).$$

Thus $u_{x,r} = u$ for the primitive fixed above.

We specify the two formulas to be implemented. In this subsection $\mathcal{C}_M$ and $\mathcal{C}_S$ denote deterministic maps of their listed anchors, with exact primitive means available as analytical nodes. The parameter σ of $\mathcal{C}_M$ is the parent noise parameter, so its construction sets $s = \sigma/2$ exactly once. Define

$$\mathcal{C}_M(x, r, \sigma; y, G) = \frac{\sigma}{2}G + \sum_j \omega_j M_*(y_j), \quad (\text{E.18})$$

$$\mathcal{C}_S(x, r, \tau; y) = \rho_e^{-1} \hat{T}_{0 \rightarrow \rho_e}(y), \quad \rho_e = (1 + \tau^2/4)^{-1/2}, \quad R \leq \tau \leq 1. \quad (\text{E.19})$$

The positions y_j in the first line are the last centering Picard iterates with anchors (y, G) , using (E.12). The second line uses the probability formula (E.3). No additional Gaussian is included in either map. The next section allocates additional noise and constructs the position anchor needed for $\mathcal{C}_M$.

A *mean dependency path* follows a mean node to a mean value used in its center expression, and then repeats. Direct expression size counts occurrences in one such expression, before expanding the center expressions of its mean nodes.

Source attribution. Adapted from OpenAI [14, Theorem 6.7].

Theorem E.7 (Finite mean and sampling formulas). *Fix $J \geq 1$, the parameters (E.1), and $N = \lceil 10(J+2) \rceil$. Choose the fixed interpolation order by (E.17). For all sufficiently large d , the maps (E.18)–(E.19) have the following properties.*

(i) Deterministic scalar formulas. *Their exact mean nodes can be ordered as $M_1, \dots, M_Q$, where*

$$M_i = u_{E_i, r_i}, \quad E_i = b_i + \sum_{j < i} a_{ij} M_j, \quad E_f = b_f + \sum_j a_{fj} M_j. \quad (\text{E.20})$$

All a_{ij} and a_{fj} are deterministic scalars. The whole node list, its scales, and these coefficients can be formed without knowing a mean value or making an oracle query.

(ii) Direct terms and scales. *Each node has $r_i = r\tau_i$ and*

$$\tau_i = \sqrt{1 - \rho_i^2} \geq R/\sqrt{5}, \quad b_i = x + r(\ell_i y + m_i G).$$

For $\mathcal{C}_S$, the G term is absent, $\rho_i \leq \rho_e$, and $(\ell_i, m_i) = (\rho_i, 0)$. For $\mathcal{C}_M$, $\rho_i \leq b_$ and the pair is either $(\rho_i, 0)$ or $(\rho_i \cos z, \rho_i \sin z)$, with $|z| \leq C_m \psi$. In particular both coefficients have absolute value at most one. A node for $M_0 = u$ has center x and scale r . Every mean occurring directly in the final expression for $\mathcal{C}_M$ is M_* at a computed position and has scale exactly rR . The final direct terms are $b_f = \sigma G/2$ and $b_f = y/\rho_e$, respectively.*

(iii) Coefficient sums and direct expression size. *There are positive declared bounds A, A_f such that $\sum_{j < i} |a_{ij}| \leq A$ and $\sum_j |a_{fj}| \leq A_f$, namely*

$$\begin{aligned} A_M &= C_1(1 + \log d)^{C_1} [(r/\sigma)\delta^{-4t-c_1 w} + r^2], & A_S &= C_1 r^2, \\ A_{f,M} &= C_1, & A_{f,S} &= C_1 r, \quad c_1 = 10. \end{aligned} \quad (\text{E.21})$$

Here $C_1 \geq 1$ depends only on the fixed parameters and orders. Every direct expression has at most $B_d \delta^{-c_1 w}$ terms. Every mean dependency path has length at most $D_0 = 3N^2 + 1$; using a larger absolute multiple of $(N + 1)^2$ is harmless. This depth bound is independent of w , m , and the number of mesh cells.

(iv) Errors on the ideal anchor laws. Under (E.2), for independent $Y \sim \xi_*$ and $G \sim \mathbf{N}(0, I_d)$,

$$W_2(\text{Law}(\mathcal{C}_M(x, r, \sigma; Y, G)), \mathbf{N}(u, \sigma^2 I_d/4)) \leq B_d \sigma \delta^J D_F(x). \quad (\text{E.22})$$

For $\mathcal{C}_S$ only $l \leq B_d \delta^b$, $b \geq 1/2$, is needed. With $Y \sim \mathbf{N}(0, I_d)$,

$$W_2(\text{Law}(\mathcal{C}_S(x, r, \tau; Y)), \nu_{x,r} * \mathbf{N}(0, \tau^2 I_d/4)) \leq B_d \delta^J D_F(x). \quad (\text{E.23})$$

The coefficient and dependency assertions are deterministic, for every choice of anchors. The distributional estimates require the specified ideal anchor laws.

Proof. We verify the affine structure, coefficient bounds, depth, and ideal-law errors in turn. In each additive expansion we stop at an earlier mean value. Its own center remains a separate node, which is necessary for the local coefficient bounds.

Step 1: direct terms in the transport formulas. A probability Picard state has its input position as direct term and mean coefficients with absolute sum at most $C_m r$. A harmonic state has direct term $\cos z$ times its input position plus $\sin z$ times its input momentum, with mean coefficients of absolute sum at most $C_m r$. The sine-weighted row sums are bounded because their integration intervals have bounded length. These assertions hold at every iteration: its additive direct term is the original input, not the preceding Picard iterate.

For a kernel evaluation, the scalar expansion (E.13) has absolute coefficient sum at most $B_d \sigma^{-1} \delta^{-4t-w}$, by the derivative-quadrature, angle-stencil, and forward-integral weight bounds. The momentum formula $(M_* - M_0)/s$ has coefficient sum at most C/σ . A centering Picard update has the original anchor as direct term and bounded outer integral rows. Its position and momentum corrections thus have the same respective coefficient bounds.

Step 2: center coefficients and scales. Consider a mean evaluation inside any of these formulas. Its physical center is $x + r \rho v$ when its normalized position is v , by (B.1). Substitute its direct starting states: a backward probability state starts at the current centering position; a harmonic state starts at that backward position and the current momentum; and a forward probability state starts at the harmonic position. Stop each substitution at the mean terms just described. Only the free terms pass through this chain. Thus their anchor coefficients are exactly $(\rho, 0)$ or $(\rho \cos z, \rho \sin z)$. For sampling there is only the first probability computation, so the coefficient is $(\rho, 0)$. Multiplication by $r \rho$, with $\rho \leq 1$, bounds the center coefficient sum in the mean case by

$$B_d(r/\sigma) \delta^{-4t-w} + C_m r^2.$$

The term $C_m r^2$ accounts for the backward, harmonic, and forward integral corrections. In the sampling case the bound is simply $C_m r^2$. Enlarging these positive upper bounds to (E.21) is convenient for the implementation. The mean output is the single quadrature in (E.18), so its coefficient sum is at most C_m and its means all have probability time b_* . For sampling, dividing a probability correction by $\rho_e \geq 2/\sqrt{5}$ gives output coefficient sum at most $C_m r$. These observations also give the stated final direct terms.

The scale at probability time ρ is $r\sqrt{1-\rho^2}$. In the mean computation $\rho \leq b_*$, giving $\tau \geq R$, including equality for the output means. In the sampling computation,

$$\sqrt{1-\rho_e^2} = \frac{\tau}{\sqrt{4+\tau^2}} \geq \frac{R}{\sqrt{5}}.$$

This proves every assertion about the anchors and scales.

Step 3: expression size and dependency depth. Let $P \leq B_d \delta^{-w}$ bound the number of nodes in any one probability mesh. Angle stencils, harmonic quadratures, and outer centering quadratures have fixed numbers of nodes once m is fixed. Formula (E.13) has at most $C_m P^2$ mean terms: one factor P comes from probability-time derivative quadrature, and the other from the forward transport integral. A direct centering position has the same bound, and its momentum has only a fixed number of terms. Substituting their free states into a backward, harmonic, or forward center adds at most one backward list, one harmonic list, and one forward list. No substitution continues through a mean value. Hence $B_d \delta^{-2w}$ already bounds the direct expression size; the exponent $c_1 w = 10w$ is a uniform conservative choice.

All Picard updates at a fixed layer depend only on earlier layers. Order an internal kernel computation as backward transport, harmonic motion, and forward transport, after its input states. A mean path within this computation adds at most $3N$ nodes above a path in those inputs. There are at most N outer Picard layers, and one final mean layer in (E.18). Thus $3N^2 + 1$ bounds the depth. The sampling path has depth at most N . This gives an acyclic order for (E.20). Meshes, weights, angles and partial integral rows depend only on the prescribed scalar parameters; no exact mean is needed to form any of them.

Step 4: error under the ideal anchors. Couple the numerical mean formula and the exact centering path integral by the same ideal anchors. Corollary E.6 bounds their RMS difference by $B_d \sigma \delta^J D_F(x)$, while Proposition D.3 identifies the exact output as $\mathbf{N}(u, \sigma^2 I_d/4)$. This coupling proves (E.22). For sampling, the exact probability transport from a standard Gaussian has law ξ_{ρ_e} . After division by ρ_e this is the law of

$$Z + \frac{\sqrt{1-\rho_e^2}}{\rho_e} G = Z + \frac{\tau}{2} G, \quad Z \sim \nu_{x,r}, \quad G \sim \mathbf{N}(0, I_d) \text{ independent.}$$

The same-anchor RMS estimate and $\rho_e \geq 2/\sqrt{5}$ prove (E.23). □

For the implementation we use $J = K_0 + 2$. The theorem fixes N and D_0 before w ; the implementation can therefore choose w using its recursion count, and then choose m by (E.17). The formulas above introduce no random failure event. Their errors are RMS estimates under the ideal input laws, and their algebraic structure holds for every input.

F Recursive evaluation using original gradients

We replace every conditional mean in the finite formulas by original-gradient queries. The center of a child mean may itself be an approximate computation, so its private Gaussian noise is absorbed into the child's seeds. We first construct the deterministic shift identity and query cap, and only then prove the distributional comparison. The complete construction and proofs in this section are adapted from [14].

F.1 Why Gaussian center noise helps

A seeded mean routine at scale $r > 0$ is a measurable deterministic map $Q(x; z_1, \dots, z_m) \in \mathbb{R}^d$, whose input is a center $x \in \mathbb{R}^d$ and m seed slots $z_j \in \mathbb{R}^d$. It may query a fixed oracle. A vector $v = (v_1, \dots, v_m) \in \mathbb{R}^m$ gives a *query-preserving translation* if, for every center, displacement and fixed seed list,

$$Q(x + \Delta; z + v \otimes \Delta/r) = Q(x; z), \quad (\text{F.1})$$

and the two computations ask exactly the same physical oracle queries. Here $(v \otimes a)_j = v_j a$. This is a deterministic identity; it does not require independent or Gaussian seeds.

Source attribution. Adapted from OpenAI [14, Lemma 7.1].

Lemma F.1 (Absorbing a Gaussian center displacement). *Let $D_s > 0$, and let Q satisfy (F.1), with $\|v\|_2 \leq D_s$. Put*

$$n = \frac{r}{2D_s}, \quad P = \left(I_m - \frac{vv^T}{4D_s^2} \right)^{1/2}.$$

For a deterministic $x \in \mathbb{R}^d$, let $Z, G_1, \dots, G_m$ be independent standard Gaussian vectors in $\mathbb{R}^d$. Then the output of $Q(x + nZ; (P \otimes I_d)G)$ has exactly the law of $Q(x; G)$.

Proof. The eigenvalues of P lie in $[\sqrt{3}/2, 1]$. The translation identity rewrites the first output as

$$Q\left(x; PG - \frac{v \otimes Z}{2D_s}\right).$$

We suppress the tensor product with I_d on slot-space matrices. The seed array in this expression is centered Gaussian and has slot covariance

$$PP^T + \frac{vv^T}{4D_s^2} = I_m.$$

Its slots are therefore independent standard Gaussian vectors. They may remain correlated with Z ; that is harmless because the rewritten call is at the deterministic center x . $\square$

The matrix P removes exactly the seed covariance supplied by the center noise. Without this adjustment, translating the center back would increase the covariance. We next construct the translation identity for every routine, so that the same calculation can be used recursively.

F.2 The numerical input and the implementation task

We retain the primitive notation of Section A: $g = \nabla F$, $\text{Lip}(g) \leq \lambda = \delta = d^{-e}$, $0 < r \leq 1$, and $\nu_{x,r}$ is the law in (A.1). Write $u_{x,r} = \mathbb{E}_{\nu_{x,r}} g(x + rZ)$, $L = \lambda r$, $l = \lambda r^2$, and $D_F(x) = \sqrt{d} + |g(x)|$. We use the uniform moment bounds (A.6) and conditional coupling rules of Lemma A.3. Only evaluations of g will be used by the implementation.

Fix K_0, t as in Theorem A.1, and retain the numerical parameters

$$R = \delta^t, \quad b_* = (1 - R^2)^{1/2}, \quad \psi = \delta^{4t}, \quad J = K_0 + 2.$$

Theorem E.7 supplies two finite acyclic circuits, $\mathcal{C}_M$ and $\mathcal{C}_S$, whose nodes are exact conditional means. We use its notation A for the positive declared bound on a center's absolute coefficient sum, A_f for the output bound, D_0 for the mean-dependency depth, and c_1 for the absolute constant in the fan-out exponent. A center has the form $x + r(\ell_i y + m_i G) + \sum_{j < i} a_{ij} u_{x_j, r_j}$, with $r_i = r \tau_i$. All

these scalar data are deterministic functions of the prescribed numerical parameters, independent of centers, anchors, and oracle responses. We will use the circuit theorem’s explicit coefficient and scale bounds when constructing shifts and propagating sizes; the circuits themselves are comparison formulas, not additional runtime oracles.

For the mean circuit the parameter supplied is the parent’s σ . Its underlying centering flow uses $s = \sigma/2$ once, so its direct output is $(\sigma/2)G$. Under the specified ideal anchor law, its error is

$$W_2(\text{Law}(\mathcal{C}_M), \mathbf{N}(u_{x,r}, \sigma^2 I_d/4)) \leq \text{polylog}(d) \sigma \delta^J D_F(x).$$

For the sampling circuit the direct output is y/ρ_e , where $\rho_e = (1 + \tau^2/4)^{-1/2}$, and the ideal target is $\nu_{x,r} * \mathbf{N}(0, \tau^2 I_d/4)$. These numerical statements require the anchor laws and size assumptions specified in Theorem E.7; we shall verify those assumptions at their uses.

We must implement every exact mean, including those appearing in the center expressions. The proof will use fixed-seed stability to compare an inaccurate center with an ideal noisy center. Gaussian absorption then identifies the ideal noisy-center call with a standalone call. Only after these two steps do we use the child’s law guarantee.

F.3 Two routines and two different kinds of depth

We construct a mean routine $\mathbf{M}(x, r, \sigma; b, p)$ targeting $u_{x,r} + \sigma G$ and a sampling routine $\mathbf{S}(x, r, \tau; b, p)$ targeting $Z + \tau G$, for independent $Z \sim \nu_{x,r}$ and standard Gaussian G . The top call is $\mathbf{S}(x, r, 1; 1, K_0)$. Throughout, a reachable call means a descendant of this call, including the call itself. The first label b records the size conditions

$$\max\{l, L/\sigma\} \leq \text{polylog}(d) \delta^b \quad (\mathbf{M}), \quad l/\tau \leq \text{polylog}(d) \delta^b \quad (\mathbf{S}), \quad (\text{F.2})$$

where $R \leq \tau \leq 1$. The second label p records the requested error power. The intended errors are $\text{polylog}(d) \sigma \delta^p D_F(x)$ and $\text{polylog}(d) \delta^p D_F(x)$, respectively. A call is terminal precisely when $p \leq b$; it is nonterminal when $p > b$. Every sampling routine reserves a final noise $\tau G/2$, unused in its own queries. Before adding that reserve, its target law is $\nu_{x,r} * \mathbf{N}(0, 3\tau^2 I_d/4)$.

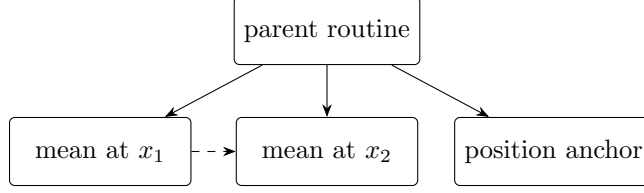
For a nonterminal sampling call, use $\mathcal{C}_S$ with a standard Gaussian position anchor, supply additional final noise $n = \tau/\sqrt{2}$, and then add the reserve. For a nonterminal mean call, pass the parent parameter σ to $\mathcal{C}_M$ and supply final noise $n = \sqrt{3}\sigma/2$. Obtain its position anchor from an $\mathbf{S}$ call at the same center and scale with $\tau = R/b_*$, multiplying that output by b_* . An ideal such anchor has exactly the required law $b_* Z + RN$. Its momentum anchor is an independent standard Gaussian. For all sufficiently large d , $R/b_* \in [R, 1]$, so the anchor call has an admissible noise parameter.

An expression dependency and a routine call play different roles. For example, suppose a parent circuit contains $E_f = a u_{x_2, r_2}$ and $x_2 = x + c u_{x_1, r_1}$. Evaluating E_f invokes two new mean routines: one to help supply x_2 , and one at that supplied center. Both are children of the *same parent routine*. Their own internal circuits begin the next routine level. The two nested affine expressions do not create two successive losses in the parent’s accuracy label. Figure 2 makes this distinction explicit.

To evaluate an expression, we expand its mean dependencies as an uncached tree, duplicating every mean occurrence together with its entire center-supplying subtree. All mean calls at any expression tier are direct children of the routine whose circuit is being evaluated.

The children of a parent labeled (b, p) receive labels as follows:

child’s role	child labels	(F.3)
term of E_f in an $\mathbf{M}$ parent	$(b + t, p)$	
term of E_f in an $\mathbf{S}$ parent	(b, p)	
any other term, or the $\mathbf{M}$ position anchor	$(b - 10t, p - 1/4)$.	



Solid arrows: routine calls. Dashed arrow: mean dependency.

Figure 2: The two depths used in the recursive implementation. A mean needed to supply another mean’s center is still a direct child of the same parent routine. Its expression dependency does not create an extra routine level. Each occurrence has its own fresh seed blocks.

In the last row the labels are always those of the parent routine, not successively decreased while descending its expression tree.

Source attribution. Adapted from OpenAI [14, Lemma 7.2].

Lemma F.2 (Bounded routine height). *Starting at $(b, p) = (1, K_0)$, the rules (F.3) give a finite routine tree of height at most a constant $H = H(K_0, t)$. Every reachable label satisfies $b > 1/2$.*

Proof. Before $4K_0 + 1$ decreasing transitions, the total loss in b is less than $10t(4K_0 + 1) < 1/4$, so $b > 3/4$. At that many transitions, $p \leq K_0 - (4K_0 + 1)/4 < 0$, forcing termination. Therefore no branch can have more such transitions. Positive transitions increase b by t , while a nonterminal satisfies $b < p \leq K_0$; their total number is consequently bounded in terms of K_0, t and the already bounded total decrease in b . A neutral transition is an S-to-M transition. Another neutral transition requires first reaching another S, which can occur only through a decreasing transition. This bounds the neutral transitions as well and proves both claims. $\square$

A routine is a deterministic function of its center, scalar parameters, oracle, and a finite ordered list of d -dimensional seed slots. Standalone use fills these slots by independent standard Gaussians. In addition, each routine will have a deterministic vector v of scalar coefficients, one per slot, with

$$\|v\|_2 \leq D_s, \quad D_s := \delta^{-t}. \quad (\text{F.4})$$

For *every* fixed seed list, changing x to $x + \Delta$ and adding $v \otimes \Delta/r$ to the seeds will preserve every physical query point. The output of M will be unchanged; the normalized output of S will decrease by Δ/r . These are pathwise properties, not statements restricted to Gaussian seeds.

Within each call write $u = u_{x,r}$ and $\nu = \nu_{x,r}$. At a terminal call define

$$\begin{aligned} \mathbf{M} &= g(x + rZ) + \sigma G, \\ \mathbf{S} &= Z - rg(x + rZ) + (\sqrt{3}\tau/2)G_1 + (\tau/2)G_2, \end{aligned}$$

with separate seed slots. The last term in S is its reserved noise. Taking $v = -1$ on the Z slot and zero elsewhere proves the shift property and $\|v\|_2 = 1$.

F.4 Evaluating an expression at a noisy center

Fix a parent circuit and its anchors, and write $M_i = u_{x_i, r_i}$ for its exact mean nodes. For each expression $E = b_0 + \sum_i a_i M_i$, the operation $\text{Eval}(E, n)$ will approximate its exact value plus an

independent $N(0, n^2 I_d)$ noise. Assign disjoint seed blocks to every term occurrence, its center-supplying computation, and each residual noise; keep all these blocks separate from the anchors. Thus the term blocks are independent conditional on the anchors in a standalone call.

Here is the recursive evaluation rule, assuming the child routines and their shift vectors have already been constructed. Take $A' = A$ for a center expression and $A' = A_f$ at the output, and put $\sigma_i = n/(2A')$. For the term of scale $r_i = r\tau_i$, first evaluate its center expression with noise

$$n_i = \frac{r_i}{2D_s}. \quad (\text{F.5})$$

At this supplied center run the child **M** with scale r_i , noise σ_i , and labels from (F.3). If its shift vector is v_i , apply to its fresh seed block G_i the deterministic slot-space transformation

$$P_i = \left(I - \frac{v_i v_i^T}{4D_s^2} \right)^{1/2}, \quad \text{and use seeds } (P_i \otimes I_d)G_i. \quad (\text{F.6})$$

Denote its output by Q_i . Return

$$b_0 + \sum_i a_i Q_i + \left(n^2 - \sum_i a_i^2 \sigma_i^2 \right)^{1/2} G_E, \quad (\text{F.7})$$

where G_E is a separate slot. The square root is defined, since

$$\sum_i a_i^2 \sigma_i^2 \leq \left(\sum_i |a_i| \right)^2 \frac{n^2}{4(A')^2} \leq n^2/4.$$

If there are no terms, return $b_0 + nG_E$. The mean-node ordering makes the expression recursion finite. At the parent's output use the value of n specified above; add the reserved noise for **S**. A routine supplied with correlated seeds performs these same deterministic operations, without any change of rule.

At an ideal supplied center $x_i + n_i Z$, Lemma F.1 now identifies this child with its standalone law at x_i . We still owe the construction of the shift vectors and a proof that the noisy supplied centers are accurate. We prove these in that order.

F.5 Query-preserving shifts and the deterministic cap

We construct the shifts required by Lemma F.1 and bound their norms. The same bottom-up construction proves that all routines are well defined and have a deterministic query cap.

For a nonterminal **S**, shift its Gaussian position anchor by $-\rho_e \Delta/r$. For a nonterminal **M**, we require its position anchor y_* to shift by $-\Delta/(b_* r)$ and leave its momentum anchor unchanged. To obtain this, apply the shift of its anchor **S** call and additionally shift that call's reserved seed by the coefficient $-2R/b_*$ times Δ/r . The ordinary shift changes its output by $-\Delta/r$. Since the reserved standard deviation is $R/(2b_*)$, the additional change is $-R^2 \Delta/(b_*^2 r)$. Multiplication by b_* makes the total change

$$-\left(b_* + \frac{R^2}{b_*} \right) \frac{\Delta}{r} = -\frac{\Delta}{b_* r}.$$

The extra slot is used in no query, so query invariance is preserved.

For a term of scale $r\tau_i$, the direct part of its supplied center then changes by $k_i \Delta$, where

$$k_i = 1 - \ell_i \rho_e \quad (\text{S}), \quad k_i = 1 - \ell_i/b_* \quad (\text{M}).$$

These coefficients satisfy $|k_i|/\tau_i \leq C$. For $\mathbf{S}$, $\rho_i \leq \rho_e$ gives $1 - \rho_i \rho_e \leq 2(1 - \rho_i) \leq 2\tau_i$. For $\mathbf{M}$, its two possible coefficients obey

$$0 \leq 1 - \rho_i \cos z/b_* \leq C(1 - \rho_i) + Cz^2,$$

with $z = 0$ when there is no cosine. Use $1 - \rho_i \leq \tau_i$, $\tau_i \geq cR$, and $z^2/R \leq C\delta^{7t} \rightarrow 0$.

Shift that child's untransformed seed block by

$$\frac{k_i}{\tau_i} P_i^{-1} v_i \otimes \frac{\Delta}{r}. \quad (\text{F.8})$$

If (F.4) holds for the child, the smallest eigenvalue of P_i is at least $\sqrt{3}/2$, so $\|P_i^{-1}\|_{\text{op}} \leq 2$. Formula (F.8) induces exactly the child's shift for center displacement $k_i \Delta$. Leave all residual noise slots fixed. Induction from leaf expressions upward shows that every mean output is unchanged, and therefore that the actual supplied-center change is precisely its direct-term change. All queries are preserved. The parent's final direct term is unchanged for $\mathbf{M}$ and changes by $-\Delta/r$ for $\mathbf{S}$, giving the required output behavior.

The center-supplying subtree and the invoked child have disjoint slots. In particular, the factor k_i/τ_i in (F.8) multiplies only the shift of that child, not the shifts of routines used to supply its center. Those are other direct children of the same parent and are charged once in its total shift vector.

Source attribution. Adapted from OpenAI [14, Lemma 7.3].

Lemma F.3 (Well-defined construction and query cap). *One can choose $w > 0$, depending only on K_0, t and the fixed Picard counts, and then choose the interpolation orders, so that the entire construction is well defined for all sufficiently large d . Every routine reached from a top call satisfies (F.4). Its total number of seed slots and oracle queries is at most $\text{polylog}(d)\delta^{-t}$ on every execution.*

Proof. Let D_0 be the dependency-depth bound in Theorem E.7, and let F_0 be its direct fan-out bound. Expanding a circuit into an expression tree costs at most $(D_0 + 1)(1 + F_0)^{D_0}$ nodes. Its size is thus $\text{polylog}(d)\delta^{-C_2 w}$, where C_2 depends on the Picard counts but not on w or the interpolation orders. This bounds its direct child invocations and its own anchor and residual slots. Expanding through the height H from Lemma F.2 gives $\text{polylog}(d)\delta^{-C_3 w}$ slots and terminal queries. Interpolation orders change only fixed multiplicative constants in these bounds, not C_2 or C_3 .

The terminal shift has norm 1. Assuming the shift bound for the children, the bounded ratios and inverse norms above show that a parent's shift norm is at most a fixed constant times one plus the sum of its direct children's shift norms, including the anchor call. The extra reserved-slot shift has bounded coefficient. The triangle inequality is sufficient even if ordinary and extra anchor shifts affect the same slot. Iterating through the fixed routine height gives the same bound $\text{polylog}(d)\delta^{-C_3 w}$, after increasing C_3 if necessary.

Choose $w > 0$ with

$$C_3 w < t/2, \quad w < t, \quad c_1 w < t,$$

and only then choose the fixed interpolation orders needed in Theorem E.7. For sufficiently large d , the last shift bound is smaller than $D_s = \delta^{-t}$. This is a valid bottom-up induction: it proves each child's bound before using its inverse transformation in the parent. The count bound is also at most $\text{polylog}(d)\delta^{-t}$.

To make the parameter order explicit, first fix the Picard counts, hence D_0, H, C_3 , then w , then the interpolation orders. The universal scalar D_s fixes every center-noise budget. The circuit shapes, labels, requested noises, and slot counts are determined by these scalar parameters before any shifts are computed. Compute child shifts before their matrices and the parent shift. No center value,

oracle response, numerical-error estimate, or assumption (F.2) enters this finite construction or the count. Only terminal routines query g , once each; hence the bound is a cap on every execution, not an expected query count. $\square$

F.6 Terminal errors and propagation of scales

Termination and query counts no longer depend on the accuracy analysis. We now prove the terminal bounds and verify that the numerical size conditions hold for every reachable child.

Source attribution. Adapted from OpenAI [14, Lemma 7.4].

Lemma F.4 (Terminal accuracy). *In standalone use, a terminal call satisfying (F.2) obeys*

$$\begin{aligned} W_2(\text{Law}(\mathbf{M}), \mathbf{N}(u, \sigma^2 I_d)) &\leq \text{polylog}(d) \sigma \delta^p D_F(x), \\ W_2(\text{Law}(\mathbf{S}_{\text{pre}}), \nu_{x,r} * \mathbf{N}(0, 3\tau^2 I_d/4)) &\leq \text{polylog}(d) \delta^p D_F(x), \end{aligned}$$

where $\mathbf{S}_{\text{pre}}$ is the output before adding its reserved noise $\tau G_2/2$. The sampling bound also holds after adding that same independent reserve to both laws. The fixed-seed center Lipschitz constant is at most λ for $\mathbf{M}$ and $r\lambda$ for $\mathbf{S}$.

Proof. For the standard Gaussian seed Z , the primitive moment bounds give

$$\|g(x + rZ) - u\|_{L^2} \leq CL(\sqrt{d} + r|g(x)|) \leq CLD_F(x).$$

Since $L/\sigma \leq \text{polylog}(d)\delta^b$ and $p \leq b$, this is within the mean error budget; couple the added Gaussian identically.

Let z_0 be the mode of $\nu_{x,r}$, so $z_0 = -rg(x + rz_0)$. Then

$$\|Z - rg(x + rZ) - (Z + z_0)\|_{L^2} \leq l\|Z - z_0\|_{L^2} \leq ClD_F(x).$$

Also $W_2(\mathbf{N}(z_0, I_d), \nu_{x,r}) \leq Cl\sqrt{d}$. Apply the synchronous comparison in Lemma A.2 with Gaussian potential $H_0(z) = |z - z_0|^2/2$ (contraction constant 1) and tilted potential $H(z) = |z|^2/2 + F(x + rz)$. Their gradient difference is $r[g(x + rz) - g(x + rz_0)]$, whose $L^2(\nu_{x,r})$ norm is at most $l\|Z - z_0\|_{L^2(\nu_{x,r})} \leq Cl\sqrt{d}$. This proves the displayed Wasserstein bound under exactly the C^2 and bounded-Hessian hypotheses of that lemma. Adding the same independent comparison noises proves the sampling error $ClD_F(x) \leq \text{polylog}(d)\tau\delta^b D_F(x)$, which is at most the requested bound because $\tau \leq 1$ and $p \leq b$. Omitting G_2 proves the pre-reserved assertion. The Lipschitz bounds follow directly from the two displayed terminal formulas. $\square$

Source attribution. Adapted from OpenAI [14, Lemma 7.5].

Lemma F.5 (Size conditions descend through the routine tree). *Starting with a top call satisfying (F.2), every descendant satisfies that condition with its assigned labels, up to a uniform polylogarithmic factor. At each nonterminal circuit,*

$$\lambda A \leq \text{polylog}(d)\delta^{1/3}. \tag{F.9}$$

For nonfinal expression terms within an $\mathbf{M}$ parent, $\sigma_i \leq C\sigma$; within an $\mathbf{S}$ parent, $r\sigma_i \leq C$.

Proof. For a final term of an M parent, the output uses only means of scale rR , so $r_i = rR$, while σ_i is a fixed positive multiple of σ . Thus L_i/σ_i gains a factor R and l_i gains R^2 , allowing labels $(b+t, p)$. For a final term of an S parent, σ_i is a fixed positive multiple of τ/r . Therefore $L_i/\sigma_i \leq Cl/\tau$ and $l_i \leq l$, allowing the unchanged labels.

For a nonfinal term, let $r\tau_{\text{prev}}$ be the scale of the node whose center is currently being supplied. Its expression is evaluated with budget $r\tau_{\text{prev}}/(2D_s)$, irrespective of the tier at which it occurs. Consequently

$$\sigma_i = \frac{r\tau_{\text{prev}}}{4D_s A}, \quad \frac{L_i}{\sigma_i} \leq \frac{CD_s \lambda A}{R}.$$

The coefficient estimates give $\lambda A \leq \text{polylog}(d)\delta^{b-4t-c_1w}$ for an M parent. For an S parent, $\lambda A = C_1 l \leq \text{polylog}(d)\delta^b$, since $\tau \leq 1$. Using $D_s/R = \delta^{-2t}$ and $c_1w < t$ costs less than the allowed $10t$ decrease in b . Also $l_i \leq l$. For the position-anchor S call of M, $l/\tau \leq l/R$, which has the same allowance. This proves the size conditions by the bounded routine height. All accumulated fixed and polylogarithmic factors are still one polylogarithmic factor.

As $b > 1/2$ and $4t + c_1w < 5t < 1/6$, the coefficient bounds imply (F.9). Finally, the positive declared bound for A includes a fixed multiple of r/σ in an M parent, so $\sigma_i \leq r/(4D_s A) \leq C\sigma$. For S, $A = C_1 r^2$ gives $r\sigma_i \leq 1/(4C_1 D_s) \leq C$. $\square$

F.7 Stability for arbitrary fixed seeds

Source attribution. Adapted from OpenAI [14, Lemma 7.6].

Lemma F.6 (Fixed-seed stability). *For every seed array, including correlated or transformed arrays, the compiled M output is a function of x with Lipschitz constant at most $\text{polylog}(d)\lambda$; the normalized S output has constant at most $\text{polylog}(d)r\lambda$. If the anchors of a nonterminal M are treated as separate inputs, its Lipschitz constant in the position anchor is at most $\text{polylog}(d)L$.*

Proof. Induct on routine height, with the terminal case supplied by Lemma F.4. First hold the anchors fixed. A direct center term has Lipschitz constant 1 in x and at most Cr in a position anchor. The center-supplying expression combines child outputs with total absolute coefficient at most A . Using the child M bound, the contribution of earlier supplied-center sensitivities is at most $\text{polylog}(d)\lambda A$ times their maximum. By (F.9), this is small. Induction through the bounded expression depth therefore bounds center sensitivities by a constant times their direct sensitivities, with an allowed polylogarithmic factor.

At the final expression the coefficient sum is A_f . Its direct term has no center dependence when anchors are held fixed; for M it also has no position-anchor dependence. Hence its center sensitivity is at most $\text{polylog}(d)A_f\lambda$, and its M position-anchor sensitivity at most $\text{polylog}(d)A_f\lambda r$. These are the asserted bounds with anchors treated as separate inputs. The S Gaussian anchor is indeed center independent. The M position anchor is supplied by S at the same center, whose center Lipschitz constant by induction is $\text{polylog}(d)r\lambda$. Its additional contribution is at most $\text{polylog}(d)Lr\lambda = \text{polylog}(d)L^2$, and $L^2 \leq \lambda$ for $r \leq 1$ and $\lambda \leq 1$. All seed transformations depend only on scalar parameters, not on x . The proof uses only fixed-seed identities and Lipschitz bounds, and therefore requires no seed independence. $\square$

F.8 Standalone accuracy and the primitive theorem

We prove the law guarantees by induction on routine height. A standalone call uses independent standard Gaussian seeds. A child in the actual computation receives transformed seeds, so we invoke its standalone bound only after applying the absorption identity.

Source attribution. Adapted from OpenAI [14, Proposition 7.7].

Proposition F.7 (Accuracy of the compiled routines). *For every call reached from the top labels, the standalone routines satisfy*

$$\begin{aligned} W_2(\text{Law}(\mathbf{M}), \mathbf{N}(u, \sigma^2 I_d)) &\leq \text{polylog}(d) \sigma \delta^p D_F(x), \\ W_2(\text{Law}(\mathbf{S}_{\text{pre}}), \nu_{x,r} * \mathbf{N}(0, 3\tau^2 I_d/4)) &\leq \text{polylog}(d) \delta^p D_F(x), \end{aligned}$$

where $\mathbf{S}_{\text{pre}}$ denotes the output before its reserved noise. The latter bound also holds after adding the reserved noise to both laws, and

$$\|\text{Law}(\mathbf{S}) - \nu_{x,r} * \mathbf{N}(0, \tau^2 I_d)\|_{\text{TV}} \leq \text{polylog}(d) \frac{\delta^p D_F(x)}{\tau}.$$

Proof. We induct on the finite routine height. The terminal case is Lemma F.4. The actual outputs have finite second moments because their computations are finite compositions of affine maps and globally Lipschitz oracle evaluations. The ideal outputs have the moments in (A.6). Thus each coupling below is well defined.

For a nonterminal $\mathbf{M}$, first couple its approximate $\mathbf{S}$ position anchor to an ideal anchor, independently of all remaining seed blocks. The child labels are $(b - 10t, p - 1/4)$. By induction and Lemma F.6, replacing this anchor costs at most

$$\text{polylog}(d) L \delta^{p-1/4} D_F(x).$$

After division by the mean error unit $\sigma \delta^p D_F(x)$, the cost is at most $\text{polylog}(d) \delta^{b-1/4}$, hence is bounded for large d . We may now assume that the position anchor has law $b_* Z + RN$, where $Z \sim \nu_{x,r}$ and N is standard Gaussian, and that the momentum anchor is an independent standard Gaussian. These are exactly the input laws in Theorem E.7.

Let x_i be the centers of the deterministic exact-mean circuit under these anchors, and put $D_i = \sqrt{d} + |g(x_i)|$. Their RMS values over anchors obey

$$\max_i \|D_i\|_{L^2} \leq \text{polylog}(d) D_F(x). \quad (\text{F.10})$$

Indeed, both position and momentum anchors have RMS at most $CD_F(x)$. Use $|g(x_i)| \leq |g(x)| + \lambda|x_i - x|$ and the circuit's direct terms. An exact primitive mean at a preceding center has magnitude at most CD_j . Their contribution to this maximum-of-RMS recurrence is at most $C\lambda A$ times its previous maximum. This is small by (F.9); the direct terms cost at most $CD_F(x)$. Induction through the node ordering proves (F.10) without a maximum inside the expectation.

Condition on the anchors. For an expression E , write ϵ_E for the W_2 error of $\text{Eval}(E, n)$ relative to its deterministic exact value plus $\mathbf{N}(0, n^2 I_d)$. Couple the supplied center of its i th term to $x_i + n_i Z$ with its conditional W_2 error ϵ_{E_i} . Use the same transformed child seeds on these two centers, drawn independently of this coupling. The fixed-seed Lipschitz bound costs at most $\text{polylog}(d) \lambda \epsilon_{E_i}$. At the ideal noisy center, Lemma F.1 identifies the child's law with its standalone law at x_i . The induction hypothesis therefore supplies the additional error $\text{polylog}(d) \sigma_i \delta^{p_i} D_i$.

Different term occurrences have disjoint blocks, so their outputs are independent conditional on the anchors. Product couplings compare them to independent noised exact means. Together with the independent residual in (F.7), the ideal sum has exactly covariance $n^2 I_d$. We conclude that

$$\epsilon_E \leq \sum_i |a_i| [\text{polylog}(d) \lambda \epsilon_{E_i} + \text{polylog}(d) \sigma_i \delta^{p_i} D_i]. \quad (\text{F.11})$$

In (F.11), the first term uses pathwise stability, and the second uses standalone accuracy after absorption. This order avoids assigning ideal laws to the actual intermediate states.

Take RMS in the anchors and expand (F.11) through the finite expression tree. At its root, direct-child labels have $p_i = p$ and

$$\sum_i |a_i| \sigma_i \leq A_f \frac{n}{2A_f} = n/2.$$

Thus their intrinsic errors fit the requested budget. At tier $j \geq 2$, absolute coefficient sums and Lipschitz factors contribute at most

$$A_f (\text{polylog}(d) \lambda A)^{j-1}.$$

All such children have the single accuracy-label loss $p_i = p - 1/4$, not one loss per expression tier. Their noise units are at most $C\sigma$ in an M parent and C/r in an S parent by Lemma F.5; these units combine with $A_f = C_1$ or $C_1 r$, respectively. Already the first factor $\text{polylog}(d) \lambda A \leq \text{polylog}(d) \delta^{1/3}$ pays for the loss $\delta^{-1/4}$, leaving a positive power $\delta^{1/12}$. The bounded expression depth allows all further tiers and polylogarithmic factors to be summed. Leaf affine expressions have zero error. This proves the required final evaluation error.

Conditionally on the anchors, the ideal final evaluation is its deterministic exact-mean circuit value plus a Gaussian of the specified fixed covariance. Its noise may therefore be taken independent of the anchors. By Lemma F.5, Theorem E.7 applies. It compares the ideal mean circuit to $\mathbf{N}(u, \sigma^2 I_d/4)$ and the ideal sampling circuit to $\nu_{x,r} * \mathbf{N}(0, \tau^2 I_d/4)$, with the respective errors $\text{polylog}(d) \sigma \delta^J D_F(x)$ and $\text{polylog}(d) \delta^J D_F(x)$. Since $J = K_0 + 2$ and $p \leq K_0$, both errors fit the requested bounds. Couple the independent evaluation noise identically in each comparison. For M, its variance $3\sigma^2/4$ raises the Gaussian target variance from $\sigma^2/4$ to σ^2 . For S, its variance $\tau^2/2$ raises the Gaussian convolution variance from $\tau^2/4$ to $3\tau^2/4$. This closes the standalone induction. Adding the common reserved noise preserves the W_2 bound; Gaussian smoothing by its standard deviation $\tau/2$ gives the stated total variation bound by (A.7). $\square$

Proof of Theorem A.1. Use the top S call with $(b, p, \tau) = (1, K_0, 1)$. Because $l = \delta r^2 \leq \delta$, its size condition holds with constant 1. Proposition F.7 gives both W_2 and total variation error at most $\text{polylog}(d) \delta^{K_0} D_F(x)$ from $\nu_{x,r} * \mathbf{N}(0, I_d)$, uniformly in x and $0 < r \leq 1$. Lemma F.3 gives at most $\text{polylog}(d) \delta^{-t}$ evaluations of g on every execution. The output has the form $X_{\text{pre}} + G/2$, where the reserved standard Gaussian G is independent of X_{pre} and all queries, and the pre-reserved conclusion of Proposition F.7 gives $W_2(\text{Law}(X_{\text{pre}}), \nu_{x,r} * \mathbf{N}(0, 3I_d/4)) \leq \text{polylog}(d) \delta^{K_0} D_F(x)$. The exact means and numerical comparison laws serve only to analyze this finite algorithm; its only oracle operations are the terminal evaluations. $\square$

G The mean-root specialization

This appendix specializes the finite mean formula and recursive evaluation of OpenAI [14, Theorem 6.7, Lemmas 7.2–7.3 and 7.5–7.6, and Proposition 7.7] to the stated mean-root output. The underlying two-type recursive construction is inherited.

We prove Proposition 6.1 from the explicit constructions in Sections E and F. This appendix records the change of root and the final Gaussian reserve, because Theorem A.1 is a noisy sampling result rather than a standalone mean theorem.

G.1 The recursive computation

We use the mean routine $M(x, r, \sigma; b, p)$ and sampling routine $S(x, r, \tau; b, p)$ as defined in [Section F](#), with label rules [\(F.3\)](#). Their mean and sampling size conditions are respectively

$$\max\{\delta r^2, \delta r/\sigma\} \leq B_d \delta^b, \quad \delta r^2/\tau \leq B_d \delta^b.$$

A call is terminal when $p \leq b$. For a nonterminal parent, final-expression mean children of a mean routine receive labels $(b + t, p)$, final-expression mean children of a sampling routine receive (b, p) , and all other children, including the mean routine's position anchor, receive $(b - 10t, p - 1/4)$. All expression-tree children are children of that parent routine; following an affine supplying expression does not repeatedly lower the same parent labels.

The ideal mean-circuit theorem, [Theorem E.7](#), produces a finite affine conditional-mean expression whose law is within $B_d \sigma \delta^{p+2} D_F(x)$ in W_2 of $N(m(x), \sigma^2 I/4)$ under the mean size condition. Its prescribed Picard and interpolation orders depend on the fixed labels and parameters. The shift and query-cap, size-propagation, fixed-seed stability, and two-type coupling results, [Lemmas F.3, F.5 and F.6](#) and [Proposition F.7](#), provide the following properties for the recursive evaluation: deterministic seed slots; the exact shift [\(6.4\)](#); shift norm at most δ^{-t} ; cap $B_d \delta^{-t}$; center Lipschitz constant at most $B_d \delta$; and the intended W_2 errors $B_d \sigma \delta^p D_F(x)$ for means and $B_d \delta^p D_F(x)$ for sampling. These properties are uniform in the center and scalar scales when their declared size conditions hold. They do not assert exact unbiasedness or a joint guarantee for exported internal records.

G.2 Starting at a mean routine

Start the same recursion with $M(x, r, \sigma; 1, K_M)$. Its root is legal by [\(6.2\)](#). To check termination independently of the old root type, consider a branch. Before $4K_M + 1$ decreasing transitions, the total reduction of b is less than $10t(4K_M + 1) < 1/4$. At that many transitions, $p < 0$ and the call is terminal. Thus $b > 3/4$ along every nonterminal part of a branch. The number of increasing transitions is bounded because each increases b by t while a nonterminal call has $b < p \leq K_M$. A neutral transition is from S to M ; another such transition requires a decreasing transition to a new S . Hence the height is bounded by a constant depending only on K_M, t .

The proofs of [Lemmas F.2 and F.3](#) depend on this height, the fixed circuit orders, and scalar size formulas, rather than on a sampling root. The size-propagation induction applies with the same root label $b = 1$. The two-type coupling induction can therefore be run at the mean root, yielding [\(6.3\)](#), with the same shift, cap, and fixed-seed stability conclusions. This changes neither a child definition nor a probability law within the recursion.

G.3 Reserving a fresh Gaussian

At a nonterminal mean root, the affine evaluator in [Section F](#) uses final Gaussian budget $n_0 = \sqrt{3}\sigma/2$. If its final scalar coefficients are a_i , with declared absolute sum A_f , each child receives noise $\sigma_i = n_0/(2A_f)$. The independent residual variance is

$$v = n_0^2 - \sum_i a_i^2 \sigma_i^2 \geq 3n_0^2/4 = 9\sigma^2/16.$$

Split it into independent variances $v - \sigma^2/4$ and $\sigma^2/4$, retaining the second until after all queries. This is positive, does not change the output law, and adds no query. At a terminal mean root, the output is $\nabla F(x + rZ) + \sigma G$, so the same splitting is immediate.

For the pre-reserve estimate, repeat the coupling proof of [Proposition F.7](#) before the last independent addition. Fixed-seed stability is unchanged on subtracting this additive noise. Every center coupling and every independent product coupling of child means is unchanged. The ideal affine evaluation covariance drops from $3\sigma^2/4$ to $\sigma^2/2$. The ideal circuit itself approximates $\mathcal{N}(m(x), \sigma^2 I/4)$ by [Theorem E.7](#). Adding the ideal evaluation noise gives covariance $3\sigma^2/4$ before the reserve, with total error $B_d \sigma \delta^{K_M} D_F(x)$. No deconvolution of a Wasserstein bound is used. Finally convolution by the reserved $\sigma G/2$ preserves the W_2 bound and gives TV error at most twice that bound divided by σ . This proves all assertions of [Proposition 6.1](#).

H The translated-posterior inequality

The following estimate is used only in the conditional-current calculation. We give its proof to make the normalization and direction of the Bregman divergence explicit.

Lemma H.1 (Translated posterior). *Let U be convex with $0 \preceq \nabla^2 U \preceq I$, and let $0 < u \leq 1/8$. Define $Q_y(dx) \propto e^{-U(x) - \|x-y\|^2/(2u)} dx$. For a fixed translation ζ , let Q'_y be $Q_{y-\zeta}$ translated by ζ . If $\mathcal{R} = \log(dQ_y/dQ'_y)$, then for every input law ν of finite entropy,*

$$\mathbb{E}_\nu \mathcal{R} \leq \frac{1}{2} \mathbb{E}_\nu D_U(X - \zeta, X) + 6u \text{KL}(\nu \| Q_y).$$

Here $D_U(v, x) = U(v) - U(x) - \nabla U(x) \cdot (v - x)$.

Proof. The likelihood ratio has $\nabla \mathcal{R} = \nabla U(x - \zeta) - \nabla U(x)$, $\text{Lip}(\nabla \mathcal{R}) \leq 1$, and $\|\nabla \mathcal{R}\|^2 \leq 2D_U(x - \zeta, x)$ by cocoercivity. The translated posterior has curvature at least $1/u$, so its logarithmic Sobolev inequality yields $\mathbb{E}_{Q_y} \mathcal{R} \leq (u/2) \mathbb{E}_{Q_y} \|\nabla \mathcal{R}\|^2$. Put $L = \text{KL}(\nu \| Q_y)$, $A = \mathbb{E}_\nu \|\nabla \mathcal{R}\|^2$, and couple $X \sim \nu$, $Y \sim Q_y$ with $\mathbb{E} \|X - Y\|^2 \leq 2uL$. Lipschitzness of $\nabla \mathcal{R}$ bounds $\mathbb{E} \|\nabla \mathcal{R}(Y)\|^2 \leq 2A + 4uL$. Taylor's upper bound at X then gives

$$\mathbb{E}_\nu \mathcal{R} \leq uA + \sqrt{2uLA} + (u + 2u^2)L \leq (u + 1/8)A + (5u + 2u^2)L.$$

Since $u \leq 1/8$, the asserted inequality follows. Truncation and the variational definition of entropy cover infinite displayed Bregman expectation without ambiguity. $\square$

For clarity, if $\Psi = -\log(e^{-U} * \mathcal{N}(0, uI))$, direct cancellation of the quadratic terms gives

$$\mathcal{R}(x) = D_U(x - \zeta, x) - \nabla U(x) \cdot \zeta + \Psi(y) - \Psi(y - \zeta).$$

Rearranging the lemma therefore gives the force-potential inequality used in [Lemma 5.3](#).

I The momentum-refresh transport estimate

This appendix proves [Lemma 5.2](#) for the generic base potential U introduced in [Section 2.2](#). The finite conditional-HMC chain applies it with $U = V_a$. Throughout, $0 < h \leq 1/4$, $m_{\text{obs}} = m_U/(1 + m_U h^2)$, $\Gamma = \Gamma_U(h)$ and $\rho = e^{-\Gamma/32}$. Write $U_s = -\log \pi_{U,s}$. All transports are quadratic-cost Brenier maps and are analytical objects, not executed operations.

In addition to the shared functional inequalities, we use the entropy subgradient inequality for log-concave q ,

$$H(\mu | q) \geq H(\nu | q) - \sqrt{I(\nu | q)} W_2(\mu, \nu). \quad (\text{I.1})$$

We use Fathi's centered symmetrized Gaussian transport inequality [9, Theorem 1.1]: if one of μ, σ is centered, $W_2^2(\mu, \sigma) \leq 2H(\mu \mid \gamma_d) + 2H(\sigma \mid \gamma_d)$. Brenier existence and optimal-plan stability are as in [16]. The remaining steps, including the finite-entropy passage, are proved here. Write $h_{\text{diff}}(\mu) = -\int \log(d\mu/dx) d\mu$ and $\text{maxcorr}(\mu, \nu) = \sup_{\lambda \in \mathcal{C}(\mu, \nu)} \int x \cdot y d\lambda$. The bridge uses a temporary half-width $\bar{h}$; the fixed smoothing variance remains a throughout the main construction.

I.1 Entropy-sized corrected currents

For $D_U(x, z) = U(x) - U(z) - \nabla U(z) \cdot (x - z)$ and $c_- I \preceq \nabla^2 U \preceq c_+ I$,

$$D_U(z, x) \leq \sqrt{c_+/c_-} D_U(x, z). \quad (\text{I.2})$$

Indeed, set $r = \sqrt{c_+/c_-}$ and $q(t) = (x - z)^\top \nabla^2 U(z + t(x - z))(x - z)$. The two costs are $\int_0^1 tq(t)dt$ and $\int_0^1 (1-t)q(t)dt$. The coefficient $t - r(1-t)$ changes sign at $r/(1+r)$. Using the lower bound on its negative interval and the upper bound on its positive interval gives $|x - z|^2(c_+ - c_-r^2)/(2(1+r)) = 0$. If $T_{\#}\rho = q \propto e^{-U}$ is the Brenier map, then

$$\mathbb{E}D_U(Y, T(Y)) \leq H(\rho \mid q). \quad (\text{I.3})$$

For regular invertible maps, change of variables and integration by parts against q express the difference as $\mathbb{E}\{\text{tr}(DT)^{-1} - d + \log \det DT\} \geq 0$. Approximation proves the inequality in general.

For a fixed label Y , let $\xi = Y - T(Y)$, where T sends its actual marginal to π_{U, h^2} , and put $\mathcal{P}_Y = \mathbb{E}[U_{h^2}(Y) - U_{h^2}(T(Y))]$. The reference conditional Z -law has density proportional to $e^{-|Z|^2/2 - U(Y + hZ)}$, mean $-h\nabla U_{h^2}(Y)$, and curvature at least one. Let Q be the auxiliary momentum, with reference law γ_d , and let R denote the actual conditional entropy of (Z, Q) against this law times γ_d , and let H_Y be label entropy. Conditional T_2 and label T_2 imply, for

$$C_\theta = \mathbb{E}[\{-\cos \theta(Z + h\nabla U_{h^2}(Y)) + \sin \theta Q\} \cdot \xi], \quad \sqrt{m_{\text{obs}}}|C_\theta| \leq 2\sqrt{RH_Y} \leq R + H_Y.$$

The corrected current is

$$L_\theta = C_\theta + h \cos \theta \mathbb{E}D_{U_{h^2}}(T(Y), Y) = \mathbb{E}[(-\cos \theta Z + \sin \theta Q) \cdot \xi] - h \cos \theta \mathcal{P}_Y.$$

Write $L_{h^2} = \sup_y \|\nabla^2 U_{h^2}(y)\|_{\text{op}} \leq 1$. Equations (I.2)–(I.3) give

$$\sqrt{m_{\text{obs}}}|L_\theta| \leq (1 + h\sqrt{L_{h^2}})(H_Y + R) \leq 2(H_Y + R), \quad L_{h^2} \leq 1. \quad (\text{I.4})$$

For the same physical law, $L_\pi = -L_0$: explicitly, $L_0 = -\mathbb{E}[Z \cdot \xi] - h\mathcal{P}_Y$ and $L_\pi = \mathbb{E}[Z \cdot \xi] + h\mathcal{P}_Y$. At boundary angles this current is independent of the auxiliary momentum, so the same comparison uses the physical entropy after discards.

I.2 Common-target compatibility in every dimension

Consider an actual pair (B, K) with reference $q \otimes \gamma_d$. For a half-width $\bar{h} > 0$, let $Z_\pm = B \pm \bar{h}K$ and let $T_\pm$ be their actual Brenier maps to the common reference label law $q * N(0, \bar{h}^2 I)$. Write $\xi_\pm = Z_\pm - T_\pm(Z_\pm)$, let $\mathcal{P}_\pm = \mathbb{E}[\Phi_{\bar{h}}(Z_\pm) - \Phi_{\bar{h}}(T_\pm(Z_\pm))]$, where $\Phi_{\bar{h}} = -\log(q * N(0, \bar{h}^2 I))$, and let $H_\pm$ be label entropies. Put $R_K = \mathbb{E}_B H(\text{Law}(K \mid B) \mid \gamma_d)$. We claim that

$$\Delta L_{\text{flip}} := \mathbb{E}[K \cdot (\xi_+ - \xi_-)] - \bar{h}(\mathcal{P}_+ + \mathcal{P}_-) \leq -\bar{h}(H_+ + H_-) + 4\bar{h}R_K. \quad (\text{I.5})$$

For the regular calculation define

$$M_{\pm} = DT_{\pm}(Z_{\pm}), \quad U_B(k) = \frac{T_+(B + \bar{h}k) - T_-(B - \bar{h}k)}{2\bar{h}},$$

$$N_B = DU_B = \frac{M_+ + M_-}{2}, \quad c = \mathbb{E}[K \cdot U_B(K)],$$

$$\mathcal{L} = 2h_{\text{diff}}(K | B) - d \log(2\pi e) + \frac{1}{2} \mathbb{E}(\log \det M_+ + \log \det M_-).$$

The identities $H_{\pm} = \mathcal{P}_{\pm} + \mathbb{E} \log \det M_{\pm}$ and $\xi_+ - \xi_- = 2\bar{h}(K - U_B(K))$ give exactly

$$\Delta L_{\text{flip}} = -\bar{h}(H_+ + H_-) + 4\bar{h}R_K - 2\bar{h}\mathcal{B}, \quad \mathcal{B} = c - d - \mathcal{L}. \quad (\text{I.6})$$

We prove $\mathcal{B} \geq 0$. Conditionally on B , write $\nu_B = \text{Law}(K | B)$, $\sigma_B = (U_B)_{\#} \nu_B$, with means $e(B)$, $s(B)$, and $b = \mathbb{E}[e \cdot s]$. The map U_B is a convex gradient, so its actual coupling maximizes correlation between ν_B and σ_B . For any conditional self-coupling $(K, K^*) | B$, both endpoint competitors $T_-(B - \bar{h}K^*)$ and $T_+(B + \bar{h}K^*)$ retain their required unconditional common target. Adding the two endpoint optimality inequalities cancels the B terms and yields

$$c + \mathbb{E}[K \cdot U_B(K^*)] \geq 0. \quad (\text{I.7})$$

Choose a conditional maximal-correlation coupling of $K \sim \nu_B$ with $Z \sim -\sigma_B$ and lift $-Z$ through U_B by disintegration to obtain $K^* \sim \nu_B$. Therefore

$$c \geq \mathbb{E}_B \text{maxcorr}(\nu_B, -\sigma_B), \quad c = \mathbb{E}_B \text{maxcorr}(\nu_B, \sigma_B).$$

Translate both laws before applying Fathi's theorem and expand the quadratic cost. For any two laws of finite entropy and second moments,

$$\text{maxcorr}(\mu, \sigma) \geq \mathbb{E}\mu \cdot \mathbb{E}\sigma + h_{\text{diff}}(\mu) + h_{\text{diff}}(\sigma) - d \log(2\pi).$$

Writing $L_0 = \mathbb{E}_B[h_{\text{diff}}(\nu_B) + h_{\text{diff}}(\sigma_B) - d \log(2\pi)]$, we obtain $c \geq L_0 \pm b$. Conditional change of variables under U_B gives

$$L_0 = d + \mathcal{L} + \Delta, \quad \Delta = \mathbb{E} \log \det N_B - \frac{1}{2} \mathbb{E}(\log \det M_+ + \log \det M_-) \geq 0.$$

The last inequality is log-determinant concavity. Hence $\mathcal{B} = c - L_0 + \Delta \geq |b| + \Delta \geq 0$, proving (I.5). Approximate measurable conditional optimal couplings suffice; disintegration avoids any invertibility assumption on U_B .

I.3 Conditional laws from one endpoint Fisher information

Suppose $q = e^{-U}/Z$ with $0 \preceq \nabla^2 U \preceq I$. For $Z_t = B + tK$, $|t| \leq \bar{h}$, $\bar{h}^2 \leq 1/16$, define

$$H = H(\text{Law}(B, K) | q \otimes \gamma_d), \quad H_t = H(\text{Law}(Z_t) | q * N(0, t^2 I)), \quad R_t = H - H_t.$$

If $I_{\bar{h}}$ is relative Fisher information of $Z_{\bar{h}}$, then

$$\sup_{|t| \leq \bar{h}} R_t \leq 4(R_{\bar{h}} + \bar{h}^2 I_{\bar{h}}). \quad (\text{I.8})$$

To prove this, retain the actual $Z_{\bar{h}}$ marginal and replace its conditional K row by the exact reference row. Denote this baseline by a bar and put $f = d\text{Law}(Z_{\bar{h}})/d(q * N(0, \bar{h}^2 I))$. At the reference,

$$q_{t,z}(dk) \propto e^{-|k|^2/2 - U(z - tk)} dk, \quad \bar{f}_t(z) = \mathbb{E}_{q_{t,z}} f(z + (\bar{h} - t)K).$$

The entropy chain rule and curvature-one conditional LSI give $H_{\bar{h}} - \bar{H}_t \leq (\bar{h} - t)^2 I_{\bar{h}}/2$. A stationary synchronous Langevin construction with drift $-k + t\nabla U(z - tk)$ realizes these conditional laws with $|K_t(z) - K_t(z')| \leq |t||z - z'|$ almost surely: the finite-past difference satisfies $d|\Delta K|/ds \leq -|\Delta K| + |t||z - z'|$. Consequently $z \mapsto z + (\bar{h} - t)K_t(z)$ has Lipschitz constant $1 + |t||\bar{h} - t|$. Differentiating the expectation and applying weighted Cauchy–Schwarz gives $\bar{I}_t \leq [1 + |t||\bar{h} - t|]^2 I_{\bar{h}}$. To make the reference-law bookkeeping explicit, put $Q = q \otimes \gamma_d$, $\Delta_t = \bar{h} - t$, and $q_t = \text{Law}_Q(Z_t)$. The replacement baseline on the full pair has relative density $d\bar{P}/dQ = f(Z_{\bar{h}})$. For the unnormalized entropy

$$\text{Ent}_\rho(g) = \mathbb{E}_\rho[g \log g] - \mathbb{E}_\rho[g] \log \mathbb{E}_\rho[g],$$

the entropy chain rule and the conditional curvature-one LSI give

$$\begin{aligned} H_{\bar{h}} - \bar{H}_t &= \int \text{Ent}_{q_t, z}(f(z + \Delta_t K)) q_t(dz) \\ &\leq \frac{\Delta_t^2}{2} \mathbb{E}_Q \left[\frac{|\nabla f(Z_{\bar{h}})|^2}{f(Z_{\bar{h}})} \right] = \frac{\Delta_t^2}{2} I_{\bar{h}}. \end{aligned}$$

For the shared-noise realization $K_t(z, W)$ above, set $\Phi_t(z, W) = z + \Delta_t K_t(z, W)$ and $L_t = 1 + |t\Delta_t|$. Differentiation and weighted Cauchy–Schwarz yield

$$\frac{|\nabla \bar{f}_t(z)|^2}{\bar{f}_t(z)} \leq L_t^2 \mathbb{E}_W \left[\frac{|\nabla f(\Phi_t(z, W))|^2}{f(\Phi_t(z, W))} \right].$$

Integration against the reference marginal q_t gives $\bar{I}_t \leq L_t^2 I_{\bar{h}}$, since, when z has law q_t independently of W , $\Phi_t(z, W)$ has the $Z_{\bar{h}}$ reference marginal. These calculations concern the replacement baseline, not the unreplaced actual conditional row.

These computations extend from smooth positive bounded f by Fisher approximation and lower semicontinuity. Conditional T_2 couples the actual and replaced conditional laws at the same actual $Z_{\bar{h}}$, so $W_2(\text{Law}(Z_t), \text{Law}(\bar{Z}_t)) \leq |\bar{h} - t| \sqrt{2R_{\bar{h}}}$. Using (I.1) with the baseline in its finite-Fisher argument now gives

$$R_t \leq R_{\bar{h}} + \frac{(\bar{h} - t)^2}{2} I_{\bar{h}} + |\bar{h} - t|(1 + |t||\bar{h} - t|) \sqrt{2R_{\bar{h}} I_{\bar{h}}}.$$

Since $|\bar{h} - t| \leq 2\bar{h}$, Young’s inequality bounds the coefficients of $R_{\bar{h}}$ and $\bar{h}^2 I_{\bar{h}}$ respectively by $1 + \sqrt{2}(1 + 2\bar{h}^2)$ and $2 + \sqrt{2}(1 + 2\bar{h}^2)$, both below four. This proves (I.8).

I.4 An exact product bridge

Now execute $P' = \rho P + \sqrt{v}G$, where $\rho = e^{-\Gamma/32}$, $v = 1 - \rho^2$, and G is independent standard Gaussian. Set

$$\beta = 1 - \rho, \quad c = (1 + \rho)/2, \quad B = X + \frac{h}{2}(P' - P), \quad K = \frac{P + P'}{2\sqrt{c}}, \quad \bar{h} = h\sqrt{c}, \quad s = h^2\beta/2.$$

At the stationary reference, $(P + P')/2$ and $(P' - P)/2$ are independent Gaussians of covariances cI and $(\beta/2)I$, independent of X . The bridge reference is thus exactly $\pi_{U,s} \otimes \gamma_d$. Its labels $B \pm \bar{h}K$ are exactly $Y_- = X - hP$ and $Y_+ = X + hP'$, and $s + \bar{h}^2 = h^2$ gives the common target π_{U,h^2} .

Let $H_{\text{pre}}, H_{\text{post}}$ denote the physical entropies relative to Π_U , let $D = H_{\text{pre}} - H_{\text{post}}$, and put $R_+ = H_{\text{post}} - H_+$. The reverse chain rule is

$$D = \mathbb{E}_{X, P'} H(\text{Law}(P | X, P') | N(\rho P', vI)). \quad (\text{I.9})$$

Indeed the actual and reference forward OU conditional kernels agree, so the triple entropy is H_{pre} . Write $H_{\text{br}} = H(\text{Law}(B, K) \mid \pi_{U,s} \otimes \gamma_d)$ and $R_{\text{br},t} = H_{\text{br}} - H(\text{Law}(B+tK) \mid \pi_{U,s+t^2})$. Data processing into (B, K) therefore gives $R_{\text{br},+} \leq D + R_+$. Put $z = \mathbb{E}[P - \rho P' \mid X, P']$. Gaussian entropy duality gives $\mathbb{E}|z|^2 \leq 2vD$; output differentiation gives $\ell_{P,\text{out}} = \rho z/v$. Thus

$$I_{P,\text{out}} \leq 2\rho^2 D/v.$$

Velocity integration by parts, without an X score, gives

$$h\ell_+(Y_+) = \mathbb{E}[\ell_{P,\text{out}} \mid Y_+] - e_+(Y_+), \quad e_+(y) = \mathbb{E}[P' \mid Y_+ = y] - h\nabla U_{h^2}(y).$$

The reference conditional law has curvature at least one and the displayed reference mean. Hence $\mathbb{E}|e_+|^2 \leq 2R_+$ and $h^2 I_+ \leq 2I_{P,\text{out}} + 4R_+$. Apply (I.8) to the bridge with $q = \pi_{U,s}$. Its potential U_s has Hessian at most I by Gaussian smoothing, so all hypotheses of that estimate hold. This gives

$$R_{\text{br},0} \leq 4(1+4c)R_+ + (4+8\rho^2/\beta)D. \quad (\text{I.10})$$

I.5 Joining the currents

Retain the bridge notation above, and write

$$L_- = \mathbb{E}[P \cdot \xi_-] + h\mathcal{P}_-, \quad L_+ = \mathbb{E}[P' \cdot \xi_+] - h\mathcal{P}_+, \quad \Delta L = L_+ - L_-.$$

Put $z = \mathbb{E}[P - \rho P' \mid X, P']$. The bridge current satisfies the following exact identity, involving only integrable first moments:

$$c \Delta L = \sqrt{c} \Delta L_{\text{br}} - \frac{1}{2} \mathbb{E}[z \cdot \xi_+]. \quad (\text{I.11})$$

Indeed, for $\varepsilon = (P' - P)/2$, direct substitution of $K = (P + P')/(2\sqrt{c})$ and $\bar{h} = h\sqrt{c}$ gives

$$\Delta L - \sqrt{c} \Delta L_{\text{br}} = \mathbb{E}[\varepsilon \cdot (\xi_+ + \xi_-)] - \frac{\beta h}{2} (\mathcal{P}_+ + \mathcal{P}_-).$$

Since ξ_- is measurable in (X, P) and ξ_+ in (X, P') ,

$$\mathbb{E}[\varepsilon \mid X, P] = -\frac{\beta}{2} P, \quad \mathbb{E}[\varepsilon \mid X, P'] = \frac{\beta}{2} P' - \frac{1}{2} z.$$

Thus the right side is $\beta \Delta L/2 - \mathbb{E}[z \cdot \xi_+]/2$; use $1 - \beta/2 = c$ to obtain the identity.

The reverse OU entropy identity and label T_2 give $\mathbb{E}|z|^2 \leq 2vD$ and $m_{\text{obs}} \mathbb{E}|\xi_+|^2 \leq 2H_+$. Consequently, with $J = \sqrt{m_{\text{obs}}} L$,

$$\frac{\sqrt{m_{\text{obs}}}}{2c} |\mathbb{E}[z \cdot \xi_+]| \leq \frac{\sqrt{vH_+D}}{c} \leq \frac{\Gamma}{16} H_{\text{post}} + \frac{4v}{c^2 \Gamma} D.$$

Apply (I.5) and (I.10) in the exact join, noting $\bar{h}/\sqrt{c} = h$, and discard the favorable H_- term. Since $H_+ = H_{\text{post}} - R_+$, this gives

$$\Delta J \leq -\frac{15\Gamma}{16} H_{\text{post}} + (17 + 64c)\Gamma R_+ + \left(16\Gamma + \frac{32\Gamma\rho^2}{\beta} + \frac{4v}{c^2\Gamma}\right) D.$$

Here $\Gamma/64 \leq \beta \leq \Gamma/32$, $1/2 \leq c \leq 1$, $v = \beta(1 + \rho) \leq \Gamma/16$, and $\Gamma \leq 1$. The conditional-entropy coefficient is at most 81 and the loss coefficient at most $16 + 2048 + 1 = 2065$. In particular the stated estimate follows:

$$\boxed{\Delta J \leq -\frac{\Gamma}{4} H_{\text{post}} + 81\Gamma R_+ + 2066D.} \quad (\text{I.12})$$

I.6 The finite-entropy domain

We extend the integrated estimate from regular densities to every finite-entropy input. For a finite-entropy physical density f relative to the strongly log-concave reference, first clip f above and below and normalize. Then approximate the resulting bounded positive function by smooth functions constant outside growing balls. The bounds above and below can be preserved for each approximation. A diagonal choice gives convergence of entropy and second moments, hence W_2 convergence. The approximants have finite relative Fisher information. Their smooth, positive endpoint densities admit locally regular Brenier maps; cutoff integration by parts justifies the identities used above. In particular, with ϱ denoting the source label density, finite label Fisher information and finite transport displacement imply finite weighted $\text{tr } DT$ through $\mathbb{E} \text{tr } DT = d + \mathbb{E}[\nabla \log \varrho \cdot (Y - T(Y))]$. Change of variables supplies integrable endpoint log determinants, and log-determinant concavity and trace bounds control the midpoint determinant. Thus the conditional entropy calculations used in the regular proof are well defined. Local Brenier regularity and optimal-plan stability here are the usual optimal-transport regularity and stability results [16].

At this regularized stage, the endpoint and reference label densities are strictly positive, have full support, and are smooth with densities and reciprocals bounded on every compact set. The local Brenier-map regularity follows from [7, Theorem 1(ii), with full Euclidean source and target domains]; in one dimension it follows from monotone rearrangement. This is a local regularity input only. The cutoff and trace estimates above and the entropy and transport limits below supply the separate tail and approximation arguments.

For the pure-flip estimate the right side is $4\bar{h}H_{\text{total}} - \bar{h}H_+ - \bar{h}H_- - 4\bar{h}H_B$. For (I.12) it is

$$2066H_{\text{pre}} - (2066 + \Gamma/4 - 81\Gamma)H_{\text{post}} - 81\Gamma H_+.$$

In each case the source entropy converges and all marginal/output entropies have negative coefficients and are lower semicontinuous. The OU kernel is W_2 continuous. The currents converge: compactness of the joint laws of the physical variables and their transported labels, together with uniqueness of each limiting Brenier plan, identifies every subsequential limit; convergence of second moments makes all quadratic products uniformly integrable. Potential differences have at most quadratic growth. Taking the upper limit of the right side proves the estimates for every finite-entropy physical law. No separated rough score, trace, or log determinant is asserted to survive this passage.

J Source correspondence and attribution

The analytic modules in [Sections A to F](#) adapt Sections 2–7 of OpenAI, *Subpolynomial query complexity for well-conditioned log-concave sampling* (September 26, 2026) [14]. The correspondence below uses the numbering of the [pinned source version](#). These inherited results and their proofs are not claimed as new results of this manuscript. OpenAI is the immediate source of the adapted statement and proof text, not necessarily the original source of every standard mathematical ingredient; the earlier-source citations are retained. The assumptions and qualifications remain in the statements; the table records provenance rather than asserting verbatim identity of the expositions.

In particular, the independent Gaussian reserve and the sampling pre-reserve comparison in [Theorem A.1](#) and [Proposition F.7](#) are already part of OpenAI’s Theorem 2.1 and Proposition 7.7. The underlying finite mean and sampling formulas in [Theorem E.7](#) are adapted from its Theorem 6.7.

The mean-root specialization in [Section G](#) starts the inherited recursion at its mean routine and proves the stated mean-output guarantees. It is a specialization of the construction in OpenAI [14, Sections 6–7], rather than an independent invention of the underlying noisy-mean or sampling machinery. The posterior routine in [Proposition 6.3](#) adapts the conditional-descent construction of [14, Section 8.2], with the arbitrary-center estimates and terminal rule stated here; related posterior descent appears in [5, Algorithm 3.3 and Section 6.3]. The manuscript-specific outer construction is the centered conditional-Hamiltonian phase, its numerical entropy comparison, and its combination with the original-gradient implementation. The large-condition-number branch retains the standard BPS algorithm and windowed-thinning analysis of Lu and Luo [12, Algorithm 1].

This manuscript	OpenAI [14]	Result or construction
Theorem A.1	Theorem 2.1	Noisy near-quadratic sampling
Lemma A.2	Lemma 2.2	Uniformly convex laws
Lemma A.3	Lemma 2.3	Conditional comparisons and smoothing
Lemma B.1	Lemma 3.1	All-split cumulant bound
Proposition B.2	Proposition 3.2	Gaussian jet bounds
Proposition B.3	Proposition 3.3	Probability transport
Lemma C.1	Lemma 4.1	Material derivative identities
Lemma C.2	Lemma 4.2	A single contraction
Definition C.3	Definition 4.3	Tensor expressions and their weights
Theorem C.4	Theorem 4.4	Tensor divergence bound
Lemma C.5	Lemma 4.5	Hilbert–Schmidt divergence estimate
Corollary C.6	Corollary 4.6	Derivatives along transported paths
Lemma D.1	Lemma 5.1	Transport derivatives
Lemma D.2	Lemma 5.2	The centering kernel
Proposition D.3	Proposition 5.3	A noisy mean from a path integral
Lemma D.4	Lemma 5.4	Stationary time derivatives
Proposition E.1	Proposition 6.1	Probability-flow circuit
Lemma E.2	Lemma 6.2	Errors in the angle computation
Lemma E.3	Lemma 6.3	Derivative quadrature at zero
Proposition E.4	Proposition 6.4	Kernel-action circuit
Proposition E.5	Proposition 6.5	Mean-flow circuit
Corollary E.6	Corollary 6.6	Accuracy without increasing dependency depth
Theorem E.7	Theorem 6.7	Finite mean and sampling formulas
Lemma F.1	Lemma 7.1	Absorbing a Gaussian center displacement

This manuscript	OpenAI [14]	Result or construction
Lemma F.2	Lemma 7.2	Bounded routine height
Lemma F.3	Lemma 7.3	Well-defined construction and query cap
Lemma F.4	Lemma 7.4	Terminal accuracy
Lemma F.5	Lemma 7.5	Size conditions descend through the routine tree
Lemma F.6	Lemma 7.6	Fixed-seed stability
Proposition F.7	Proposition 7.7	Accuracy of the compiled routines

Adaptation and license notice. The adapted exposition reorganizes the source material, reconciles notation, namespaces labels and citations, and omits unrelated outer-sampler material. These are modifications of the cited OpenAI exposition; the additional mean-root, affine-composition, and outer-sampler arguments are identified separately above. The source correspondence records the immediate provenance of the inherited statements and proofs.

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